
IBD CLINICAL KNOWLEDGE RULEBOOK

AI Model Training Edition

15

Guidelines Extracted

667

Clinical Rules Captured

2019

to 2025 Publication Range

Source: IBD-Guidelines.pdf

Societies: ECCO, ACG, IOIBD, APMA, CCFI

Purpose: Structured clinical rules for AI/LLM ingestion

Citation format: Vancouver (first author et al.)

All recommendations include evidence level / GRADE strength where available

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Lichtenstein GR et al. *Am J Gastroenterol*. 2025;120(6):1225-1264. doi:10.14309/ajg.0000000000003465

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Turner D et al. *Gastroenterology*. 2021;161(5):1551-1569. doi:10.1053/j.gastro.2021.06.051

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Akyuz F et al. *Intest Res*. 2024. doi:10.5217/ir.2024.00089

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European Crohn's and Colitis Guidelines on Sexuality, Fertility, Pregnancy, and Lactation

File: 1. ECCO_IBD and Fertility etc.pdf

Torres J et al. *J Crohns Colitis*. 2023;17(1):1-27. doi:10.1093/ecco-jcc/jjac115

36 rules extracted

Statement 1

In patients with IBD there is an increased risk of sexual dysfunction, particularly in females and in those with active disease or perianal disease [EL3]

Evidence: [EL3]

Statement 2

The efficacy of oral contraceptives does not seem to be reduced in women with IBD [EL5]. Oral contraceptives do not seem to be associated with increased probability of IBD flares [EL2]. Oral contraceptives are generally low-risk in women with IBD; nevertheless, a careful assessment of thrombotic risk is recommended before prescription[EL5]

Evidence: [EL5], [EL2], [EL5]

Statement 4

Pre-conception counselling in IBD is associated with improved pregnancy outcomes [EL3]. Individuals with IBD who are planning pregnancy should undergo prepregnancy counselling to address parental concerns, to aim for disease remission, and to discuss medication use during pregnancy [EL5]

Evidence: [EL3], [EL5]

Statement 5

Patients with IBD, particularly with CD, are more likely to choose voluntary childlessness than healthy controls [EL2]. Lack of pregnancy-specific IBD knowledge may impact on voluntary childlessness. Therefore, appropriate education on pregnancy and family planning for all patients with IBD of childbearing age is recommended[EL5]

Evidence: [EL2], [EL5]

Statement 3

Two studies have directly addressed the impact of preconception care on pregnancy-related outcomes. One study in IBD patients compared outcomes in 155 patients with an active pregnancy desire, who were referred to a dedicated IBD pre-conception clinic, with 162 patients who attended the clinic only after they became pregnant. Counselling improved medication compliance and smoking cessation, reduced flares during pregnancy, and lowered the risk of delivering a LBW infant.⁵⁶ Another study reported outcomes from a multidisciplinary, single-centre clinic created to manage women with IBD and their neonates ['IBD MOM' clinic]. Patients with severe IBD were more likely to be referred to the clinic. Outcomes of 90 women who attended the clinic were compared with 206 IBD patients who attended community clinics and 61 689 controls. Pregnant women with moderate or severe IBD, who attended the 'IBD MOM' clinic, achieved similar perinatal outcomes as women with milder forms of IBD and were comparable to the general population regarding pregnancy complications, birthweight, and caesarean section [C-section] rates, but had significantly higher rates of preterm delivery, likely attributable to disease severity.⁵⁷ Based on these studies, early referral to specialist centres with a multidisciplinary team [including IBD physicians, maternal-fetal medical specialists, psychologists, and colorectal surgeons] should be considered.

Statement 6

Active disease is associated with decreased fertility in women with IBD [EL3]. Achieving clinical remission may increase the probability of successful conception. Active disease is also associated with decreased fertility in men with IBD [EL4]

Evidence: [EL3], [EL4]

Statement 7

Most of the commonly used IBD drugs have no demonstrated impact on fertility, in particular sperm quality [EL4]. Sulphasalazine is associated with reversible oligospermia and asthenozoospermia [EL2]

Evidence: [EL4], [EL2]

Statement 8

IBD-related pelvic surgeries lead to decreased fertility and fecundity rates in women [EL2]. Laparoscopic surgical approaches may lower the risk of infertility [EL2]

Evidence: [EL2], [EL2]

Statement 9

In women who have had restorative proctocolectomy with IPAA for UC, IVF procedures are three times more likely than in women without restorative proctocolectomy. However, both groups have a similar probability of having a live birth after IVF [EL3]

Evidence: [EL3]

Statement 10

Pregnancy may increase the risk of relapse or worsening disease in patients with UC and complications in patients with CD, specially if disease is active at conception [EL3]. IBD remission before conception is recommended [EL2]

Evidence: [EL3], [EL2]

Statement 11

Faecal calprotectin can reliably monitor disease activity during pregnancy [EL2]. Some blood parameters, such as haemoglobin and CRP, are affected by pregnancy and may not be reliable, although trends can be helpful [EL5]

Evidence: [EL2], [EL5]

Statement 12

During pregnancy, endoscopy can be performed when needed to guide clinical decision making [EL3]. Capsule endoscopy during pregnancy is considered a contraindication [EL5]. Ultrasonography [EL4] and MRI without the use of gadolinium [EL4] are radiation-free and are recommended instead of a CT scan [EL5]

Evidence: [EL3], [EL5], [EL4], [EL4], [EL5]

Statement 13

Pregnancy and IBD increase the risk of VTE events [EL2]. VTE risk assessment should be performed before conception and during pregnancy, especially in patients with active disease [EL5]. LMWH is recommended after C-section, during hospital admission for disease flares, or when other risk factors of VTE are present [EL5]

Evidence: [EL2], [EL5], [EL5]

Statement 14

A multidisciplinary team that includes an experienced gastroenterologist, obstetrician, and surgeon may be valuable in helping to optimise outcomes in pregnant women with a disease flare [EL5]. The choice of therapy for a flare during pregnancy should consider the severity of disease activity and the gestational age [EL5]

Evidence: [EL5], [EL5]

Statement 15

Pregnant women experiencing a flare should be managed according to current guidelines for non-pregnant patients, with 5-ASA, steroids, ciclosporin, anti-TNF agents [EL4], ustekinumab, or vedolizumab [EL5]. Initiating monotherapy with a thiopurine is generally not recommended due to the slow onset of action and the potential risk of adverse events [EL5]. Currently, JAK inhibitors and S1P receptor modulators should be avoided during pregnancy [EL5]

Evidence: [EL4], [EL5], [EL5], [EL5]

Statement 16

In case of a flare beyond gestational week 37, early delivery could be considered prior to initiation of medical therapy [EL5]

Evidence: [EL5]

Statement 17

Indications for surgery in pregnant women with IBD are the same as for non-pregnant patients [EL5]. The indication for IBD-related surgery during pregnancy should be determined promptly, based on IBD severity and general maternal conditions. Urgent surgery should be performed if clinically indicated, regardless of the gestational age [EL5]. The surgical management should be discussed in a multidisciplinary team involving gastroenterologists, colorectal surgeons, obstetricians, and neonatal specialists, as required [EL5]

Evidence: [EL5], [EL5], [EL5]

Statement 18

Patients on thiopurine monotherapy can continue treatment throughout pregnancy [EL5]. When thiopurines are used as combination therapy with biologics, thiopurine discontinuation may be considered on an individual basis if the patient is in long-term remission [EL5]. Demonstration of adequate serum anti-TNF levels may be helpful in this setting [EL5]

Evidence: [EL5], [EL5], [EL5]

Statement 19

For women with active disease just before or during pregnancy, or with disease that is difficult to control, continuation of anti-TNF [EL3] or non-TNF biologics [EL5] throughout pregnancy is recommended. The last dose of anti-TNF in the third trimester should be timed in accordance with the presumed due date to reduce foetal exposure [EL5]

Evidence: [EL3], [EL5], [EL5]

Statement 20

For women in remission, discontinuing anti-TNF prior to the third trimester is not recommended, as it may increase the risk of relapse [EL3] and lead to unfavourable pregnancy outcomes [EL3]. However, if a pregnant patient in long-term remission wishes to discontinue anti-TNF prior to the third trimester, resumption of anti-TNF shortly after delivery is recommended [EL5]

Evidence: [EL3], [EL3], [EL5]

Statement 21

For women in remission treated with non-TNF biologic agents [ustekinumab, vedolizumab], an individualised - decision on discontinuing treatment should be made, considering the risk of relapse and the limited data on the consequences of fetal exposure [EL5]

Evidence: [EL5]

Statement 22

Pregnant women with IBD seem to have a higher risk of gestational diabetes, stillbirth, preterm pre-labour rupture of membranes, preterm delivery, small for gestational age, and low birthweight newborns [EL2]. Disease activity during pregnancy is a risk factor, as it is associated with preterm pre-labour rupture of membranes, preterm birth, and low birthweight in IBD pregnant women. Disease activity during pregnancy also increases the risk of stillbirth and low Apgar score in CD [EL3]

Evidence: [EL2], [EL3]

Statement 23

Most drugs used for the treatment of IBD are considered low-risk during pregnancy [EL3]

Evidence: [EL3]

Statement 24

ASA is regarded as low-risk during pregnancy [EL3]

Evidence: [EL3]

Statement 25

Although short-term corticosteroids, including budesonide, can in general be regarded as low-risk during pregnancy for managing flares, the risk of maternal-fetal complications should be considered [EL3]

Evidence: [EL3]

Statement 26

In case of perianal or abdominal sepsis or pouchitis, metronidazole and ciprofloxacin can be administered in patients with IBD during pregnancy. However, given the risk of arthropathies in children, ciprofloxacin should be avoided if possible during the first trimester of pregnancy [EL4]

Evidence: [EL4]

Statement 28

Anti-TNF antibodies are regarded as low-risk during pregnancy [EL3]. Data for vedolizumab and ustekinumab are limited, but no increased risk of adverse pregnancy outcomes has been identified [EL4]

Evidence: [EL3], [EL4]

Statement 27

Thiopurines are not associated with significant neonatal adverse outcomes in pregnant patients with IBD [EL3]. Methotrexate [EL3], thalidomide [EL3], JAK inhibitors, and ozanimod are contraindicated during pregnancy [EL5]. Data for ciclosporin and tacrolimus in patients with IBD during pregnancy are limited

Evidence: [EL3], [EL3], [EL3], [EL5]

Statement 29

The mode of delivery does not seem to influence outcome of patients with IBD regarding development or worsening of inactive perianal disease [EL2] and anal sphincter damage [EL2]

Evidence: [EL2], [EL2]

Statement 30

Mode of delivery should be guided by obstetric considerations. In patients with active perianal disease, prior rectovaginal fistula, and after restorative proctocolectomy, C-section is recommended after multidisciplinary discussion involving gastroenterologists, obstetricians, and IBD surgeons [EL5]

Evidence: [EL5]

Statement 31

The risk of serious infections requiring hospital admission in the first 5 years of life does not seem to be increased in children exposed to anti-TNF agents or thiopurines during pregnancy [EL4]. However, there are no data beyond 5 years of follow-up on the effect of in utero exposure of IBD drugs on the developing immune system and neurodevelopment [EL4]. Safety data on children exposed to vedolizumab, ustekinumab, and tofacitinib are limited [EL5]

Evidence: [EL4], [EL4], [EL5]

Statement 32

There is no increased risk of postpartum relapse in patients with CD compared with non-pregnant patients with CD. In UC, there may be an increased risk of relapse postpartum [EL3]

Evidence: [EL3]

Statement 33

For patients who continue biologics during the entire pregnancy, the treatment should be continued uninterrupted in the postpartum phase unless there is a contraindication to their use [EL5]. For patients who interrupted treatment during pregnancy, the treatment should be resumed after delivery as soon as possible [EL5]. Re-induction or continuation of previous maintenance therapy is dependent on clinical circumstances [EL5]

Evidence: [EL5], [EL5], [EL5]

Statement 34

Inactivated vaccines are recommended according to national guidelines. In children exposed in utero to biologics, live attenuated vaccines should be withheld within the first year of life or until the biologic is no longer detectable in the infant's blood [EL3]

Evidence: [EL3]

Statement 35

Drugs that are considered low-risk during pregnancy are also considered low-risk during breastfeeding and thus can be continued [EL3]

Evidence: [EL3]

Statement 36

Breastfeeding does not seem to influence disease activity of patients with IBD [EL4]

Evidence: [EL4]

ACG Clinical Guideline Update: Ulcerative Colitis in Adults

File: 10. ACG_clinical_guideline_update__ulcerative_colitis.13.pdf

Rubin DT et al. *Am J Gastroenterol*. 2025;120(6):1187-1224. doi:10.14309/ajg.0000000000003463

145 rules extracted

Key Concept 1

UC should be suspected in patients presenting with hematochezia, increased stool frequency, or bowel urgency. Infectious aetiologies must be excluded at the time of initial diagnosis.

Key Concept 2

Colonoscopy with ileal intubation and biopsy from multiple segments (including the terminal ileum) is required to confirm the diagnosis of UC. Review by a pathologist with gastrointestinal expertise is preferred.

Key Concept 3

Disease extent categories in UC: (i) Proctitis — inflammation within 18 cm of the anal verge; (ii) Left-sided colitis — inflammation from the sigmoid colon to the splenic flexure; (iii) Extensive colitis — inflammation beyond the splenic flexure (includes pancolitis).

Key Concept 4

If the terminal ileum is normal on colonoscopy: no upper endoscopy or small-bowel imaging is required unless clinical findings otherwise suggest Crohn's disease.

Key Concept 5

Endoscopic severity must be reported using a validated scoring scale: the Mayo Endoscopic Score (MES) or the Ulcerative Colitis Endoscopic Index of Severity (UCEIS).

Key Concept 6

Disease monitoring may use fecal calprotectin (FC), C-reactive protein (CRP), flexible sigmoidoscopy or colonoscopy, or intestinal ultrasound. Select the monitoring modality based on clinical context, disease extent, and patient preference.

Key Concept 7

Patients with mildly-to-moderately active UC who have poor prognostic factors should be treated with therapies appropriate for moderate-to-severe disease. Poor prognostic factors include: extensive colitis, deep ulcers, elevated CRP or FC, prior hospitalization, young age at diagnosis, and need for corticosteroids at first flare.

Key Concept 8

Patients with mildly-to-moderately active UC should be formally reassessed at 8 weeks to determine response to induction therapy. Adjust or escalate therapy based on clinical, biochemical, and endoscopic response.

Key Concept 9

There is insufficient evidence to recommend routine probiotics, prebiotics, or fecal microbiota transplantation (FMT) as primary induction therapy in mild-to-moderate UC.

Key Concept 10

5-ASA therapy for maintenance is likely LESS effective in patients who previously had severely active UC compared with those who had moderately active UC. Consider advanced therapy for maintenance after severe flares.

Key Concept 11

Budesonide MMX has NOT been studied for maintenance of remission in moderate-to-severely active UC and should not be used for this indication.

Key Concept 12

Most clinical trials demonstrate benefit from using the same agent that induced remission as the maintenance therapy. Switching mechanism of action for maintenance without clinical justification is not recommended.

Key Concept 13

There is insufficient evidence supporting proactive therapeutic drug monitoring (TDM) in ALL unselected UC patients in remission. Reactive TDM (in response to loss of response) is recommended.

Key Concept 14

There is no evidence to recommend routine measurement of serum concentrations of vedolizumab, ustekinumab, guselkumab, mirikizumab, or risankizumab. TDM for these agents remains investigational.

Key Concept 15

Obtain surgical consultation in patients with moderate-to-severe UC who are refractory or intolerant to medical therapy. Surgical decision-making should involve a multidisciplinary team.

Key Concept 16 (Anti-TNF: TDM in non-response)

In UC patients who are non-responsive to anti-TNF therapy: assess trough drug levels and antibody formation to identify the mechanism of non-response and guide the decision to optimise existing therapy versus switch to an alternative mechanism.

Key Concept 17 (Anti-TNF: primary non-response)

PRIMARY non-responders to anti-TNF agents (lack of clinical benefit despite adequate serum drug levels) should be evaluated for an alternative mechanism of action. Do NOT cycle to another anti-TNF agent within the same class in primary non-responders.

Key Concept 18 (Biosimilars)

Biosimilars to anti-TNF agents and ustekinumab are acceptable alternatives to reference biologics. No delays in switching are necessary; patients and clinicians should be notified of any biosimilar substitution.

Key Concept 19 (SC vs IV biologics)

Subcutaneous infliximab and subcutaneous vedolizumab are equivalent to IV maintenance dosing. Equivalence for induction dosing or escalated doses has not been robustly established.

Key Concept 20 (No predictive biomarker)

There are currently no validated therapeutic biomarkers or companion diagnostic tests to predict treatment response in active UC. Therapy selection is based on clinical factors, disease severity, and patient preference.

Key Concept 21 (Infliximab preferred anti-TNF)

Infliximab is the PREFERRED anti-TNF therapy for patients with moderate-to-severe UC, based on the largest evidence base and established efficacy.

Key Concept 22 (Infection risk: vedolizumab or anti-IL-23)

Patients at higher risk for infectious complications may benefit from vedolizumab or anti-IL-23 agents (risankizumab, guselkumab, mirikizumab) over more systemically immunosuppressive options such as anti-TNF or JAK inhibitors.

Key Concept 23 (EIM-guided therapy selection)

Initial and subsequent therapies for moderate-to-severe UC may be chosen based on extraintestinal manifestations (EIMs). Select therapies with demonstrated efficacy for both UC and the specific EIM (e.g., anti-TNF for peripheral arthritis, ankylosing spondylitis, pyoderma gangrenosum).

Key Concept 24 (First therapy higher response rates)

First-line therapies for UC have higher response and remission rates than therapies used after prior failure. Optimise first-line therapy before escalating.

Key Concept 25 (Primary vs secondary non-response)

Distinguishing between primary non-response (inadequate benefit from start) and secondary non-response (loss of initially adequate response) is critical for selecting the next therapy. Primary non-responders should switch mechanism; secondary non-responders may benefit from dose optimisation or switching within the same class.

Key Concept 26 (ASUC: sigmoidoscopy timing)

Flexible sigmoidoscopy should be performed within 72 hours (preferably within 24 hours) of ASUC admission — to assess endoscopic severity and obtain biopsies to exclude CMV colitis.

Key Concept 27 (ASUC: toxic megacolon)

Assess for toxic megacolon in ALL patients presenting with ASUC. Toxic megacolon criteria: colonic dilatation >6 cm on plain radiograph plus systemic features (fever >38°C, HR >120 bpm, neutrophilia, anaemia, hypotension, dehydration).

Key Concept 28 (ASUC: medications to avoid)

NSAIDs, narcotic analgesics, and anticholinergic agents should be AVOIDED in patients with ASUC — they can precipitate or worsen toxic megacolon.

Key Concept 29 (ASUC: Day 3 reassessment)

If failing to adequately respond to IV corticosteroids by Day 3, OR if toxicity is suspected: obtain surgical consultation and consider rescue therapy (infliximab or cyclosporine).

Key Concept 30 (ASUC: infliximab vs cyclosporine)

Choice between infliximab and cyclosporine for ASUC rescue therapy should be based on: provider experience with each agent, history of prior failure of immunomodulator or anti-TNF therapy, and serum albumin level (low albumin <2.5 g/dL favours infliximab with dose intensification).

Key Concept 31 (ASUC: surgery indications)

Absolute indications for surgery in ASUC: toxic megacolon, colonic perforation, severe refractory haemorrhage, and refractoriness to all available medical therapy.

Key Concept 32 (ASUC: prior biologic not contraindication to surgery)

Infliximab and cyclosporine do NOT increase postoperative complications of colectomy. Surgery should NOT be deferred in a patient requiring urgent colectomy based solely on prior or recent exposure to these agents.

Key Concept 33 (ASUC: IFX dose intensification)

In ASUC patients initiating infliximab rescue therapy: consider dose intensification (10 mg/kg) in patients with low serum albumin (<2.5 g/dL) — due to increased drug clearance in severe inflammation and protein loss.

Key Concept 34 (ASUC: tofacitinib/upadacitinib insufficient data)

There are INSUFFICIENT DATA to routinely recommend tofacitinib or upadacitinib in ASUC failing to adequately respond to IV corticosteroids or infliximab induction. These agents should not be used for this indication outside of clinical trials.

Recommendation 1

We recommend stool testing to rule out *Clostridioides difficile* in patients suspected of having UC. (Strong recommendation, very low quality of evidence)

Recommendation 2

We recommend against serologic antibody testing to establish or rule out a diagnosis of UC. (Strong recommendation, very low quality of evidence)

Recommendation 3

We recommend against serologic antibody testing to determine the prognosis of UC. (Strong recommendation, very low quality of evidence)

Key Concept 1

The diagnosis of UC should be suspected in patients with hematochezia, increased stool frequency, or bowel urgency.

Key Concept 2

Infectious etiologies should be excluded at the time of diagnosis.

Key Concept 3

Colonoscopy with intubation of the ileum and biopsies of affected and unaffected areas should be obtained to confirm the diagnosis of UC, with mucosal biopsies interpreted by a pathologist, preferably one with expertise in gastrointestinal pathology.

Key Concept 4

Categories of disease extent include (i) proctitis (within 18 cm of anal verge, distal to rectosigmoid junction), (ii) left-sided colitis (extending from sigmoid to splenic flexure), (iii) extensive colitis (beyond splenic flexure which includes those with involvement of the entire colorectum [pancolitis]).

Key Concept 5

If the terminal ileum is normal, further evaluation of the stomach and small bowel by upper endoscopy and cross-sectional imaging is not needed unless there are other symptoms or findings to suggest proximal gastrointestinal involvement or a diagnosis of CD rather than UC.

Key Concept 6

Definitions of disease severity are needed to guide treatment decisions; definitions should be based on (i) patient-reported outcomes (bleeding, normalization of bowel habits, bowel urgency), (ii) the inflammatory burden (endoscopic assessment including extent and severity, and markers of inflammation including fecal calprotectin [FC], C-reactive protein [CRP], and serum albumin), (iii) disease course (need for hospitalization, need for steroids, failure to respond to medications), and (iv) disease impact (HRQoL and social functioning).

Key Concept 7

Endoscopic severity should be reported using a validated endoscopic scale such as the Mayo Endoscopic Score (MES) or the UC Endoscopic Index of Severity (UCEIS).

Key Concept 8

Disease assessment and monitoring in response to therapy and during maintenance and periods of suspected relapse may be performed with FC, CRP, endoscopic assessment with flexible sigmoidoscopy or colonoscopy, and/or intestinal ultrasound.

Recommendation 4

We recommend treating patients with UC to achieve endoscopic improvement (defined as resolution of inflammatory changes [MES 0 or 1]) to increase the likelihood of sustained steroid-free remission and to prevent hospitalizations and surgery. (Strong recommendation, moderate quality of evidence)

Recommendation 5

We recommend the use of FC in UC to assess response to therapy, to evaluate suspected relapse, and during maintenance. (Strong recommendation, moderate quality of evidence)

Key Concept 9

UC is a chronic condition for which therapy is required to induce and maintain remission, therapeutic decisions should be categorized into those for (i) induction and (ii) maintenance, with goals of obtaining and maintaining a steroid-free remission and obtaining biological response through reduction in biomarkers or endoscopic improvement.

Key Concept 10

Strategies for management of UC should reflect the patient's and provider's goals and recognize the chronic nature of the disease.

Key Concept 11

Symptomatic remission relates to improvement in PROs while endoscopic healing is defined as restoration of intact mucosa without friability. Deep remission is a combination of symptomatic remission and endoscopic healing and is a preferred goal of management. Corticosteroid-free remission is defined based on symptoms and endoscopic findings without corticosteroid use for a sustained period of time (usually more than 12 weeks).

Key Concept 12

Initial treatment of UC should focus on restoration of normal bowel frequency and control of the primary symptoms of bleeding and bowel urgency. An endoscopically healed mucosa is associated with sustained remission and reduced risk of colectomy.

Key Concept 13

Histologic remission is associated with some improved clinical outcomes but has not yet been validated prospectively as a preferred target for treatment.

Key Concept 14

Control of mucosal inflammation may reduce dysplasia risk.

Key Concept 15

Given the chronic nature of UC and the therapies for UC, monitoring for disease-related and drug-related complications is important. This should incorporate preventive strategies as outlined here and in a separate guideline from the ACG (100).

Key Concept 16

Routine visits are recommended to monitor for relapse and address health maintenance needs.

Key Concept 17

Patients with UC should be screened for coexistent anxiety and depressive disorders, and when identified, patients should be provided with resources to address these conditions.

Recommendation 6

In patients with mildly to moderately active ulcerative proctitis, we recommend rectal 5-aminosalicylate acid (5-ASA) therapies at a dose of 1 g/daily for induction of remission. (Strong recommendation, moderate quality evidence)

Recommendation 7

For patients with mildly to moderately active proctitis not responsive to topical 5-ASA, we suggest tacrolimus suppository or beclomethasone suppository over no treatment. (Conditional recommendation, low quality of evidence)

Recommendation 8

For patients with mildly to moderately active proctitis or left-sided colitis, we suggest use of topical corticosteroids (suppository, foam, enema), over no treatment. (Conditional recommendation, very low quality of evidence)

Recommendation 9

In patients with mildly to moderately active proctitis or left sided colitis, we recommend rectal 5-ASA enemas at a dose of at least 1 g/daily preferred over rectal steroids for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 10

In patients with mildly to moderately active left-sided UC, we suggest rectal 5ASA enemas at a dose of at least 1 g/daily combined with oral 5-ASA at a dose of at least 2.0 g/daily compared with oral 5-ASA therapy alone for induction of remission. (Conditional recommendation, low quality of evidence)

Recommendation 11

In patients with mildly to moderately active left-sided UC, who are intolerant or nonresponsive to oral and rectal 5-ASA at appropriate doses (oral at least 2.0 g daily and rectal at least 1 g daily), we recommend oral budesonide Multi Matrix System (MMX) 9 mg/d for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 12

In patients with mildly to moderately active extensive colitis, oral 5-ASA at a dose of at least 2.0 g daily is recommended to induce remission. (Strong recommendation, moderate quality of evidence)

Recommendation 13

In patients with UC of any extent who fail to respond to 5-ASA therapy, we recommend oral systemic corticosteroids to induce remission. (Strong recommendation, low quality of evidence)

Recommendation 14

In patients with mildly to moderately active UC who fail to reach remission with appropriately dosed 5-ASA (at least 2–4.8 g daily oral mesalamine and/or at least 1 g daily rectal mesalamine), we suggest against changing to an alternate 5-ASA formulation to induce remission. Alternative therapeutic classes should be considered. (Conditional recommendation, low quality of evidence)

Recommendation 15

In patients with mildly active UC of any extent, we suggest using a low dose (2.0–2.4 g) of 5-ASA, in comparison with a higher dose (4.8 g), because there is no difference in remission rate. (Conditional recommendation, very low quality of evidence)

Recommendation 16

In patients with mildly to moderately active UC of any extent not responding to oral 5-ASA, we recommend the addition of budesonide MMX 9 mg/d to induce remission. (Strong recommendation, moderate quality of evidence)

Recommendation 17

In patients with mildly to moderately active UC of any extent using 5-ASA to induce remission, we recommend either once daily or more frequently dosed oral 5-ASA based on patient preference to optimize adherence because efficacy and safety are no different. (Strong recommendation, moderate quality evidence)

Recommendation 18

In patients with mildly active ulcerative proctitis, we recommend rectal 5-ASA at a dose of 1 g daily for maintenance of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 19

In patients with mildly active left-sided or extensive UC, we recommend oral 5ASA therapy (at least 1.5 g/d) for maintenance of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 20

We recommend against systemic, budesonide MMX, or topical corticosteroids for maintenance of remission in patients with UC. (Strong recommendation, moderate quality of evidence)

Key Concept 18

Patients with mildly to moderately active UC and a number of prognostic factors associated with an increased risk of hospitalization or surgery should be treated with therapies for moderate-to-severe disease (Table 8). Each prognostic factor carries a different weight and must be discussed in a shared decision-making fashion with the patient. For example, age alone is a weaker prognostic factor than severe endoscopic activity. However, young age combined with another factor may represent sufficient criteria to treat using therapies with proven efficacy in patients with moderate-to-severe UC.

Key Concept 19

Patients with mildly to moderately active UC should be reassessed to determine response to induction therapy within 8 weeks.

Key Concept 20

There is not sufficient evidence for routine use of probiotics, prebiotics, or other alternative therapies as primary induction therapy for patients with mildly to moderately active UC.

Key Concept 21

There is not sufficient evidence of an optimal approach to fecal microbial transfer as a primary induction treatment for patients with mildly to moderately active UC.

Key Concept 22

Patients with previously mildly to moderately active UC who have achieved remission should be treated with maintenance therapy with demonstrated efficacy in prevention of relapse.

Key Concept 23

In patients with previously mildly to moderately active UC who have achieved remission, there is insufficient evidence to recommend the use of a probiotic as primary or adjunctive therapy for maintenance of remission.

Recommendation 21

In patients with moderately active UC, we recommend oral budesonide MMX for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 22

In patients with moderately to severely active UC of any extent, we recommend oral systemic corticosteroids to induce remission. (Strong recommendation, low quality of evidence)

Recommendation 23

In patients with moderately to severely active UC, we recommend against monotherapy with thiopurines or methotrexate for induction of remission. (Strong recommendation, low quality of evidence)

Recommendation 24

In patients with moderately to severely active UC, we recommend sphingosine-1-phosphate (S1P) receptor modulators, ozanimod and etrasimod, for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 25

In patients with moderately to severely active UC, we recommend the interleukin (IL)-12/23p40 antibody ustekinumab for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 26

In patients with moderately to severely active UC, we recommend the IL23p19 inhibitor guselkumab, mirikizumab, or risankizumab for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 27

In patients with moderately to severely active UC, we recommend vedolizumab for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 28

In patients with moderately to severely active UC, we recommend anti-tumor necrosis factor (TNF) therapy using infliximab for induction of remission. (Strong recommendation, high quality of evidence)

Recommendation 29

In patients with moderately to severely active UC, we recommend anti-TNF therapy using adalimumab or golimumab for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 30

In patients with moderately to severely active UC, we recommend the Janus kinase (JAK) inhibitor tofacitinib for induction of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 31

In patients with moderately to severely active UC, we recommend the JAK inhibitor upadacitinib for induction of remission. (Strong recommendation, high quality of evidence)

Recommendation 32

In patients with moderately to severely active UC who have failed 5-ASA therapy and in whom advanced therapies with biologics or JAK inhibitors are used for induction of remission, we suggest against using 5-ASA for added clinical efficacy. (Conditional recommendation, very low quality of evidence)

Recommendation 33

When infliximab is used as induction therapy for patients with moderately to severely active UC, we recommend combination therapy with a thiopurine. (Strong recommendation, moderate quality of evidence for azathioprine)

Key Concept 24

Patients with mildly to moderately active UC who are not responsive (or are intolerant) to 5-ASA therapies should be treated as patients with moderate-to-severe disease.

Key Concept 25

Strategies for the management of the nonhospitalized patient with moderately or severely active UC are similar with the exception of a few considerations in which the data exist specifically for a patient with moderately active UC: a. 5-ASA therapy could be used as monotherapy for induction of moderately but not severely active UC. b. In patients with moderately active UC, consider nonsystemic corticosteroids such as budesonide MMX before the use of systemic steroid therapy. c. In patients with severely active UC, consider systemic corticosteroids rather than topical corticosteroids. d. Corticosteroids may be avoided entirely when other effective induction strategies are planned.

Key Concept 26

The extent of bowel involvement in moderately to severely active UC should not limit the choice of advanced therapies for these patients. This includes patients with moderately to severely active isolated proctitis who should have access to and be treated with therapies with demonstrated efficacy in patients with more extensive UC of similar activity.

Key Concept 27

Data on combination anti-TNF and immunomodulators in moderately to severely active UC only exist for infliximab and thiopurines.

Key Concept 28

The patient with nonresponse or loss of response to anti-TNF therapy should be assessed with trough serum concentrations of drug to identify the reason for lack of response and whether to optimize the existing therapy or to select an alternate therapy.

Key Concept 29

Patients who are primary nonresponders to an anti-TNF (defined as lack of therapeutic benefit after induction and despite sufficient serum drug concentrations) should be evaluated and considered for alternative mechanisms of disease control (e.g., in a different class of therapy) rather than cycling to another drug within the anti-TNF class.

Key Concept 30

Biosimilars to anti-TNF therapies and to ustekinumab are acceptable substitutes for originator therapies. Delays in switching should not occur and patients and clinicians should be notified about such changes.

Key Concept 31

Subcutaneous infliximab and vedolizumab are considered equivalent to the standard intravenous maintenance dosing of these agents. The equivalence of the subcutaneous formulations for induction or as substitution for escalated doses of these therapies has not been robustly established.

Key Concept 32

Obtain consultation with a surgeon and consider colectomy in patients with moderately to severely active UC who are refractory or intolerant to medical therapy.

Recommendation 34

In patients with prior moderately to severely active UC who have achieved remission but previously failed 5-ASA therapy and are now on anti-TNF therapy, we suggest against using concomitant 5-ASA for efficacy of maintenance of remission (162). (Conditional recommendation, low quality of evidence)

Recommendation 35

In patients with prior moderately to severely active UC, we recommend against systemic corticosteroids for maintenance of remission. (Strong recommendation, moderate quality of evidence)

Recommendation 36

For patients with prior moderately to severely active UC now in remission because of corticosteroid induction, we suggest thiopurines for maintenance of remission as compared with no treatment or corticosteroids. (Conditional recommendation, low quality of evidence)

Recommendation 37

In patients with prior moderately to severely active UC now in remission, we suggest against using methotrexate for maintenance of remission. (Conditional recommendation, low quality of evidence)

Recommendation 38

We recommend continuing S1P receptor modulators ozanimod or etrasimod for maintenance of remission as compared with no treatment after induction of remission with these agents. (Strong recommendation, moderate quality of evidence)

Recommendation 39

We recommend continuing ustekinumab for maintenance of remission as compared to no treatment in patients who responded to the induction dose of this medication. (Strong recommendation, moderate quality of evidence)

Recommendation 40

We recommend continuing guselkumab, mirikizumab, or risankizumab as compared with no treatment for maintenance of remission in patients who respond to the induction dosing of the same treatment. (Strong recommendation, moderate quality of evidence)

Recommendation 41

We recommend continuing vedolizumab as compared with no treatment for maintenance of remission (intravenous [IV] or subcutaneous [SC] dosing) in patients with prior moderately to severely active UC now in remission after vedolizumab induction. (Strong recommendation, moderate quality of evidence)

Recommendation 42

We recommend continuing anti-TNF therapy using adalimumab, golimumab, or infliximab (IV or SC dosing) for maintenance of remission after anti-TNF induction in patients with prior moderately to severely active UC. (Strong recommendation, moderate quality of evidence)

Recommendation 43

We recommend continuing tofacitinib or upadacitinib as compared with no treatment for maintenance of remission in patients with prior moderately to severely active UC now in remission after induction with tofacitinib or upadacitinib. (Strong recommendation, moderate quality of evidence)

Key Concept 33

5-ASA therapy for maintenance of remission is likely not as effective in prior severely active UC as compared with prior moderately active UC (140).

Key Concept 34

Budesonide MMX has not been studied for maintenance of remission of prior moderately to severely active UC.

Key Concept 35

Most clinical trials and available data demonstrate a benefit of using the steroid-sparing therapy that induces remission to maintain that remission.

Key Concept 36

There is insufficient evidence supporting a benefit for proactive therapeutic drug monitoring in all unselected patients with UC in remission.

Key Concept 37

There is insufficient evidence to recommend assessment of serum concentrations of vedolizumab, ustekinumab, guselkumab, mirikizumab, or risankizumab.

Key Concept 38

Patients with moderately to severely active UC who do not maintain remission despite optimized medical therapy should be considered for elective proctocolectomy.

Key Concept 39

A patient with moderately to severely active disease regardless of the extent of bowel involvement (including isolated proctitis) should be treated with therapies that have demonstrated efficacy for the activity and severity of the disease.

Recommendation 44

In patients with moderately to severely active UC who are responders to anti-TNF therapy and now losing response, we suggest measuring serum drug levels and antidrug antibodies (if there is not sufficient drug present) to assess reason for loss of response. (Conditional recommendation, very low quality of evidence)

Recommendation 45

In patients with moderately to severely active UC, we recommend vedolizumab as compared with adalimumab for induction and maintenance of remission. (Strong recommendation, moderate quality of evidence)

Key Concept 40

There are no validated therapeutic biomarkers or companion diagnostic tests to enhance selection or predict response to treatment for the patient with active UC.

Key Concept 41

Patients with UC should have available all medical options as recommended by their doctor and healthcare team. Third-party payers and requirements for step therapy should not come between the patient and their healthcare team in making decisions about treatment for UC.

Key Concept 42

Patients with moderately to severely active UC have higher rates of response and remission with their first therapies than after failure of one or more other advanced therapies.

Key Concept 43

Given the expanding number of therapies per mechanistic class, a distinction between primary nonresponse and secondary nonresponse is important to select the next therapeutic option.

Key Concept 44

Post hoc subgroup analyses and network meta-analyses provide hypothesis-generating data but are not sufficient to stratify therapies for individual patients.

Key Concept 45

Infliximab is the preferred anti-TNF therapy for patients with moderately to severely active UC.

Key Concept 46

Some patients with moderately to severely active UC who are at higher risk for infectious complications may benefit from vedolizumab or an anti-IL-23 strategy over more systemically immunosuppressive medical options.

Key Concept 47

Initial and subsequent therapies for moderately to severely active UC may be chosen based on extraintestinal manifestations, including the involvement of joints or skin, in which therapies which have efficacy in both UC and the extraintestinal organ is known.

Recommendation 1

In patients with acute severe UC (ASUC), we recommend testing for CDI. (Strong recommendation, moderate quality of evidence)

Recommendation 2

In patients with ASUC, we recommend pharmacologic DVT prophylaxis as compared with no prophylaxis to prevent VTE. (Strong recommendation, low quality of evidence)

Recommendation 3

We recommend against routine use of broad-spectrum antibiotics in the management of ASUC. (Strong recommendation, low quality of evidence)

Recommendation 4

We suggest against total parenteral nutrition for the purpose of bowel rest in ASUC. (Conditional recommendation, very low quality of evidence)

Recommendation 5

In patients with ASUC, we recommend a total of 60 mg/d of methylprednisolone or hydrocortisone 100 mg 3 or 4 times per day to induce remission. (Strong recommendation, low quality of evidence)

Recommendation 6

In patients with ASUC failing to adequately respond to intravenous corticosteroids (IVCS) by 3 days, we recommend medical rescue therapy with infliximab or cyclosporine. (Strong recommendation, moderate quality of evidence)

Recommendation 7

In patients with ASUC who achieve remission with infliximab treatment, we recommend maintenance of remission with the same agent. (Strong recommendations, moderate quality of evidence)

Recommendation 8

In patients with ASUC who achieve remission with cyclosporine treatment, we suggest maintenance of remission with thiopurines. (Conditional recommendation, low quality of evidence)

Recommendation 9

In patients with ASUC who achieve remission with cyclosporine treatment, we suggest maintenance of remission with vedolizumab. (Conditional recommendation, very low quality of evidence)

Key Concept 48

All patients with ASUC should undergo a flexible sigmoidoscopy within 72 hours and preferably within 24 hours of admission. This should be used to assess endoscopic severity of inflammation and to obtain biopsies to evaluate for cytomegalovirus (CMV) colitis.

Key Concept 49

All patients with ASUC should be assessed for the presence of toxic megacolon.

Key Concept 50

Response in patients with acute severe UC should be monitored using stool frequency, rectal bleeding, physical examination, vital signs, and serial CRP measurements.

Key Concept 51

Nonsteroidal anti-inflammatory drugs (NSAIDs), narcotics, and medications with anticholinergic side effects should be avoided in ASUC.

Key Concept 52

In patients with ASUC failing to adequately respond to medical therapy by 3 days or with suspected toxicity, surgical consultation should be obtained.

Key Concept 53

In patients with ASUC, the choice between infliximab and cyclosporine should be based on provider experience with the agent, history of prior failure of immunomodulator or anti-TNF therapy, and serum albumin.

Key Concept 54

Toxic megacolon, colonic perforation, severe refractory hemorrhage, and refractoriness to medical therapy are indications for surgery in patients with ASUC.

Key Concept 55

Infliximab and cyclosporine do not increase postoperative complications of colectomy, and surgery should not be deferred based on this exposure.

Key Concept 56

In patients with ASUC failing to adequately respond to IVCS by 3 days or to infliximab induction, there are insufficient data to routinely recommend treatment with tofacitinib or upadacitinib.

Key Concept 57

In patients with ASUC initiating infliximab, dose intensification should be considered for those patients with low serum albumin (<2.5 g/dL).

ACG Clinical Guideline: Management of Crohn's Disease in Adults

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Lichtenstein GR et al. *Am J Gastroenterol.* 2025;120(6):1225-1264. doi:10.14309/ajg.0000000000003465

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Recommendation 39 (No maintenance in silent mild CD)

No maintenance treatment is an option for some patients with asymptomatic (silent), mild Crohn's disease. However, routine monitoring is still required to identify disease progression — no treatment does not mean no follow-up. (Conditional recommendation, very low quality evidence)

Recommendation 40 (Sulfasalazine for mild colonic CD only)

Sulfasalazine should only be considered for patients with symptomatic mild COLONIC Crohn's disease. Sulfasalazine is NOT effective for ileal or ileocolonic CD. (Conditional recommendation, moderate quality evidence)

Recommendation 41 (Antibiotics not effective for luminal CD)

Antibiotics are NOT an effective treatment for luminal inflammatory Crohn's disease and should NOT be used as primary therapy for luminal CD. Antibiotics remain indicated for infectious complications (abscesses, perianal sepsis). (Strong recommendation, moderate quality evidence)

Recommendation 42 (Diet-based strategies in mild low-risk CD)

For adult patients with mild CD and low risk of disease progression, diet-based strategies (e.g., Mediterranean-type diet, CD exclusion diet) with careful monitoring may be considered as part of a comprehensive management approach. (Conditional recommendation, low quality evidence)

Recommendation 44 (Biosimilars for de novo CD)

Biosimilar infliximab, adalimumab, and ustekinumab are effective for de novo induction and maintenance therapy in patients with moderate-to-severe Crohn's disease. (Strong recommendation, moderate quality evidence)

Recommendation 45 (Biosimilar switching in stable CD)

There are safety and efficacy data supporting transition or switching to biosimilar infliximab or adalimumab in patients with Crohn's disease who are in stable remission on maintenance therapy. (Conditional recommendation, moderate quality evidence)

Recommendation 47 (IV corticosteroids in hospitalized CD)

For hospitalized patients with severe-to-fulminant Crohn's disease, IV corticosteroids may be used to control inflammatory burden while evaluating steroid-sparing options. IV corticosteroids are a bridge to definitive therapy, not a long-term solution. (Conditional recommendation, very low quality evidence)

Recommendation 48 (Anti-TNF in severe inpatient CD)

Anti-TNF agents are effective for severely active Crohn's disease. Infliximab may be administered in the inpatient setting for severe-to-fulminant CD, including as rescue therapy. (Conditional recommendation, low quality evidence)

Recommendation 49 (Antibiotics for simple perianal fistula)

Antibiotics (imidazoles: metronidazole or ciprofloxacin) can be considered for patients with SIMPLE perianal fistulas as primary therapy. Antibiotics provide symptomatic relief but do not achieve fistula healing — advanced therapy is required for definitive management. (Conditional recommendation, very low quality evidence)

Recommendation 50 (Drainage + seton before advanced therapy)

Drainage of perianal abscesses AND appropriate seton placement to facilitate drainage should be undertaken BEFORE commencing advanced therapy (biologics) for perianal fistulising Crohn's disease. Initiating biologic therapy in the presence of undrained sepsis is contraindicated. (Strong recommendation, moderate quality evidence)

Recommendation 51 (Stricture: triple assessment)

For CD patients with strictures, symptom assessment, radiological evaluation (MRE or CTE), and endoscopic assessment are ALL necessary to guide clinical approach. Distinguish between inflammatory strictures (amenable to medical therapy) and fibrostenotic strictures (requiring endoscopic or surgical intervention). (Strong recommendation, low quality evidence)

Recommendation 52 (Inflammatory stricture: medical therapy)

Symptomatic CD strictures WITH endoscopic or radiological evidence of active inflammation may respond to advanced therapies (biologics, JAK inhibitors). Treat the inflammatory component before considering endoscopic or surgical intervention. (Conditional recommendation, very low quality evidence)

Recommendation 53 (Fibrostenotic stricture: dilation or surgery)

Symptomatic CD strictures with predominantly fibrostenotic features (minimal active inflammation) may benefit from endoscopic balloon dilation (for short, accessible strictures) or surgical resection. Endoscopic dilation has acceptable short-term outcomes but high recurrence rates. (Conditional recommendation, low quality evidence)

Recommendation 54 (Prophylactic treatment post-resection)

Prophylactic treatment is recommended after small intestinal resection in patients with RISK FACTORS for postoperative CD recurrence. Risk-stratify all patients prior to initiating postoperative prophylaxis. (Strong recommendation, moderate quality evidence)

Recommendation 55 (Risk factors for postoperative recurrence)

Risk factors for postoperative CD recurrence that should inform treatment decisions: smoking, prior intestinal resection, penetrating disease behaviour, extensive small bowel involvement, perianal disease, and failure of prior immunosuppressive or biologic therapy. (Strong recommendation, moderate quality evidence)

Recommendation 56 (Surgery for localized short-segment CD)

Surgery may be considered as a primary therapeutic option for patients with symptomatic CD localized to a short segment of bowel (particularly terminal ileal disease), where benefits of resection may outweigh long-term medical therapy risks. (Conditional recommendation, low quality evidence)

Recommendation 57 (Surgery for enteric complications)

Surgery is required to treat enteric complications of Crohn's disease, including: intestinal obstruction unresponsive to medical therapy, fistulae causing significant morbidity, and abdominal abscesses not amenable to percutaneous drainage. (Strong recommendation, low quality evidence)

Recommendation 58 (Intestinal resection most common surgery)

Resection of a segment of diseased intestine is the most common surgical procedure for Crohn's disease. Bowel-sparing techniques (strictureplasty) should be prioritised where possible to preserve intestinal length. (Strong recommendation, low quality evidence)

Recommendation 59 (Abdominal abscess: drainage then reassess)

CD patients who develop abdominal abscess should initially undergo surgical or radiologically guided percutaneous drainage. Some patients may respond to medical therapy after drainage; others will require subsequent surgical resection of the involved segment. (Conditional recommendation, very low quality evidence)

Key Concept: Biologic dose optimisation

Biologic therapy dose optimisation (escalating dose or shortening interval) may be considered for patients with inadequate or loss of response to any biologic agent. Therapeutic drug monitoring should guide optimisation where available.

Key Concept: Thiopurine + anti-TNF combination (immunogenicity)

Azathioprine, 6-mercaptopurine, or methotrexate may be used as adjunctive therapy to reduce immunogenicity of anti-TNF agents. Combination therapy reduces anti-drug antibody formation and improves trough levels — this indication is independent of active disease control.

Key Concept: TPMT testing required before thiopurine

Thiopurine methyltransferase (TPMT) activity testing OR TPMT/NUDT15 genotyping is REQUIRED before the initial use of azathioprine or 6-mercaptopurine. Low or absent TPMT activity is a contraindication to standard-dose thiopurine therapy.

Recommendation 4

We recommend against the use of oral mesalamine for induction or maintenance in patients with mildly to moderately active CD (strong recommendation, moderate level of evidence).

Recommendation 5

We recommend controlled ileal release budesonide at a dose of 9 mg daily for induction of symptomatic remission in patients with mildly to moderately active ileocecal CD (strong recommendation, moderate level of evidence).

Recommendation 6

We recommend against the use of ileal release budesonide for maintenance of remission in patients with mildly to moderately active ileocecal CD (strong recommendation, low level of evidence).

Recommendation 39

No maintenance treatment is a treatment option for some patients with asymptomatic (silent), mild CD; however, routine monitoring is recommended to identify disease progression.

Recommendation 40

Sulfasalazine should only be considered for patients with symptomatic mild colonic CD.

Recommendation 41

Antibiotics are not an effective treatment for luminal inflammatory CD and should not be used as a primary therapy.

Recommendation 42

For adult patients with mild CD and low risk of progression, dietbased strategies along with careful monitoring for inadequate symptom relief, worsening inflammation, or disease progression may be considered.

Recommendation 7

We recommend oral corticosteroids for short-term induction of remission in patients with moderately to severely active CD (strong recommendation, low level of evidence).

Recommendation 8

We recommend against azathioprine (at doses of 1.5–2.5 mg/kg/d) and 6-mercaptopurine (at doses of 0.75–1.5 mg/kg/d) for induction of remission in moderately to severely active CD (strong recommendation, moderate level of evidence).

Recommendation 9

We suggest azathioprine (at doses of 1.5–2.5 mg/kg/d) and 6mercaptopurine (at doses of 0.75–1.5 mg/kg/d) for maintenance of remission in patients with moderately to severely active CD who had induction of remission with corticosteroids (conditional recommendation, low level of evidence).

Recommendation 10

We recommend TPMT testing before initial use of azathioprine or 6-mercaptopurine to treat patients with CD (strong recommendation, low level of evidence).

Recommendation 11

We suggest methotrexate (up to 25 mg once weekly intramuscular or subcutaneous) for maintenance of remission in patients with moderately to severely active CD who had induction of remission with corticosteroids (conditional recommendation, moderate level of evidence).

Recommendation 43

Azathioprine, 6-mercaptopurine, or methotrexate may be used in treatment of active CD and as adjunctive therapy for reducing immunogenicity associated with anti-TNF therapy.

Recommendation 12

We recommend anti-TNF agents (intravenous infliximab, subcutaneous adalimumab, subcutaneous certolizumab pegol) for induction and maintenance of remission for moderately to severely active CD (strong recommendation, moderate level of evidence).

Recommendation 13

We recommend combination therapy of intravenous infliximab with immunomodulators (thiopurines) as compared with treatment with either immunomodulators alone or intravenous infliximab alone in patients with CD who are naive to those agents (strong recommendation, moderate level of evidence).

Recommendation 14

We recommend subcutaneous infliximab as an option for maintenance of remission in patients with moderately to severely active CD who respond to intravenous induction with infliximab (strong recommendation, moderate level of evidence).

Recommendation 44

Biosimilar infliximab, adalimumab, and ustekinumab are effective treatments for patients with moderate-to-severe CD and can be used for de novo induction and maintenance therapy.

Recommendation 45

There are data to support the safety and efficacy of transitioning or switching to biosimilar infliximab or adalimumab for with patients CD in stable disease maintenance.

Recommendation 15

We recommend intravenous vedolizumab for induction and maintenance of symptomatic remission in patients with moderately to severely active CD (strong recommendation, moderate level of evidence).

Recommendation 16

We recommend subcutaneous vedolizumab as an option for maintenance of remission in patients with moderately to severely active CD who respond to 2 intravenous induction doses of vedolizumab (strong recommendation, moderate level of evidence).

Recommendation 17

We recommend ustekinumab in patients with moderate-to-severe CD for induction and maintenance of remission (strong recommendation, moderate level of evidence).

Recommendation 46

Biologic therapy (including anti-IL-12/23 therapy, anti-TNF therapy, and anti-integrin therapy) dose optimization may be considered for patients with inadequate or loss of response to that specific biologic agent's induction and maintenance.

Recommendation 18

We recommend the use of risankizumab for induction and maintenance of remission in patients with moderate to severely active CD (strong recommendation, moderate level of evidence).

Recommendation 19

We recommend the use of risankizumab as compared with ustekinumab in patients with moderate-to-severe CD and prior exposure to anti-TNF therapy (conditional recommendation low level of evidence).

Recommendation 20

We recommend the use of mirikizumab for induction and maintenance of remission in patients with moderate to severely active CD (strong recommendation, moderate level of evidence).

Recommendation 21

We recommend the use of intravenous guselkumab for induction followed by subcutaneous guselkumab for maintenance of remission in patients with moderate to severely active Crohn's disease (strong recommendation, moderate level of evidence).

Recommendation 22

We recommend the use of subcutaneous guselkumab for induction and maintenance of remission in patients with moderate to severely active Crohn's disease (strong recommendation, moderate level of evidence).

Recommendation 23

We recommend upadacitinib for induction and maintenance of remission for patients with moderately to severely CD who have previously been exposed to anti-TNF agents (strong recommendation, moderate level of evidence).

Recommendation 47

For hospitalized patients presenting with severe to fulminant CD, intravenous corticosteroids may be used to control inflammatory burden while evaluating steroid-sparing treatment options.

Recommendation 48

Anti-TNF agents are effective for severely active CD and infliximab may be administered in the inpatient setting for patients with severe to fulminant disease.

Recommendation 24

We recommend infliximab for induction of remission of perianal fistulizing CD (strong recommendation, moderate level of evidence).

Recommendation 25

We suggest adalimumab for induction of remission of perianal fistulizing CD (conditional recommendation, low level of evidence).

Recommendation 26

We suggest the use of antibiotics combined with infliximab or adalimumab to improve clinical response in perianal fistulizing CD (conditional recommendation, very low level of evidence)

Recommendation 27

We suggest vedolizumab for induction of remission of perianal fistulizing CD (conditional recommendation, very low level of evidence).

Recommendation 28

We suggest ustekinumab for induction of remission of perianal fistulizing CD (conditional recommendation, very low level of evidence).

Recommendation 29

We suggest upadacitinib use for induction of remission of perianal fistulizing CD (conditional recommendation, very low level of evidence).

Recommendation 49

Antibiotics (imidazoles) can be considered for patients with simple perianal fistulas as a primary therapy.

Recommendation 50

Drainage of perianal abscesses with appropriate placement of setons to facilitate drainage should be undertaken before treating perianal fistulizing disease with advanced therapy to increase treatment effectiveness.

Recommendation 51

For patients with stricturing CD, symptom, radiologic, and endoscopic assessments are necessary to help guide treatment approach.

Recommendation 52

Patients with CD with symptomatic strictures and evidence of active inflammation may respond to advanced therapies.

Recommendation 53

Patients with CD with symptomatic strictures plus endoscopic or radiologic features indicating more of a fibrostenotic predominant picture may benefit from endoscopic dilation or surgery.

Recommendation 30

In patients with surgically induced remission of CD, we suggest postoperative endoscopic assessment at 6–12 months over no monitoring (conditional recommendation, moderate level of evidence).

Recommendation 31

In patients with CD with low-risk recurrence of postoperative disease, we suggest continued observation as compared with immediate initiation of medical therapy for CD (conditional recommendation, very low level of evidence).

Recommendation 54

Prophylactic treatment is recommended after small intestinal resection in patients with risk factors for recurrence.

Recommendation 32

We suggest imidazole antibiotics (metronidazole) at doses between 1 and 2 g/d after small intestinal resection in CD patients to prevent recurrence (conditional recommendation, low level of evidence).

Recommendation 33

In patients with high-risk CD, we recommend anti-TNF therapy to prevent postoperative endoscopic recurrence (strong recommendation, moderate level of evidence).

Recommendation 34

In patients with high-risk CD, we recommend vedolizumab therapy to prevent postoperative recurrence (conditional recommendation, low level of evidence).

Recommendation 55

Risk factors for postoperative CD recurrence should be considered when deciding on treatment.

Recommendation 56

Surgery may be considered for patients with symptomatic CD localized to a short segment of bowel.

Recommendation 57

Surgery is required to treat enteric complications of CD.

Recommendation 58

A resection of a segment of diseased intestine is the most common surgery for a patient with CD.

Recommendation 35

We suggest that an intra-abdominal abscess (>2 cm) be treated with antibiotics and a drainage procedure, and immunosuppression held until drainage is achieved, either radiographically or surgically (conditional recommendation, low level of evidence).

Recommendation 59

Patients with CD who develop an abdominal abscess should undergo a surgical resection. However, some may respond to medical therapy after radiologically guided drainage.

Recommendation 1

Disease Classification: There is a need to define novel IBD subtypes and phenotypes based on molecular markers. Artificial intelligence (AI) and machine learning (ML) have been proposed to standardize and improve endoscopic, histological, and radiological assessments. A longitudinal approach, with diverse ethnic representation, is emphasized for validating biomarkers and phenotypes.

Recommendation 2

Endpoints: Consensus on objective and reproducible clinical endpoints is critical. AI and ML can help automate and enhance these assessments, and efforts should focus on developing molecular biomarkers that correlate with disease progression and allow for frequent monitoring and timely treatment adjustments.

Recommendation 3

Longitudinal Assessment: Ideal clinical trials would use prospective cohort designs with serial biomarker sampling. The American Journal of GASTROENTEROLOGY

Recommendation 4

Clinical Translation: Companion diagnostic tests, commonly used in oncology, are essential for predicting therapeutic responses in CD. Industry-backed research programs are needed to ensure the development of validated, reproducible, and globally accessible biomarkers, which will facilitate biomarker-driven trials.

Recommendation 5

Therapies: The success of precision medicine depends on effective therapeutic agents. While numerous biologics and small molecules are under development, it is crucial to optimize the use of existing agents. AI and ML can assist in optimizing drug choices and dosing, while combination therapies (biologics or biologics combined with small molecules) may be necessary for some patients. Personalized approaches, including therapeutic drug monitoring, will help improve treatment outcomes by enhancing our understanding of pharmacokinetics and pharmacodynamics.

STRIDE-II: Selecting Therapeutic Targets in Inflammatory Bowel Disease - Update

File: 15. STRIDE consensus.pdf

Turner D et al. *Gastroenterology*. 2021;161(5):1551-1569. doi:10.1053/j.gastro.2021.06.051

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STRIDE-II Recommendation 1

Clinical response is an immediate treatment target. Consider changing treatment if this target has not been achieved.

STRIDE-II Recommendation 2

Clinical response should be defined as: a) CD: decrease of at least 50% in PRO2 (abdominal pain and stool frequency), and in children decrease in PCDAI of at least 12.5 points and in wPCDAI at least 17.5 points b) UC: decrease of at least 50% in PRO2 (rectal bleeding and stool frequency), and in children decrease in PUCAI of at least 20 points

STRIDE-II Recommendation 3

Clinical remission is an intermediate (ie, medium-term) treatment target. Consider changing treatment if this target has not been achieved.

STRIDE-II Recommendation 4

Clinical remission should be defined as: a) CD: PRO2 (abdominal pain ≤ 1 and stool frequency ≤ 3) or HBI ≤ 5 ; in children by PCDAI (<10 points or <7.5 excluding the height item) or wPCDAI (<12.5 points) b) UC: PRO2 (rectal bleeding ≤ 0 and stool frequency ≤ 0) or partial Mayo (<3 and no score >1), and in children PUCAI <10 points

STRIDE-II Recommendation 5

Clinical response or remission are insufficient to be used as long term treatment targets

STRIDE-II Recommendation 6

In children, restoration of normal growth is a long-term treatment target. Consider changing treatment if this target has not been achieved.

STRIDE-II Recommendation 7

Endoscopic healing is a long-term target. Consider changing treatment if this target has not been achieved.

STRIDE-II Recommendation 8

Assessment of endoscopic healing can be achieved by sigmoidoscopy or colonoscopy. When not feasible, alternatives in CD can be capsule endoscopy or balloon enteroscopy.

STRIDE-II Recommendation 9

Endoscopic healing should be measured by: a) CD: SES-CD <3 points or absence of ulcerations (e.g. SES-CD ulceration subscores ≤ 0) b) UC: Mayo endoscopic subscore ≤ 0 points, or UCEIS ≤ 1 points

STRIDE-II Recommendation 10

Histologic remission is not a treatment-target in either CD or UC. Nonetheless, in UC it could be used as an adjunct to endoscopic remission to represent a deeper level of healing.

STRIDE-II Recommendation 11

Transmural healing (assessed by CTE, MRE, or bowel ultrasound) is not a treatment-target in either CD or UC. Nonetheless, in CD it should be used as an adjunct to endoscopic remission to represent a deeper level of healing.

STRIDE-II Recommendation 12

Normalization of CRP (to values under the upper limit of normal) and fecal calprotectin (to 100–250 mg/g)^b is an intermediate treatment target in UC and CD. Consider changing treatment if this target has not been achieved.

STRIDE-II Recommendation 13

Absence of disability and normalized health-related quality of life are long-term treatment targets. Consider changing treatment if this target has not been achieved.

Diet and Inflammatory Bowel Disease: The Asian Working Group Guidelines

File: 2. Diet in IBD _Asian working group.pdf

Sood A et al. *Indian J Gastroenterol.* 2019;38(3):220-246. doi:10.1007/s12664-019-00976-1

59 rules extracted

Statement 12 (Malnutrition screening)

All patients with IBD are at higher risk of malnutrition; all patients should be screened for malnutrition at presentation using a validated tool (e.g., MUST, NRS-2002). Malnutrition must be identified and addressed early in the disease course.

Statement 13 (Malnutrition prevalence)

The prevalence of malnutrition in patients with IBD depends on disease subtype (CD > UC), severity, extent, and duration of disease. Patients with CD, active disease, and extensive involvement are at highest risk.

Statement 14 (BMI insufficient for nutritional assessment)

Body mass index (BMI) alone is NOT sufficient for nutritional assessment of a patient with IBD. Validated multidimensional tools (e.g., MUST, NRS-2002, global clinical assessment) and laboratory parameters should be used.

Statement 15 (Dietician involvement)

Dietitians and/or nutritionists should be ACTIVELY INVOLVED in the nutritional care of all patients with IBD — both during active disease and in remission — as part of the multidisciplinary team.

Statement 16 (Acute severe UC: oral intake preferred)

For admitted patients with acute severe UC, adequate ORAL caloric intake is preferred over bowel rest. Bowel rest is NOT recommended as routine management in ASUC; it does not improve outcomes.

Statement 17 (ASUC: reintroduce standard diet after stabilisation)

After stabilisation of acute severe UC, a standard diet should be gradually reintroduced. Routine use of oral nutritional supplements is NOT recommended in this setting unless oral intake remains inadequate.

Statement 18 (Active CD: oral high-protein diet preferred over TPN)

For active inflammatory Crohn's disease, oral diet with adequate HIGH PROTEIN content is preferred to total parenteral nutrition (TPN). Enteral nutrition (exclusive enteral nutrition or supplemental enteral nutrition) is preferred over TPN when oral intake is insufficient.

Statement 19 (Remission: no dietary restriction needed)

Once in remission, no specific diet modification or food restriction is required — patients can continue a normal balanced diet. Excessive dietary restriction in remission is not evidence-based and may worsen nutritional status.

Statement 20 (No specific food causes relapse)

No specific dietary item has been established to cause disease relapse in patients with IBD who are in remission. Patients should not be advised to restrict specific foods on the assumption that they trigger relapse, unless there is a clear individual intolerance.

Statement 21 (Milk: not routinely restricted)

Milk should NOT be routinely restricted in all IBD patients. Restriction is indicated only in patients with confirmed severe hypolactasia (lactase deficiency). Unnecessary dairy restriction may worsen calcium and vitamin D status.

Statement 22 (Gluten-free diet not indicated in IBD)

A gluten-free diet is NOT of proven value in patients with IBD (unless coeliac disease coexists). Routine adoption of a gluten-free diet is not recommended and may have adverse nutritional consequences.

Statement 23 (Low-FODMAP diet for IBS-like symptoms only)

A low-FODMAP diet may help alleviate IBS-like symptoms (bloating, urgency, pain) associated with IBD. It is NOT a treatment for IBD inflammation and should NOT be used as primary therapy for active luminal disease.

Statement 24 (Protein and caloric requirements)

Protein requirements are INCREASED in active IBD vs. remission (1.2–1.5 g/kg/day in active disease vs. 1.0 g/kg/day in remission). Caloric requirements in remission are similar to those of healthy individuals. Energy requirements increase during active disease.

Statement 32 (Pre-surgical nutritional optimisation)

Nutritional status should be OPTIMISED before elective IBD surgery to reduce postoperative complications. If a patient is malnourished, a period of nutritional support (7–14 days) before surgery is recommended.

Statement 33 (Pre-surgical enteral nutrition if oral insufficient)

If nutritional goals cannot be met with oral diet alone before elective IBD surgery, oral nutritional supplements or enteral nutrition should be initiated BEFORE surgery. Parenteral nutrition is reserved for patients in whom the gastrointestinal tract cannot be used.

Statement 34 (ERAS protocol in elective IBD surgery)

In elective IBD surgery, the Enhanced Recovery After Surgery (ERAS) protocol should be followed in the perioperative period. Key ERAS elements: preoperative carbohydrate loading, avoidance of prolonged preoperative fasting, early postoperative feeding.

Statement 35 (Postoperative: restart oral diet early)

Oral diet or enteral nutrition should be started as soon as the patient can tolerate postoperatively (within 24 hours of surgery where possible). Early postoperative feeding reduces complications and length of stay.

Statement 36 (Postoperative: nutritional support if oral not resumed in 7 days)

If oral diet cannot be resumed within 7 days postoperatively, enteral nutrition (preferred) OR parenteral nutrition should be initiated to prevent further nutritional deterioration.

Statement 37 (Fistula: nutrition type based on location)

In CD patients with fistula, the type of nutritional support depends on fistula location: oral feeds can be given for distal (low ileal or colonic) and low-output fistulas; partial or exclusive parenteral nutrition is required for proximal (high jejunal) or high-output fistulas.

Statement 38 (Pregnancy nutrition in IBD)

IBD patients who are pregnant should be specifically evaluated for iron deficiency and folate deficiency, with supplementation as required. Recommended Dietary Allowances for pregnancy (protein, micronutrients) should be followed. Folate supplementation (at least 400–800 mcg/day) should be given to all pregnant women with IBD, especially those on sulfasalazine (which requires 2 mg/day folate).

Statement 1

Diet has an important role in the pathogenesis of inflammatory bowel disease (IBD), both ulcerative colitis (UC) and Crohn's disease (CD).

Statement 2

Epidemiological studies indicate that adoption of Western diet (low in fruits and vegetables, rich in fats, ω -6 fatty acids, red meat, and processed foods) contributes to the increasing incidence of inflammatory bowel disease in developing countries.

Statement 3

Dietary constituents like maltodextrins and emulsifiers may have a role in the development of inflammatory bowel disease.

Statement 4

Vitamin D may have a protective role in the natural history of inflammatory bowel disease.

Statement 5

Breastfeeding may have a protective role in the development of inflammatory bowel disease.

Statement 6

Exclusive enteral nutrition (EEN) is as effective as steroids in inducing remission in children with luminal Crohn's disease.

Statement 7

Exclusive enteral nutrition is effective in adult Crohn's disease but is inferior to corticosteroids for inducing remission.

Statement 8

There is no difference between elemental and polymeric formulae in terms of efficacy.

Statement 9

Partial enteral nutrition has been documented to be useful for maintenance of remission in luminal Crohn's disease along with pharmacotherapy.

Statement 10

More evidence is required, before elimination diets such as specific carbohydrate diet (SCD), Crohn's disease exclusion diet (CDED), semi-vegetarian diet (SVD), antiIBD diet, or low fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAP) can be recommended as a therapy for Crohn's disease.

Statement 11

There is no specific role for exclusive (EEN) or partial enteral nutrition (PEN) for induction or maintenance of remission in patients with ulcerative colitis.

Statement 12

Patients with inflammatory bowel disease are at a higher risk of malnutrition; hence, all the patients should be screened for it at presentation.

Statement 13

The prevalence of malnutrition in a patient with inflammatory bowel disease depends upon disease subtype, severity, extent, and duration.

Statement 14

Body mass index alone is not sufficient for nutritional assessment of a patient with inflammatory bowel disease.

Statement 15

Dietitians/nutritionists should be involved in nutrition care of patients with inflammatory bowel disease.

Statement 16

Dietary recommendations in inflammatory bowel disease: for admitted patients with acute severe ulcerative colitis, adequate oral caloric intake is preferred to bowel rest.

Statement 17

After stabilization of acute severe ulcerative colitis, a standard diet should be gradually introduced and oral nutritional supplements should not be a routine.

Statement 18

For active inflammatory Crohn's disease, oral diet with high protein is preferred to total parenteral nutrition.

Statement 19

Once in remission, there is no need for diet modification or restriction and the patients can continue a normal diet as other family members.

Statement 20

No dietary item in particular is established to cause relapse of disease activity in a patient in remission.

Statement 21

Milk should not be routinely restricted in all patients with inflammatory bowel disease unless patient has severe hypolactasia.

Statement 22

A gluten-free diet is not of a proven value in patients with inflammatory bowel disease.

Statement 23

A low FODMAP diet may help in alleviating irritable bowel syndrome -like symptoms associated with IBD.

Statement 24

Patients with inflammatory bowel disease should receive adequate calories, proteins, and fats in their diet. The calorie and protein requirement of a patient with IBD in remission is similar to that of a healthy individual. However, the protein requirement is increased in a patient with active disease.

Statement 25

Patients with inflammatory bowel disease who have anemia should be evaluated appropriately for the cause of anemia and adequately treated.

Statement 26

Proactive screening for osteopenia and its treatment should be done as per guidelines.

Statement 27

Patients should be screened for micronutrient deficiency including calcium, phosphate, magnesium, iron, folic acid, and vitamin B12 in an appropriate clinical context.

Statement 28

Except for patients with stricturing Crohn's disease, there is no evidence for recommending either a low or a high fiber diet for patients with inflammatory bowel disease.

Statement 29

Patients with inflammatory bowel disease should refrain from alcohol as it may worsen the symptoms of the disease.

Statement 30

Patients of inflammatory bowel disease should be encouraged to refrain from smoking.

Statement 31

There is no scientific evidence to recommend probiotics as a food supplement.

Statement 32

The nutritional status of patients with inflammatory bowel disease should be optimized prior to elective surgery for a better outcome.

Statement 33

If the nutritional goals cannot be met with an oral diet alone, oral nutritional supplements (ONS) or enteral nutrition should be initiated prior to surgery/ perioperative phase.

Statement 34

In elective surgery, the early/enhanced recovery after surgery protocol should be followed in the perioperative period

Statement 35

Oral diet/enteral nutrition should be started as soon as the patient can tolerate in the postoperative period

Statement 36

In the postoperative period, if oral diet cannot be resumed within 7 days then enteral/ parenteral nutrition should be initiated.

Statement 37

In Crohn's disease patients with a fistula, the type of diet depends upon the location of fistula—oral feeds for distal (low ileal or colonic) and low output fistula, and partial or exclusive parenteral nutrition for proximal and high output fistula.

Statement 38

Inflammatory bowel disease patients with pregnancy should be specifically evaluated for iron and folate deficiency and replacement done accordingly. Recommended Dietary Allowances (RDA) for pregnancy and lactation should be followed.

Statement 46

Fernández-Banares F, Cabré E, Esteve-Comas M, Gassull MA. How effective is enteral nutrition in inducing clinical remission in active Crohn's disease? A meta-analysis of the

ECCO Guidelines on Therapeutics in Crohn's Disease: Medical Treatment

File: 3. ECCO and Crohns.pdf

Gordon H et al. *J Crohns Colitis*. 2024;18(10):1531-1555. doi:10.1093/ecco-jcc/jjae091

31 rules extracted

Statement 1.1

We recommend against the use of 5-aminosalicylic acid for induction of remission of CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 1.2

We recommend against the use of oral 5-aminosalicylic acid as maintenance therapy in CD [strong recommendation, low-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, low-quality evidence]

Statement 2.1

We recommend budesonide for the induction of clinical remission in patients with active, mild-to-moderate CD limited to the ileum and/or ascending colon [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 2.2

We suggest systemic corticosteroids can be used as induction therapy in patients with active, moderate-to-severe CD [weak recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 3.1

We recommend against the use of thiopurine monotherapy as induction therapy for CD [strong recommendation, very low-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, very low-quality evidence]

Statement 3.2

We suggest thiopurine monotherapy can be used as maintenance therapy in CD [weak recommendation, low-quality evidence]. [Consensus: 95%]

Evidence: [weak recommendation, low-quality evidence]

Statement 4.1

We suggest parenteral methotrexate can be used as induction therapy in moderate-to-severe CD [weak recommendation, moderate-quality evidence]. [Consensus: 94%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 4.2

We suggest parenteral methotrexate monotherapy can be used as maintenance therapy in moderate-to-severe CD [weak recommendation, low-quality evidence]. [Consensus: 97%]

Evidence: [weak recommendation, low-quality evidence]

Statement 5.1

We recommend infliximab as induction therapy with moderate-to-severe active CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 5.2

We recommend infliximab as maintenance therapy in moderate-to-severe CD [strong recommendation, low-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, low-quality evidence]

Statement 5.3

We recommend combination therapy with a thiopurine when starting infliximab as induction therapy in patients with moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 5.4

We recommend combination therapy with infliximab and thiopurines for a minimum of 6–12 months when using infliximab as maintenance therapy in patients with CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 5.5

In patients with CD who have achieved long-term remission with the combination of anti-TNF and thiopurines, we suggest de-escalation to anti-TNF monotherapy and withdrawal of thiopurines [weak recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 6.1

We recommend adalimumab as induction therapy in patients with moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 6.2

We recommend adalimumab monotherapy as maintenance therapy in moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 6.3

We suggest adalimumab monotherapy should be used over combination therapy with thiopurines as induction therapy in patients with moderate-to-severe CD naïve to biologics [weak recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 6.4

We suggest adalimumab monotherapy should be used over combination with an immunomodulator as maintenance therapy in patients with moderate-to-severe CD naïve to biologics [weak recommendation, low-quality evidence]. [Consensus: 98%]

Evidence: [weak recommendation, low-quality evidence]

Statement 7.1

We suggest certolizumab can be used as induction therapy in patients with moderate-to-severe CD [weak recommendation, moderate-quality evidence]. [Consensus: 97%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 7.2

We suggest certolizumab can be used as maintenance therapy in moderate-to-severe CD [weak recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 8.1

There is insufficient evidence to recommend the use of proactive therapeutic drug monitoring, compared with reactive therapeutic drug monitoring or standard of care, when using anti-TNF agents [weak recommendation, very low-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, very low-quality evidence]

Statement 9.1

We recommend ustekinumab as induction therapy in moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 9.2

We recommend ustekinumab as maintenance therapy in moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 10.1

We suggest adalimumab or ustekinumab are equally as effective as induction therapy in biologic-naïve patients with moderate-to-severe CD [weak recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 10.2

We suggest adalimumab and ustekinumab are equally as effective as maintenance therapy in biologic-naïve patients with moderate-to-severe CD [weak recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [weak recommendation, moderate-quality evidence]

Statement 11.1

We recommend risankizumab as induction therapy in moderate-to-severe CD [strong recommendation, high-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, high-quality evidence]

Statement 11.2

We recommend risankizumab as maintenance therapy in moderate-to-severe CD [strong recommendation; high-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation; high-quality evidence]

Statement 12.2

We recommend vedolizumab as maintenance therapy in moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 13.1

We recommend upadacitinib as induction therapy in moderate-to-severe CD [strong recommendation; high-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation; high-quality evidence]

Statement 13.2

We recommend upadacitinib as maintenance therapy in moderate-to-severe CD [strong recommendation; moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation; moderate-quality evidence]

Statement 12.1

We recommend vedolizumab as induction therapy in moderate-to-severe CD [strong recommendation, moderate-quality evidence]. [Consensus: 100%]

Evidence: [strong recommendation, moderate-quality evidence]

Statement 14.1

We suggest exclusive enteral nutrition can be used as induction therapy in patients with mild-to-moderate CD who are motivated to adhere to dietary therapy, have access to dietetic support, and prefer to avoid corticosteroids. [Weak recommendation, very low-quality evidence.] [Consensus: 100%] [Weak recommendation, very low-quality evidence.]

ACG Clinical Guideline Update: Preventive Care in Inflammatory Bowel Disease

File: 4. acg_clinical_guideline_update__preventive_care_in.15.pdf

Farraye FA et al. *Am J Gastroenterol.* 2025;120(7):1447-1473. doi:10.14309/ajg.0000000000003541

21 rules extracted

Recommendation 4

In all adults 50 and older with IBD, we suggest vaccination against HZ with the 2-dose inactive recombinant HZ vaccine (Conditional recommendation, low certainty of evidence)

Recommendation 5

In adults 19 and above with IBD on immune-modifying therapy or planning to start therapy, we suggest vaccination against HZ with the 2-dose inactive recombinant HZ vaccine (Conditional recommendation, very low certainty of evidence)

Recommendation 7

Adults with IBD should be evaluated for varicella immunity before initiating immune-modifying therapy, through either a documented history of chickenpox infection or completion of a 2-dose varicella vaccine series. Serologic testing is not recommended in previously vaccinated individuals because of the high false-negative rates. If nonimmune status is confirmed, vaccination should be completed before initiating immunomodifying therapy when possible.

Recommendation 6

In adults with IBD, we suggest vaccination against SARS-CoV-2 in accordance with national guidelines (Conditional recommendation, very low certainty of evidence)

Recommendation 8

Household members of patients receiving immune-modifying therapies can receive live vaccines with certain precautions.

Recommendation 1

MMR, varicella, and rotavirus vaccines should be administered to contacts when indicated. They recommend the following precautions for varicella and rotavirus vaccine recipients. If a varicella vaccine recipient develops a rash postimmunization, they should avoid contact with a person with altered immunocompetence until the rash resolves.

Recommendation 9

In patients with IBD (both UC and CD), we suggest annual screening for melanoma independent of the use of biologic therapy (Conditional recommendation, very low certainty of evidence)

Recommendation 10

In patients with IBD on immune-modifying therapies (6mercaptopurine, azathioprine, methotrexate, JAK inhibitors, or S1P receptor modulators), we suggest annual screening for NMSC while using these agents, particularly older than the age of 50 (Conditional recommendation, very low certainty of evidence)

Recommendation 11

All patients with CD and UC should be screened for depression and anxiety at baseline and annually. Patients who screen positive for anxiety and/or depression should be referred for counseling/therapy.

Recommendation 11

In adults with IBD and conventional risk factors for abnormal bone mineral density, we suggest screening for osteoporosis with bone mineral density testing at the time of diagnosis and periodically after diagnosis (Conditional recommendation, very low certainty of evidence)

Recommendation 1

Parathyroid hormone excess

Recommendation 2

Primary and secondary thyroxine excess, endogenous and exogenous

Recommendation 3

Cortisol excess, endogenous and exogenous

Recommendation 1

Estrogen deficiency (premenopausal state may be linked with anorexia, bulimia, athletic amenorrhea, premature menopause, prolactinoma, or hypopituitarism)

Recommendation 2

Estrogen deficiency, postmenopausal

Recommendation 3

Testosterone deficiency, primary and secondary testicular failure

Recommendation 4

Vitamin D metabolite deficiency, inadequate intake, or malabsorption Miscellaneous (not necessarily mediated by hormonal abnormalities):

Recommendation 1

Medical conditions, including postgastrectomy states, idiopathic hypercalciuria, systemic mastocytosis, and prolonged immobilization (paraplegia and quadriplegia)

Recommendation 2

Lifestyle factors, including cigarette smoking, excessive ingestion of caffeine, and excessive sodium intake (promotes hypercalciuria) Medications:

Recommendation 5

Tacrolimus Most of these conditions/states associated with low bone mass are not influenced by changes in diet or lifestyle. Each of the circumstances listed could have an impact on the skeleton at any time in an individual's life. The effect is accentuated if it occurs in conjunction with low estrogen levels after menopause. In addition, as with low BMD, risk factors for bony fractures may be considered modifiable or nonmodifiable, as follows: Modifiable risk factors for abnormal BMD include current tobacco smoking, below-normal body weight (especially those with low body weight that is less than 129 pounds), deficiency of estrogen, menopause before age 45, bilateral ovariectomy, premenopausal amenorrhea for .1 year, lifelong low intake of calcium, excessive consumption of ethanol, visual impairment despite adequate correction (may increase risk of falling), repeated falls, inadequate physical activity, frailty, or poor overall physical condition. Nonmodifiable risk factors include history of fracture as an adult, family history of fractures, especially among first-degree relatives, White, female sex, dementia, elderly, frailty, or poor overall physical condition.

Recommendation 12

In adults with IBD who smoke, we suggest that they should be counseled to quit (Strong recommendation, low certainty of evidence)

ECCO Guidelines on Inflammatory Bowel Disease and Malignancies

File: 5. ECCO_IBD and malignancies.pdf

Gordon H et al. *J Crohns Colitis*. 2023;17(6):827-854. doi:10.1093/ecco-jcc/jjac187

55 rules extracted

Statement 3a (Non-polypoid dysplasia: EMR/ESD)

Non-polypoid dysplastic lesions without stigmata of invasive cancer can be treated by endoscopic mucosal resection (EMR) or endoscopic submucosal dissection (ESD) by a trained endoscopist, provided there is no evidence of multifocal or invisible dysplasia elsewhere in the colon. Complete en-bloc resection is preferred.

Evidence: [EL3]

Statement 3b (Post-dysplasia resection: surveillance)

After endoscopic resection of a dysplastic lesion in IBD: perform surveillance colonoscopy with high-definition chromoendoscopy or virtual chromoendoscopy at 3–6 months, then ANNUALLY based on the grade of dysplasia and completeness of resection.

Evidence: [EL4]

Statement 3c (Surgery when endoscopic resection not feasible)

Patients with IBD-related dysplastic lesions that CANNOT be endoscopically resected en bloc (due to size, location, or multifocality) should be offered surgery (colectomy).

Evidence: [EL4]

Statement 4a (Anal cancer risk in perianal CD)

Patients with anal and/or perianal Crohn's disease are at INCREASED RISK of anal cancer, particularly fistula-related adenocarcinoma. Chronic active perianal fistulising disease may be associated with advanced cancer stage at diagnosis.

Evidence: [EL3]

Statement 4b (Change in anal symptoms: re-evaluate immediately)

ANY change in anal symptoms in patients with chronic perianal Crohn's disease (e.g., new pain, altered discharge, or fistula tract changes) should trigger IMMEDIATE RE-EVALUATION including MRI and biopsy to exclude malignancy.

Evidence: [EL5]

Statement 5a (IPAA: risk factors for pouch dysplasia)

Preoperative diagnosis of dysplasia or cancer of the colon or rectum is a risk factor for pouch dysplasia or adenocarcinoma after ileal pouch-anal anastomosis (IPAA).

Evidence: [EL1]

Statement 5b (IPAA: annual surveillance if prior dysplasia/cancer or PSC)

IBD patients with IPAA who had preoperative colonic dysplasia or cancer, AND patients with IBD + primary sclerosing cholangitis (PSC): ANNUAL endoscopic surveillance of the pouch is required.

Evidence: [EL3]

Statement 6a (Cholangiocarcinoma: higher risk in IBD+PSC)

IBD patients have a higher risk of cholangiocarcinoma compared with the general population, particularly patients with UC + PSC. Risk is highest in the first year after PSC diagnosis.

Evidence: [EL3]

Statement 6b (Cholangiocarcinoma surveillance: imaging every 6-12 months)

Cholangiocarcinoma surveillance should be considered in ALL patients with IBD + PSC, regardless of disease stage. Appropriate imaging (MRI/MRCP or ultrasound) should be performed every 6–12 months.

Evidence: [EL4]

Statement 7a (Small bowel adenocarcinoma risk in CD)

CD patients with small-bowel involvement have an increased risk of small-bowel cancers. The most common: small-bowel adenocarcinoma in areas of chronic inflammation, predominantly the distal jejunum and ileum.

Evidence: [EL1]

Statement 7b (Consider small bowel adenocarcinoma in refractory stricturing CD)

Consider small-bowel adenocarcinoma in the differential diagnosis of patients with refractory, long-standing, stricturing Crohn's disease or relevant new symptoms (obstruction, weight loss, anaemia). Biopsy and imaging are required for diagnosis.

Evidence: [EL5]

Statement 7c (No routine small bowel cancer surveillance in CD)

Routine surveillance with imaging or endoscopy specifically for small-bowel cancer in CD is NOT currently recommended due to insufficient evidence. Clinical vigilance for new or worsening symptoms is required.

Evidence: [EL5]

Statement 10a (Thiopurine: lymphoproliferative and myeloproliferative risk)

Thiopurine monotherapy is associated with increased risk of lymphoproliferative disorders [EL1] and myeloproliferative disorders [EL3]. Risk is increased in OLDER patients (>65 years). Risk decreases gradually after thiopurine withdrawal.

Statement 10b (EBV screening before thiopurine in young males)

EBV serology should be checked in young adult MALES before starting thiopurine therapy [EL5]. If EBV IgG is NEGATIVE (seronegative): medications other than thiopurines should be considered, as primary EBV infection on thiopurines carries risk of aggressive fatal lymphoma.

Evidence: [EL5]

Statement 12 (Anti-TNF: lymphoma and melanoma risk; skin surveillance)

Anti-TNF monotherapy may increase the risk of lymphoma and melanoma in IBD patients [EL2]. Annual skin cancer surveillance and sun protection advice are required for all patients on anti-TNF therapy.

Statement 14a (Combination anti-TNF+thiopurine: HSTCL risk)

The combination of anti-TNF + thiopurine therapy is associated with lymphoma risk GREATER than either monotherapy [EL2]. HEPATOSPLENIC T-CELL LYMPHOMA (HSTCL) risk is significantly increased, particularly in MALES UNDER 30 YEARS receiving combination therapy for more than 2 years. HSTCL is rare but typically fatal.

Statement 14b (Combination: no additional solid-organ or skin cancer risk)

Combined anti-TNF + thiopurine therapy does NOT confer additional solid-organ or skin cancer risk beyond the risks associated with either monotherapy [EL3]. No additional surveillance beyond standard monotherapy recommendations is required.

Statement 15 (Methotrexate: NMSC risk)

Methotrexate may be associated with increased risk of non-melanoma skin cancer (NMSC) [EL5]. Insufficient evidence exists for additional screening beyond general population NMSC surveillance. Sun protection is advised.

Statement 16 (JAK inhibitors: no increased cancer risk currently)

JAK inhibitors: no current evidence of increased overall cancer risk in IBD [EL4]. Long-term safety data remain limited. Exercise caution in patients with prior malignancy.

Statement 17 (Vedolizumab: no increased cancer risk)

Vedolizumab is not associated with increased cancer risk [EL4]. Long-term safety data are accumulating and are reassuring. Vedolizumab is preferred for patients with prior malignancy or high cancer risk.

Statement 18 (Anti-IL-12/23: no increased cancer risk)

Anti-IL-12/23 agents (ustekinumab, risankizumab, guselkumab, mirikizumab): no evidence of increased overall cancer risk [EL4]. Long-term data are reassuring.

Statement 19 (Cancer therapy interactions with IBD)

Hormone therapy for cancer may increase IBD relapse risk. Radiation and chemotherapy do NOT appear to increase IBD relapse [EL3]. Checkpoint inhibitors (anti-PD1/PD-L1) are associated with immune-mediated colitis and HIGHER IBD reactivation risk [EL4] — close monitoring is essential.

Statement 20 (Thiopurines in prior malignancy: no increased recurrence)

Thiopurine use in IBD patients with a PAST malignancy does NOT appear to increase cancer recurrence risk beyond known class risk, based on available data. Most data come from patients starting thiopurines more than 5 years after cancer treatment.

Evidence: [EL5]

Statement 21 (Thiopurine withdrawal in active cancer)

Thiopurines should PREFERABLY be withdrawn in patients with ACTIVE cancer [EL5]. Patients with non-aggressive basal cell carcinoma (BCC) or preneoplastic cervical lesions may continue thiopurine with close oncological monitoring. Decisions should be made in a multidisciplinary setting.

Evidence: [EL5]

Statement 22 (Methotrexate: insufficient data in prior malignancy)

Insufficient data exist on the safety of methotrexate in IBD patients with a prior malignancy [EL5]. Use with caution; case-by-case multidisciplinary decision required.

Statement 23 (Anti-TNF in prior/current cancer: MDT decision)

Anti-TNF agents may be used in IBD patients with current or previous cancer. All decisions must be made on a case-by-case basis in a multidisciplinary oncology setting, weighing IBD disease activity, cancer type and prognosis, and available alternative IBD therapies.

Statement 24 (Vedolizumab/ustekinumab in prior malignancy: safe)

Vedolizumab [EL3] and ustekinumab [EL4] do NOT appear to increase cancer recurrence in IBD patients with a prior malignancy. These agents are preferred in patients with prior malignancy where biologic therapy is required. Insufficient evidence for JAK inhibitors in prior or active malignancy.

Statement 1

Patients with IBD of the colon should be informed that they are at increased risk of developing colorectal cancer [CRC] [EL1]. The risk of colorectal cancer is highest in patients with ulcerative colitis [UC] with extensive disease, and increases significantly 8–10 years after diagnosis or when dysplasia is detected on colonic biopsies [or both], particularly high-grade dysplasia [EL1]. Risk factors include male sex, young age at UC diagnosis, family history of CRC, and the presence of colonic strictures or primary sclerosing cholangitis [EL1]. Consensus: 100%

Evidence: [EL1], [EL1], [EL1]

Statement 2.1

Screening colonoscopy should be performed in all IBD patients 8 years after onset of first symptoms to assess disease extent and exclude dysplasia [EL4] Consensus: 95%

Evidence: [EL4]

Statement 2.2

In IBD patients with disease limited to the rectum without evidence of previous or current endoscopic or microscopic inflammation proximal to the rectum or with isolated small-bowel disease, no subsequent additional screening

Statement 2.3

In IBD patients with concurrent PSC, an annual surveillance colonoscopy should be performed following the diagnosis of PSC, irrespective of disease activity, extent, and duration [EL3] Consensus: 96.3%

Evidence: [EL3]

Statement 2.4

Patients with high-risk features [family history of CRC in a first-degree relative ≤ 50 years of age, colonic stricture or dysplasia, PSC, extensive colitis with severe active inflammation] should have their next surveillance colonoscopy scheduled for 1 year. [EL4] Patients with intermediate risk factors should have their next surveillance colonoscopy scheduled for 2–3 years. Patients with neither intermediate nor high-risk features should have their next surveillance colonoscopy scheduled for 5 years [EL5] Consensus: 87.5%

Evidence: [EL4], [EL5]

Statement 2.5

Colonoscopic surveillance is best performed when IBD is in remission, as it is otherwise difficult to discriminate between dysplasia and inflammation on mucosal biopsies [EL5] Consensus: 96.2%

Evidence: [EL5]

Statement 3

Polypoid dysplastic lesions can be treated by endoscopic resection when the lesion can be excised entirely en bloc, provided there is no evidence of multifocal or invisible dysplasia elsewhere in the colon [EL2] Non-polypoid dysplastic lesions without stigmata of invasive cancer can be treated by endoscopic mucosal resection or submucosal dissection by trained endoscopists in selected cases, provided there is no evidence of multifocal or invisible dysplasia elsewhere in the colon. [EL3] Ideally, complete en-bloc resection should be achieved. Surveillance colonoscopy with high-definition dye chromoendoscopy or virtual electronic chromoendoscopy should be performed after 3–6 months and then annually after endoscopic resection, based on the grade of dysplasia detected [EL4] Patients should be offered surgery when dysplastic lesions cannot be endoscopically resected. Factors that may render

Evidence: [EL2], [EL3], [EL4]

Statement 4

Patients with anal or perianal Crohn's disease [CD] or both are at increased risk for anal cancer, particularly fistularelated adenocarcinoma. [EL3] Chronic active perianal fistulising disease may be associated with advanced cancer stage at the time of diagnosis. [EL3] In case of any change in anal symptoms, patients with chronic perianal CD should be re-evaluated [EL5] Consensus: 100%

Evidence: [EL3], [EL3], [EL5]

Statement 5

In patients with IBD and IPAA, a preoperative diagnosis of dysplasia or cancer of the colon or rectum is a risk factor for pouch dysplasia or adenocarcinoma. [EL1] In these patients and in patients with PSC, annual pouch surveillance should be performed [EL3] Consensus: 100%

Evidence: [EL1], [EL3]

Statement 6

Patients with IBD are at higher risk for cholangiocarcinoma compared with the general population, particularly patients with UC and concomitant PSC [EL3] Surveillance for cholangiocarcinoma should be considered in all patients with IBD and PSC regardless of disease stage, and is most relevant in the first year after diagnosis [EL3] Surveillance for cholangiocarcinoma should include appropriate imaging and should be performed every 6 to 12 months [EL4] Consensus: 100%

Evidence: [EL3], [EL3], [EL4]

Statement 7

Patients with CD, particularly those with small-bowel involvement, have an increased risk of small-bowel cancers [EL1] Small-bowel adenocarcinoma is the most common subtype and is found in areas of inflammation, predominantly the distal jejunum and ileum. [EL3] The diagnosis should be considered in patients with refractory, long-standing, stricturing disease or relevant symptoms [EL5] At present, routine surveillance with imaging or endoscopy is not recommended for small-bowel cancers [EL5] Consensus: 95.0%

Evidence: [EL1], [EL3], [EL5], [EL5]

Statement 8

There is a small increased risk of non-gastrointestinal solidorgan tumours in IBD compared with the general population. [EL2] Since there is no evidence to recommend a different approach to prevention and early diagnoses of extraintestinal cancers, patients with IBD should be encouraged to follow the same primary and secondary prevention programmes as the general population, based on individual risks [EL5] Consensus: 100%

Evidence: [EL2], [EL5]

Statement 9

There is an increased risk of haematological malignancy in patients with IBD compared with the general population. However, there is insufficient evidence to determine whether the risk is independent of the effects of therapy or other risk factors [EL3] Consensus: 95.5%

Evidence: [EL3]

Statement 10

Patients on thiopurine monotherapy are at increased risk for lymphoproliferative disorders [EL1] and myeloproliferative disorders. [EL3] The risk is increased for older patients. Before starting therapy, screening for Epstein-Barr virus [EBV] infection should be considered in young adult male patients; [EL5] in patients negative for EBV IgG, medications other than thiopurines may be considered. [EL5] This should inform treatment selection, particularly in older patients or those with other risks for lymphoma [EL5] Consensus: 100%

Evidence: [EL1], [EL3], [EL5], [EL5], [EL5]

Statement 11

Thiopurines and overall cancer risk There is insufficient evidence to conclude that there is an increased overall risk of solid-organ cancer or site-specific cancer other than skin cancer or cervical neoplasia in patients with IBD on thiopurine monotherapy [EL3]

Evidence: [EL3]

Statement 12

There is no evidence of an overall increase in the risk of cancer in IBD patients treated with TNF α -antagonist monotherapy, although the risk of lymphoma and melanoma may be increased. [EL2] Skin-cancer surveillance and sun protection measures tailored to individual risk should be encouraged. [EL5] However, there are insufficient data to recommend additional screening measures beyond those recommended for the general population. Consensus: 100%

Evidence: [EL2], [EL5]

Statement 13

There is no evidence of an additional increase in risk of solid-organ or skin cancer in patients with IBD treated with combination therapy with an TNF α antagonist and a thiopurine or methotrexate [EL3] compared with the risks associated with monotherapy with either of these agents. Surveillance measures in addition to those recommended for monotherapy are not required [EL5] Consensus: 90.0%

Evidence: [EL3], [EL5]

Statement 14

The risk of lymphoma associated with combined TNF α antagonist and thiopurine therapy is greater than that of thiopurine or TNF α antagonist monotherapy. [EL2] This should inform treatment selection, particularly in older patients or those with other risks for lymphoma [EL5] Combined TNF α antagonist and thiopurine therapy significantly increases the risk of a rare hepatosplenic T cell lymphoma, particularly in male patients <30 years of age with CD [EL4] Insufficient data exist on the risk of lymphoma in IBD patients exposed to TNF α antagonists in combination with methotrexate. Consensus: 100%

Evidence: [EL2], [EL5], [EL4]

Statement 15

Patients treated with methotrexate may be at increased risk of NMSC. [EL5] There is insufficient evidence to support additional screening for skin cancer beyond that recommended for the general population [EL5] Consensus: 100%

Evidence: [EL5], [EL5]

Statement 16

There is no evidence that the overall risk of cancer is increased in patients with IBD treated with JAK inhibitors. [EL4] However, long-term data are lacking in patients with IBD. Consensus: 100%

Evidence: [EL4]

Statement 17

Current evidence does not show an increased risk of malignancy in patients with IBD treated with vedolizumab. [EL4] However, long-term data are lacking. Consensus: 100%

Evidence: [EL4]

Statement 18

Current evidence does not show an increased overall risk of cancer in IBD patients treated with anti IL-12/23 agents. [EL4] However, long-term data are lacking in IBD. Consensus: 100%

Evidence: [EL4]

Statement 20

Current evidence suggests that there is no additional risk of incident cancer with thiopurine use in patients with IBD and a past history of malignancy beyond the known risk associated with this class. However, most observational data are from patients starting treatment with thiopurines more than 5 years after cancer resolution, and in patients with a low risk of cancer recurrence [EL4] Consensus: 100%

Evidence: [EL4]

Statement 19

There is limited evidence to suggest that hormone therapy increases the risk of relapse in patients with IBD in remission at treatment initiation, whereas radiation and chemotherapy do not. [EL3] Data on the use of checkpoint inhibitors in patients with IBD are limited but indicate higher rates of IBD reactivation. [EL4] Patients with IBD being treated for cancer should be monitored closely for potential flares [EL5] Consensus: 100%

Evidence: [EL3], [EL4], [EL5]

Statement 21

Thiopurines should preferably be withdrawn in patients with an active cancer diagnosis. [EL5] Patients with non-aggressive BCC or preneoplastic lesions of the cervix may continue thiopurine therapy with close monitoring [EL5] Consensus: 100%

Evidence: [EL5], [EL5]

Statement 22

There are insufficient data to make recommendations regarding the safety of methotrexate use in patients with prior malignancies [EL5] Consensus: 100%

Evidence: [EL5]

Statement 23

TNF α antagonists may be used in patients with IBD and current or previous cancer. However, data on individual cancer types and timing of treatment with TNF α antagonists are lacking. [EL4] Decisions should be made on a case-by-case basis in a multidisciplinary setting involving oncologists, and should consider factors including current and recent IBD activity and alternative treatment options [EL5] Consensus: 100%

Evidence: [EL4], [EL5]

Statement 24

IBD patients with a history of prior malignancy do not appear to have an increased risk of cancer recurrence or new cancer when treated with vedolizumab [EL3] or ustekinumab. [EL4] There is insufficient evidence to make recommendations on the use of JAK inhibitors for patients with current or prior malignancy [EL5] There are insufficient data regarding the safety of vedolizumab, ustekinumab, or JAK inhibitors in patients with active malignancy. Decisions should be made on a case-by-case basis in a multidisciplinary setting involving oncologists, and should consider factors including current and recent IBD activity and alternative treatment options [EL5] Consensus: 100%

Evidence: [EL3], [EL4], [EL5], [EL5]

ECCO Guidelines on Therapeutics in Ulcerative Colitis: Medical Treatment

File: 6. ECCO_UC and medical treatment.pdf

Raine T et al. *J Crohns Colitis*. 2022;16(1):2-17. doi:10.1093/ecco-jcc/jjab178

19 rules extracted

Recommendation 1

We recommend 5-aminosalicylates at a dose of ≥ 2 g/ day [d] to induce remission in patients with mildly-to-moderately active UC [strong recommendation; quality of evidence low]

Evidence: [strong recommendation; quality of evidence low]

Recommendation 3

We suggest the use of oral 5-ASA [≥ 2 g/d] combined with topical [rectal] 5-ASA over oral 5-ASA monotherapy for induction of remission in adult patients with active UC of at least rectosigmoid extent [weak recommendation; very low-quality evidence]

Evidence: [weak recommendation; very low-quality evidence]

Recommendation 4

We recommend using topical [rectal] steroids for the induction of remission in patients with active distal colitis [strong recommendation, very low-quality evidence] The use of topically administered steroids has been long established for the induction of remission in patients with proctitis and distal colitis. Topically applied steroids offer the advantage over systemic

Evidence: [strong recommendation, very low-quality evidence]

Recommendation 5

We suggest treatment with topical [rectal] 5-ASAs over topical [rectal] steroids for induction of remission in patients with active distal UC [weak recommendation, very low quality of evidence]

Evidence: [weak recommendation, very low quality of evidence]

Recommendation 6

We suggest the use of colonic-release corticosteroids for induction of remission in patients with active mild-to-moderate UC [weak recommendation, low quality of evidence]

Evidence: [weak recommendation, low quality of evidence]

Recommendation 7

We suggest against the use of thiopurines as monotherapy for the induction of remission in patients with active UC [weak recommendation, very low quality of evidence]

Evidence: [weak recommendation, very low quality of evidence]

Recommendation 8

We recommend the use of oral 5-ASA at a dose ≥ 2 g/day for maintenance of remission in UC patients [strong recommendation; very low quality of evidence]

Evidence: [strong recommendation; very low quality of evidence]

Recommendation 9

We suggest the use of topical [rectal] 5-ASA for the maintenance of remission in patients with distal UC [weak recommendation, very low-quality evidence]

Evidence: [weak recommendation, very low-quality evidence]

Recommendation 10

We recommend monotherapy with thiopurines for the maintenance of remission in patients with steroid-dependent UC or who are intolerant to 5-ASA [strong recommendation, moderate quality of evidence] We identified four placebo-controlled RCTs on maintenance treatment with azathioprine in patients with UC who were steroid-dependent or intolerant to 5-ASA [SoF Table 10, available as Supplementary data at ECCO-JCC online].^{78,79,89,90} In 232 patients followed for 1 year, azathioprine was superior [56%] to placebo [35%] for the maintenance of clinical remission [RR: 1.59; 95% CI: 1.19–2.11]. No placebo-controlled data on endoscopic or histological remission, sustained clinical remission, or SAEs were available. In contrast to current clinical trials, different disease activity indices and endpoint definitions were used. Hence, indirect comparisons with novel and potentially more potent agents are difficult. Nevertheless, large-scale cohort studies highlighted the apparent clinical benefit of thiopurine monotherapy.⁹¹ Since we do not recommend the use of thiopurines for induction of remission, it is important that any maintenance strategy with thiopurines is planned alongside an effective induction agent. We did not identify any RCTs of thiopurines other than azathioprine, but due to their closely related pharmacology, we extend our recommendation across the drug class. Significant safety concerns do exist with the use of thiopurines. This is particularly true in patients aged >65 years; use of thiopurines should be discouraged in this age group.^{8,92–94} No evidence supports the use of methotrexate for the maintenance of remission in UC.⁹⁵ An RCT of methotrexate against placebo

Evidence: [strong recommendation, moderate quality of evidence]

Recommendation 11

We recommend oral prednisolone for induction of remission in non-hospitalised patients with moderately-to-severely active UC [strong recommendation; very low quality of evidence] Despite a limited evidence base, the use of systemic corticosteroids for the induction of remission in moderately-to-severely active UC is well established in clinical practice. The limited evidence is due in part to the large effect size and limited alternative options at the time of the original RCTs.^{97,98} A previous meta-analysis⁹⁹ included five placebo-controlled RCTs, although only two of them^{97,98} used standard systemic corticosteroids. Therefore, we performed a metaanalysis of just these two studies and calculated an RR of 2.83 [95% CI: 1.79–4.46] for the induction of clinical remission. The quality of evidence was rated as very low, due to a serious risk of bias, indirectness, and imprecision [in part since the number of patients included in each study was low] [SoF Table 11, available as Supplementary data at ECCO-JCC online]. No information regarding AEs with steroid treatment was available in these two studies. Other studies established the side-effect profile of corticosteroids in both short courses and also longer-term exposure in both UC and Crohn's disease.^{21,100} Due to the potential for side effects, some of which are irreversible, corticosteroid-free remission represents a desired outcome for patients.^{101,102} Overall, we believe that the ample experience with systemic steroids in clinical practice and the favourable balance between their potential benefits and harms [when used over limited periods] support the recommendation of oral prednisolone [or another equivalent systemic steroid agent, such as methylprednisolone or prednisone] as an option for induction of remission in patients with moderately-to-severely active UC. For these reasons, our recommendation is graded as strong, despite the quality of evidence being very low. A previous meta-analysis identified six RCTs that compared systemic prednisolone with budesonide, and found a significantly higher chance of induction of remission but increased steroid-related AEs with prednisolone.⁹⁹ However, none of these RCTs used a colonic-release budesonide formulation. We restrict our recommendations for budesonide MMX in mild-to-moderately active disease, and prednisolone in moderately-to-severely active UC, to reflect the study populations of the RCTs identified and the likely risk-benefit profile in these different populations. It is important to note that there are no efficacy data supporting the use of corticosteroids as maintenance therapies, and very limited data on the ability of these drugs to achieve endoscopic response. Additionally, longer-term corticosteroid exposure is associated with significant safety concerns. Due to this, along with the availability of drugs with proven ability to maintain corticosteroid-free remission,

Evidence: [strong recommendation; very low quality of evidence]

Recommendation 13

We recommend treatment with vedolizumab for the induction of remission in patients with moderately-to-severely active UC who have inadequate response or intolerance to conventional therapy [strong recommendation, low quality of evidence]

Evidence: [strong recommendation, low quality of evidence]

Recommendation 14

We recommend treatment with tofacitinib to induce remission in patients with moderate-to-severe UC who have inadequate response or intolerance to conventional therapy [strong recommendation, moderate quality of evidence]

Evidence: [strong recommendation, moderate quality of evidence]

Recommendation 15

We recommend treatment with ustekinumab for the induction of remission in patients with moderately-to-severely active UC with inadequate response or intolerance to conventional therapy. [strong recommendation, moderate quality of evidence] A single RCT compared ustekinumab with placebo for induction therapy in patients with moderately-to-severely active UC [SoF

Evidence: [strong recommendation, moderate quality of evidence]

Recommendation 16

We recommend anti-TNF agents [infliximab, adalimumab, or golimumab] for the maintenance of remission in patients with UC who responded to induction therapy with the same drug [strong recommendation, high-quality evidence]

Evidence: [strong recommendation, high-quality evidence]

Recommendation 17

In UC patients who have lost response to an anti-TNF agent, there is currently insufficient evidence to recommend for or against the use of therapeutic drug monitoring to improve clinical outcomes

Recommendation 18

We recommend vedolizumab for maintenance of remission in patients with UC who responded to induction therapy with vedolizumab [strong recommendation, moderate-quality evidence] We identified three RCTs that included 441 patients treated with intravenous vedolizumab or placebo, which reported on maintenance of clinical remission and sustained clinical remission in adult patients with moderately-to-severely active UC who responded to induction therapy [SoF Table 18, available as Supplementary data at ECCO-JCC online].^{122,123,143,144} Patients in these trials were followed up for 52–60 weeks. We performed a meta-analysis of results from these trials. Clinical remission was more common in inductionresponders who subsequently received vedolizumab compared with placebo [RR: 2.37; 95% CI: 1.74–3.23]. Likewise, sustained clinical remission was also more common in patients receiving vedolizumab maintenance therapy compared with placebo [20.7% vs 9.4%; RR: 2.16; 95% CI: 1.34–3.50]. The quality of evidence for these outcomes was moderate to high. The rate of SAEs across five studies involving 1288 patients was not significantly different between vedolizumab and placebo [RR: 0.71; 95% CI: 0.39–1.30]. The quality of evidence for this outcome was moderate due to serious imprecision arising from sparse data. Nevertheless, the safety profile of vedolizumab has been established in a large cohort study.¹¹² In

Evidence: [strong recommendation, moderate-quality evidence]

Recommendation 19

We suggest the use of vedolizumab rather than adalimumab for the induction and maintenance of remission in patients with moderately-to-severely active ulcerative colitis [weak recommendation, low level of evidence]

Evidence: [weak recommendation, low level of evidence]

Recommendation 20

We recommend tofacitinib for maintaining remission in patients with UC who responded to induction therapy with tofacitinib [strong recommendation, moderate quality of evidence]

Evidence: [strong recommendation, moderate quality of evidence]

Recommendation 21

We recommend ustekinumab for the maintenance of remission in patients with UC who responded to induction therapy with ustekinumab [strong recommendation, moderate quality of evidence]

Evidence: [strong recommendation, moderate quality of evidence]

ACG Clinical Guideline Update: Preventive Care in IBD - Summary Chart

File: 7. ACG_Preventive care chart.pdf

Farraye FA et al. *Am J Gastroenterol.* 2025;120(7):1447-1473. doi:10.14309/ajg.0000000000003541

5 rules extracted

General

Knowledge base - Society guidelines/7. ACG_Preventive care chart.pdf

ACG GUIDELINE

GUIDELINE Highlights | Preventive Care in Inflammatory Bowel Disease | Concept and Content: Erica Duh, MD | Reviewer: Freddy Caldera, DO, MS, PhD, FACP | PCV 21 or PCV 20

INTERVAL

(immunocompromised) | If PCV naïve: PCV 21 or PCV 20 | If incomplete PCV: PCV 13 $\geq$ 1 y ■ PCV 21 or PCV 20 | If PCV naïve: PCV 21 or PCV 20 | If incomplete PCV: PCV 13 + PPSV23 $\geq$ 5 y ■ PCV 21 or PCV 20 | If PCV naïve: PCV 21 or PCV 20 | If incomplete PCV: $\geq$ 5 yrs since last PCV ■ PCV 21 or PCV 20 | • 2-dose inactive recombinant herpes zoster vaccine. | • 0-1 month apart in immunocompromised; 2-6 months in all others. | Follow recommendations from Centers for Disease Control and Prevention. | Rotavirus vaccine | Infants exposed to | biologics in-utero | Administer rotavirus vaccine to infants exposed to biologics in utero. | • $\geq$ 65 years ■ recombinant, adjuvant or high dose influenza vaccine. | • Adults 19-64 years who are liver transplant recipients OR | on anti-TNF monotherapy ■ high dose influenza vaccine | • All other adults 19-64 ■ standard dose influenza vaccine | • All adults $\geq$ 75 years | • Adults 50-74 years with certain medical conditions | (including patients with IBD and liver transplant recipients) | Administer according to Advisory Committee on Immunization Practice Recommendations | 1. Evaluate for seroprotection status (Check for HbsAb $>$ 10 mIU/mL). | 2. If negative and previously immunized ■ provide challenge dose | 3. Not immune ■ provide Hepatitis B vaccine 2 or 3 dose series. | Do NOT check serologic titers to evaluate for | immunity to MMR or varicella. | Examples of completed | pneumonia vaccine | • PCV 13 and 2 doses | • PCV 15 and PPSV 23 | • PCV 20 or PCV 21 | Provide age-appropriate vaccines before initiation of immunemodifying therapies when possible.

INTERVAL

Women with IBD on immunemodifying therapy | • Within a year of onset of sexual activity | • Continue annually | • If $<$ 30, continue for 3 consecutive years before increasing to every 3 years | • Assure receipt of HPV vaccine series | • Annual screening for melanoma (independent of use of biologic therapy) | • Annual screening for NMSC (on immune-modifying therapies ex: 6-MP, | azathioprine, methotrexate, JAK inhibitors, or S1P receptor modulators)

INTERVAL

Adults with IBD and conventional risk | factors for abnormal bone mineral density | At time of diagnosis and periodically after diagnosis | IBD patients who smoke | Counseling for smoking cessation | Screen for depression and anxiety at baseline and annually. Refer for | counseling/therapy. | 6-MP = 6-mercaptopurine

ECCO Guidelines on the Prevention, Diagnosis, and Management of Infections in Inflammatory Bowel Disease

File: 8. ECCO_IBD and infections.pdf

Kucharzik T et al. *J Crohns Colitis*. 2021;15(6):879-913. doi:10.1093/ecco-jcc/jjab052

51 rules extracted

Immunosuppression classification: live vaccine rule

Live vaccines are CONTRAINDICATED in patients with moderate-to-severe immunosuppression (any biologic monotherapy, combination immunosuppression, prednisolone >20 mg/day for >2 weeks, or within 3 months of corticosteroid use at this level). For patients with selective or low-degree immunosuppression only, live vaccines can be discussed case-by-case when benefit outweighs risk.

Evidence: [EL5]

Post-immunosuppression: delay before live vaccine

Wait at least 1–6 months after STOPPING immunosuppressive therapy before administering a live vaccine — minimum 1 month after corticosteroids, minimum 3 months after conventional immunosuppressants, minimum 6 months after biologics.

Evidence: [EL5]

Statement 7.1 (Travel counselling for immunosuppressed patients)

IBD patients on immunosuppressive therapy cannot be routinely advised against travelling to countries with increased infection rates. However, pre-travel counselling regarding safety measures — including appropriate vaccinations, food and water hygiene, insect protection, and malaria prophylaxis — is STRONGLY RECOMMENDED before travel to endemic areas.

Evidence: [EL5]

Statement 7.2 (Malaria risk not increased in IBD on immunosuppression)

IBD patients on immunosuppressive therapy do NOT appear to be at increased risk for acquiring malaria or developing more severe malarial disease compared with the general population. Standard malaria prevention guidelines (chemoprophylaxis, insect bite prevention) should be followed.

Evidence: [EL5]

Statement 5.3 (Salmonella/Listeria: withhold immunosuppression)

Immunosuppressive therapy should be withheld temporarily until resolution of active Salmonella or Listeria infection. Anti-TNF agents carry particular risk for Listeria monocytogenes infection. Prevention: food hygiene education — avoid raw eggs, unpasteurised milk, raw-milk cheeses, and undercooked meat.

Evidence: [EL4]

Statement 5.7 (Nocardia risk on corticosteroids)

Patients on immunosuppressive therapy — particularly corticosteroids — are at risk for systemic and cutaneous Nocardia infections. Nocardia should be considered in the differential diagnosis of pulmonary or cutaneous infiltrates in immunosuppressed IBD patients.

Evidence: [EL5]

Statement 6.1 (Systemic fungal infection: chemoprophylaxis not routine)

Risk of systemic fungal infection in IBD is low but mortality is HIGH when it occurs. Apart from *Pneumocystis jirovecii* pneumonia (PCP) prophylaxis in high-risk patients, routine chemoprophylaxis for systemic fungal infections is NOT indicated. After systemic fungal infection: secondary prophylaxis and risk-benefit discussion with infectious disease specialist are required.

Evidence: [EL5]

Statement 7.3 (Probiotics probably safe on anti-TNF; caution with beta-haemolytic strains)

Probiotics in IBD patients on anti-TNF agents are probably safe, but safety may be a concern for probiotic preparations with known beta-haemolytic activity. Routine probiotic use is not recommended in severely immunosuppressed patients.

Evidence: [EL5]

Statement 8.2 (Pre-treatment infection/vaccination checklist)

Before initiation of immunosuppressive treatment (preferably at the time of IBD diagnosis): complete a standardised checklist on infection risk, vaccination status, and travel history. Vaccinate against VPDs before starting immunosuppression where possible.

Evidence: [EL5]

Statement 8.3 (Healthcare workers with IBD: TB testing and vaccination)

Healthcare workers with IBD on anti-TNF agents should undergo regular tuberculosis (TB) testing (IGRA annually or per local occupational health guidelines). They should closely follow occupational vaccination programmes, including influenza and hepatitis B.

Evidence: [EL5]

Statement 2.1

IBD patients at risk for opportunistic infections are those treated with immunosuppressive agents, particularly in combination [EL1]. Further predictive factors are malnutrition, obese body mass index [BMI], comorbidities, active disease, and older age [EL3]

Evidence: [EL1], [EL3]

Statement 2.2

Immunosuppressive agents should be classified according to mechanism of action, dose, duration, and route of administration [EL5]

Evidence: [EL5]

Statement 3.1*

Serological screening for hepatitis A, B, C, HIV, Epstein-Barr virus, cytomegalovirus, varicella zoster virus, and measles virus [in the absence of documented past infection or vaccination for the latter two] is recommended for all IBD patients at baseline [EL4] and especially before or during immunosuppressive treatment [EL1]. A Pap smear for human papillomavirus screening is also recommended [EL1]

Evidence: [EL4], [EL1], [EL1]

Statement 3.2

In non-immune patients, vaccination for HAV should be considered before commencement of immunosuppressive treatment [EL5]

Evidence: [EL5]

Statement 3.3*

Patients with IBD should be vaccinated against hepatitis B to achieve an anti-HBs antibody level >10 IU/L [EL1]

Evidence: [EL1]

Statement 3.4*

Patients with IBD and chronic hepatitis B infection should be treated with specific antiviral nucleos[t]ide analogues [EL1]

Evidence: [EL1]

Statement 3.5*

Prophylactic treatment with antiviral agents is not recommended in patients with IBD and previous HBV infection [HB core Ab-positive, HBsAg-negative] [EL3]

Evidence: [EL3]

Statement 3.6*

Patients with IBD and hepatitis C should be treated in accordance with national and international guidelines [EL5]. Patients with IBD and hepatitis C should be closely monitored for disease exacerbation when being treated with direct-acting antiviral agents [DAAs] [EL5]

Evidence: [EL5], [EL5]

Statement 3.7*

IBD patients with HIV infection can be treated with immunosuppressive therapy when on antiretroviral therapy with stable CD4 counts and undetectable viral load. The CD4 count should be closely monitored [EL4]

Evidence: [EL4]

Statement 3.8*

Recombinant herpes zoster vaccine [RZV] is the preferred vaccine for patients with IBD disease, given its efficacy and safety [EL3]. If RZV is not available, a live zoster vaccine [ZVL] is recommended in immunocompetent patients with IBD aged ≥ 50 years [EL4] RZV remains recommended for patients with IBD receiving immunosuppressive therapy [EL4]. If RZV is unavailable, ZVL may be considered in patients on low-dose immunosuppression [EL3]

Evidence: [EL3], [EL4], [EL4], [EL3]

Statement 3.9*

Concurrent CMV colitis worsens the prognosis of active IBD. Patients with refractory IBD should be tested for CMV colitis [EL3], especially if they are not responding to immunosuppressive therapy [EL2]

Evidence: [EL3], [EL2]

Statement 3.10*

Immunohistochemistry [IHC], possibly tissue polymerase chain reaction [PCR], or both, are essential for confirming active CMV infection [colitis] in IBD and should be the standard tests [EL2]. Findings and potential interventions should be discussed in the clinical context

Evidence: [EL2]

Statement 3.11*

Immunosuppressive therapy should not be discontinued in IBD patients with intestinal CMV reactivation in general [EL3]. Steroids should be tapered [EL4]. Antiviral therapy should be considered in steroid-refractory IBD patients with CMV colitis [EL3]. Discontinuation of immunosuppressive therapy is recommended in symptomatic disseminated CMV infection [EL 4]

Evidence: [EL3], [EL4], [EL3]

Statement 3.12*

EBV is associated with an increased risk of lymphoma in EBV-negative patients on immunosuppressive therapy, primarily thiopurines [EL4]. Use of thiopurines in EBV-IgG negative patients should be carefully considered [EL5]

Evidence: [EL4], [EL5]

Statement 3.13*

Patients on immunosuppressive therapy are considered to have an enhanced risk for development of severe influenza infection [EL5]. Annual influenza vaccination of patients on immunosuppressive therapy is recommended according to national guidelines [EL5]. Live vaccines should not be administered to immunosuppressed patients

Evidence: [EL5], [EL5]

Statement 3.14*

Immunosuppressive therapy should be discontinued in severe cases of varicella infection, disseminated HSV and VZV, symptomatic infectious mononucleosis, EBV-related mucocutaneous ulceration, and severe influenza [EL4]. Immunosuppressive therapy should be withheld in cases of measles [EL5]

Evidence: [EL4], [EL5]

Statement 3.15*

Immunosuppressed IBD patients with an ongoing HSV, VZV, or influenza infection should receive the appropriate antiviral treatment [EL4]

Evidence: [EL4]

Statement 3.16*

Immunosuppressed female IBD patients should undergo annual cervical cancer screening [EL3]

Evidence: [EL3]

Statement 3.17*

Routine prophylactic HPV vaccination is recommended for both young female and young male patients with IBD [EL2]

Evidence: [EL2]

Statement 3.18

During the SARS-CoV-2 pandemic, management of IBD should follow usual standards of care [EL5]

Evidence: [EL5]

Statement 3.19

When COVID-19 is clinically suspected, or when a patient tests positive for SARS-CoV-2 [symptomatic or asymptomatic], continuation of 5-ASA and immunosuppressive therapy should be considered on a case-by-case basis according to current knowledge [EL 4]

Statement 4.1*

The reactivation risk of latent tuberculosis infection [LTBI] in patients treated with biologics or JAK inhibitors is increased, and the disease can be more severe than in the background population [EL2]. Before its start and, ideally, before any immunosuppression, IBD patients should be screened for LTBI [EL1]. Consider re-screening patients previously exposed to biologics and JAK inhibitors before switch or swap [EL3]. Under special conditions, re-screening during anti-TNF agent therapy and JAK inhibitors should be considered [EL5]

Evidence: [EL2], [EL1], [EL3], [EL5]

Statement 4.2

LTBI should be diagnosed by a combination of patient clinical data and epidemiological factors, chest X-ray, and tuberculin skin test [TST] or interferon-gamma release assay [IGRA] [or both] according to local availability and national recommendations [EL5]

Evidence: [EL5]

Statement 4.3

Patients diagnosed with LTBI before biologic or smallmolecule therapy or prolonged high-dose systemic steroids should be treated with a complete therapeutic regimen for LTBI [EL1]. In other situations, specialist advice should be sought. When there is LTBI and active IBD, biologic or small-molecule therapy should be delayed for at least 4 weeks after chemotherapy, except in cases of greater clinical urgency and with specialist advice [EL5]

Evidence: [EL1], [EL5]

Statement 5.1*

Pneumococcal vaccination should be recommended for all patients with IBD [EL3]

Evidence: [EL3]

Statement 5.2*

Patients with IBD on immunosuppressive therapy with pneumonia should be tested for *Legionella pneumophila* [EL4]. In case of *Legionella pneumophila* infection, immunosuppressive agents should be temporarily withheld until resolution of active infection [EL5]

Evidence: [EL4], [EL5]

Statement 5.3*

Patients receiving immunosuppressive agents are at risk of more severe infections with *Salmonella enteritidis* and *S. typhimurium* [EL4] and systemic and central neurological infections with *Listeria monocytogenes*. [EL4] The incidence of *L. monocytogenes* infections appears higher in patients treated with anti-TNF agents compared with other immunosuppressive agents [EL4]. Immunosuppressive therapy should be temporarily withheld until resolution of the active infection [EL5] For IBD patients, invasive *Salmonella* spp. infections related to immunosuppressive therapy have been reported.^{327–337} Definitive diagnosis is made by isolating *Salmonella* spp. from blood, stool, or urine. Salmonellosis is treated with antibiotics such as fluoroquinolones or third-generation cephalosporins. In cases of *S. typhimurium* osteomyelitis,³³⁸ aortitis,³³⁹ or septic arthritis,^{340,341} a combination of antibiotics and surgical treatment may be required. Immunosuppressants should be temporarily withheld until resolution of active infection. Immunosuppressive therapy is considered to confer a high risk for intestinal or systemic *Salmonella* spp. infections.^{342–345} Immunosuppressive therapy is considered to confer a high risk for *L. monocytogenes* infection, which causes primarily severe septicaemia and meningitis accompanied with considerable mortality.³⁴⁶ Compared with other immunosuppressants, anti-TNF agents appear to confer a particular risk for serious infection.^{327,347–361} Given that *L. monocytogenes* infections after infliximab treatment frequently occur after three or fewer infusions, reactivation of latent infection could be considered. Treatment for *L. monocytogenes* consists of ampicillin, amoxicillin, or case of allergy to penicillin, trimethoprim/ sulphamethoxazole [TMP-SMX]. Prevention of *Salmonella* spp. and *L. monocytogenes* infections consists of food hygiene and careful food choices [such as avoidance of raw eggs, unpasteurized milk, raw-milk cheese, and insufficiently cooked or raw meat]. Diagnosis is made by appropriate microbiological blood and cerebrospinal fluid Gram staining and cultures. A high index of suspicion is appropriate for patients on immunosuppressive therapy who present with signs and symptoms of meningitis or other neurological symptoms. Comprehensive investigation, including lumbar puncture, should be performed as soon as such symptoms develop.³⁵³ This may lead to early diagnosis and treatment, which is important given the pathogenicity of *L. monocytogenes*. No conclusive data are available on whether immunosuppressive should be temporarily or indefinitely withheld in the event of active infection. Nevertheless,

Evidence: [EL4], [EL4], [EL4], [EL5]

Statement 5.4*

Screening for *C. difficile* infection [CDI] is recommended at every disease flare in patients with IBD and especially in patients receiving immunosuppressive therapy [EL3]

Evidence: [EL3]

Statement 5.5

Diagnosis of CDI requires documentation of toxigenic *C. difficile* in stool accompanied with diarrhoea. A two-step algorithm with a highly sensitive test such as glutamate dehydrogenase [GDH] antigen enzyme immunoassay or nucleic acid amplification tests should be used initially, followed by a second test with high specificity, such as toxin A/B enzyme immunoassays [EL3]

Evidence: [EL3]

Statement 5.6*

Oral vancomycin and fidaxomicin for 10 days are equally effective in treating non-severe CDI [EL1]. For severe CDI, intravenous metronidazole should be added to oral vancomycin for 10 days [EL3]. Treatment of CDI recurrence includes oral vancomycin, fidaxomicin, faecal microbiota transplantation [EL3], and bezlotoxumab [EL5]. In CDI, use of immunosuppressants can be maintained after careful risk-benefit evaluation and clinical judgement [EL5]

Evidence: [EL1], [EL3], [EL3], [EL5], [EL5]

Statement 5.7

Background: see Supplementary material, available as Supplementary data at ECCO-JCC online.

Statement 5.8

Meningococcal vaccination should be administered to patients with IBD as per regional or national recommendations for the general population [EL5]

Evidence: [EL5]

Statement 6.1

The risk of fungal infection in IBD is low. Systemic infections are exceptional, but mortality is high [EL4]. Apart from *Pneumocystis jirovecii*, chemoprophylaxis is not indicated. Chemoprophylaxis following systemic fungal infection should be discussed with an infectious disease specialist [EL5]

Evidence: [EL4], [EL5]

Statement 6.2

Screening for parasitic or fungal infections should be considered in residents of endemic areas or with relevant travel history [EL5]

Evidence: [EL5]

Statement 6.3

For patients with IBD on triple immunosuppressive therapy [including steroids, methotrexate, thiopurines, biologics], standard prophylaxis with TMP-SMX should be strongly considered [EL4]. For those on double immunosuppressive therapy, prophylactic TMP-SMX may also be considered, especially if one of these is a calcineurin inhibitor [EL4]. TMP-SMX should also be considered for any combination of high-dose corticosteroids, low lymphocyte count, or JAK inhibitors [EL5]

Evidence: [EL4], [EL4], [EL5]

Statement 7.3

Intake of probiotics in patients receiving anti-TNF agents is probably safe, but safety may be a concern for probiotics with beta-haemolytic activity [EL5]

Evidence: [EL5]

Statement 7.1*

Given the lack of data, it is currently not possible to advise against travelling to countries with increased infection rates. However, pre-travel counselling regarding safety measures is strongly recommended for patients under immunosuppression travelling to endemic areas [EL4]. Specific travel recommendations from national authorities and the World Health Organization should be consulted [EL5]

Evidence: [EL4], [EL5]

Statement 7.2

Patients with IBD, including those on immunosuppressive therapy, do not appear to be at increased risk for acquiring malaria or for a more severe disease course and should follow standard guidelines for prevention [EL5]

Evidence: [EL5]

Statement 8.1*

Live vaccines in patients with IBD receiving immunosuppressive therapy are generally considered unsafe. It is recommended to wait for at least 1–6 months after termination of immunosuppressive therapy before administration of a live vaccine [EL5]. The decision to administer any live vaccine should be considered on a case-by-case basis [EL5]

Evidence: [EL5], [EL5]

Statement 8.2

Before initiation of treatment, preferably at the time of IBD diagnosis, a standardised checklist regarding infection risk and immunisation status should be completed [EL5]

Evidence: [EL5]

Statement 8.3

Risk of *M. tuberculosis* infection is increased in healthcare workers with IBD on anti-TNF agents [EL4]. Regular testing for TB is advised for healthcare workers [EL5]. Vaccination programmes for routine and specific vaccines should be closely followed [EL5]

Evidence: [EL4], [EL5], [EL5]

ECCO Consensus on Management of Inflammatory Bowel Disease in Low- and Middle-Income Countries

File: 9. ECCO_IBD in low and middle income countries.pdf

El-Hussuna A et al. *J Crohns Colitis*. 2025. doi:10.1093/ecco-jcc/ijaf125

37 rules extracted

Statement 1

Although reliable data on the incidence and prevalence of IBD in LMICs is limited, the rates suggest an increasing incidence in regions where data are available [EL3]. Supporting text Incidence and prevalence rates (PR) of IBD in low- and middle-income countries (LMICs) are typically reported by country or region. Between 1986 and 2005, Brazil reported incidence rates (IR) per 100,000 person-years ranging from 0.74–6.76 for ulcerative colitis (UC), 0.24–3.50 for Crohn's disease (CD), and 0.42–1.75 for unclassified inflammatory bowel disease (IBDU) (12). By 2015, the IR had increased to 8.00 for UC and 5.48 for CD (7). In Colombia, the UC IR increased from 5.59 in 2010 to 6.3 in 2017, and the CD IR decreased from 1.29 in 2010 to 0.74 in 2018.(13) Mean annual IR of UC in Argentina and Panama was

Evidence: [EL3]

Statement 2

Limited data from LMICs show regional differences in IBD hospitalization rates and mortality. Mortality rates are influenced by access to timely surgical interventions and complications like sepsis and thromboembolism [EL4].

Evidence: [EL4]

Statement 3

Targeted public health interventions focusing on modifiable risk factors, such as antibiotic use, smoking, lifestyle habits, and dietary patterns, may be helpful in addressing the rising incidence and prevalence of IBD in LMICs [EL4]. Supporting text Modifiable factors Diet In Iran, higher total fat intake was associated with an increased UC risk (odds ratio [OR]: 2.46, 95% confidence interval [CI]: 1.37–6.35),(34) while higher dietary calcium intake was protective [OR: 0.08, 95% CI: 0.02-0.27].(35) A prospective cohort study conducted in 21 low-, middle-, and high-income countries revealed that higher intake of ultra-processed foods was associated with an increased risk of IBD (hazard ratio [HR]: 1.82, 95% CI: 1.22–2.72 for ≥ 5 servings/day).(36) Hygiene

Evidence: [EL4]

Statement 4

Disease and sociodemographic characteristics of adult and paediatric IBD populations vary across LMICs [EL3]. Conducting population-based studies on IBD populations from LMICs will help improve awareness of potential differences in these characteristics [EL3]. Supporting text Studies from the US showed that the UC:CD ratio was similar in African Americans compared with whites, but a higher proportion of UC was observed among Hispanic and Asian populations, ranging from 52–80%. (55, 56) UC was more frequent than CD in Latin America (57, 58), but CD was more frequent in the Caribbean region. (58) In India, UC is more common in the North and CD in the South. (59) Age at diagnosis Individuals from Asia, Middle East, and Latin America were older at CD or UC diagnosis compared with whites and African Americans. (55, 56, 60, 61) In southeast Asia, the median age at UC diagnosis was 35–42 years, which is later than CD [31–34 years]. (62) In Latin America, the mean age at diagnosis ranged from 31–41.7 years. (57, 58) In Turkey, the mean age at diagnosis was 42.6 years for UC and 37.4 years for CD. (63) The mean age at IBD onset was 9.07 years [UC: 7.73 years, CD: 10.18 years; $p < 0.001$] in Asian children. (64) Sex distribution IBD was significantly more prevalent in African American females compared with whites and Hispanics [63% vs 51.5% and 45.6%, respectively], but no sex differences were observed among whites and Hispanics. (55, 56) Studies from South Asia, Southeast Asia, and the Middle East reported more male than female patients with IBD. In India, Bangladesh, and Nepal, the male: female ratio was 1.6:1. (62) The proportion of males ranged from 27.4–47.1% in Latin America (57) and the Caribbean. (58) Disease location and behaviour in CD

Evidence: [EL3], [EL3]

Statement 5

There is a need to develop educational programs for primary healthcare providers to facilitate prompt referral of patients to specialist care, with the aim of preventing diagnostic delay and starting effective treatment to prevent IBD complications [EL4]. Supporting text In many LMICs, healthcare curricula lack training on IBD, leaving practitioners with limited knowledge. As a result, IBD may be frequently misdiagnosed and clinical supervision and laboratory facilities necessary for proper patient monitoring may be lacking. A study reviewed online web-based e-learning tools with easy access to up-to-date IBD educational materials as resources for physicians and patients worldwide. It showed that locally trained physicians have limited knowledge on IBD. IBD is not included in the training Core Curriculum in most of the participating countries. The healthcare approaches, both at the primary care and referral levels, many times lack the essential regular clinical guidance. (78) A common deficit in knowledge on IBD complications was found in countries such as Iran and Sri Lanka. Unemployed or lesser-educated patients in Sri Lanka had significantly poorer knowledge on IBD than those employed or better educated. (79, 80) For example, the former had lack of knowledge on colorectal cancer risk and surgical interventions. In LMICs, public health may prioritize infectious diseases over IBD. Most diarrhoeal diseases are presumed to be of infectious cause at the primary healthcare level, which delays referral to a specialist and diagnosis of IBD. (10) The disparity between population size and health expenditures may for example result in a very low health expenditure per

Evidence: [EL4]

Statement 6a

Available guidelines for the management of IBD are based on studies from HICs [EL1]. There is a need to develop locally applicable guidelines to address the specific needs of IBD care in LMIC settings [EL4]. Supporting text In 2010, the World Gastroenterology Organization Practice Guidelines for the Diagnosis and Management of IBD provided recommendations to account for variations observed in some LMICs. (81) In Latin America, guidelines were developed for IBD diagnosis and management, endoscopy, special situations, and quality indicators for IBD clinics. (82–86) In Asia, recommendations and consensus statements were established on definitions, diagnosis, medical management, and monitoring. (87–89) Consensus statements were prepared in India focused on diagnosis and treatment of IBD. (90–95) As existing guidelines are primarily based on studies from HICs, they may, therefore, insufficiently consider the unique challenges and resource constraints faced by LMICs. Region-specific guidelines adapted to LMIC settings would improve their relevance and applicability to address local needs, manage resource limitations, and improve outcomes.

Evidence: [EL1], [EL4]

Statement 6b

There have been good attempts to build IBD registries in LMICs [EL3]. Financial and technical support from governmental and non-governmental organizations to develop IBD registries in LMICs can aid in understanding the burden of IBD and enhance IBD quality of care [EL3]. Supporting text In Iran, a project was started in 2019 to gather detailed data on patients with IBD.(96) As the first IBD registry in the Middle East, this may become a reliable resource for national and international research on IBD.(97) In Africa, a prospective observational registry was started in 2020 in Harare, Zimbabwe, to provide data on the burden, disease phenotype, and outcomes of IBD.(98) In Asia, a multicentre registry of paediatric IBD from Thailand showed clinical and demographic characteristics of this disease.(99) The

Evidence: [EL3], [EL3]

Statement 7

The incidence and prevalence of opportunistic infections are high in IBD patients from LMICs treated with biologics, immunosuppressive drugs, and steroids [EL3]. There is a need to develop pharmacovigilance strategies to mitigate these risks [EL5]. Supporting text A meta-analysis of six cohort studies revealed a pooled IR of herpes zoster in IBD of 10.41. IBD was associated with an increased risk of developing herpes zoster [incidence rate ratio: 1.68, 95% CI: 1.53–1.84].(101) Elderly IBD patients treated with biologics were at increased risk of developing serious infections [RR: 2.70, 95% CI: 1.56–4.66] and opportunistic infections [RR: 3.16, 95% CI: 1.09–9.20].(102) A study from China showed current and past HBV infection was detected significantly more often in individuals with IBD [41.2%] than in the general population.(103) Another study from China found that severe disease activity, elevated levels of faecal calprotectin [FC], and immunosuppressive medications, especially when used in combination, were risk factors for opportunistic infections in patients with IBD. (104) Two studies from Brazil reported risk factors associated with latent tuberculosis [TB], such as patient age, disease duration, gender, hospitalization in the last 12 months, workplace [enclosed], and duration of medication use [anti-TNF agents such as adalimumab or infliximab].(105, 106) A study from Mexico found that the frequency of cytomegalovirus [CMV] infection was 17% in toxic megacolon and steroid-refractory UC.(107) In the Middle East, a study from Iran reported a prevalence rate of microsporidia infection of 12.7%, with IBD patients undergoing treatment with immunosuppressive, corticosteroid, or immunomodulator agents more susceptible to intestinal microsporidia infection.(108) A study from India reported a high risk of TB during infliximab therapy despite TB screening.(109) A meta-analysis found a pooled prevalence of TB in IBD of 0.08% [95%CI: 0.05–0.10%]. The pooled prevalence was 0.02% [95% CI: 0.02–0.03%], 0.21% [95% CI: -0.02–0.43%], and 1.59% [95% CI: 1.19–

Evidence: [EL3], [EL5]

Statement 8a

Infectious aetiologies that mimic IBD, in particular TB, amoebiasis, Salmonella, Shigella, Yersinia, Strongyloides stercoralis, and HIV-related infections, should be considered and excluded in the diagnosis of IBD in LMICs [EL3]. Supporting text Infectious colitis should be considered when diagnosing IBD in LMICs due to the high prevalence of infections, such as TB, amebiasis, and others, which can mimic IBD. The overlap of symptoms, particularly between CD and GITB, presents a significant diagnostic challenge. Misdiagnosis can lead to severe consequences, including inappropriate treatments that could exacerbate the condition. Physicians can avoid delays in appropriate diagnosis and treatment by excluding key infections like TB, Yersinia, and Strongyloides. However, in recent years, there has been an increase in the number of patients with IBD, including CD, in newly industrialized countries like China and India.(112) A recent study from India reported that IBD accounts for more than 5% of patients presenting with lower gastrointestinal symptoms, a rate higher than that of infectious colitis.(113) Therefore, while infectious causes should be excluded, this approach must be balanced with timely IBD diagnosis to avoid unnecessary delays in care. A detailed patient history, including recent travel, antibiotic use, and exposure to infectious agents, is crucial. Intestinal infections can cause colitis, and ileocolonic ulcers can be seen in GITB(114) Ileitis, resulting from Yersinia and Salmonella infections.(114) TB is considered endemic in most of LMICs, and

Evidence: [EL3]

Statement 8b

In LMICs, infectious causes should be considered for each disease relapse. We suggest all patients with acute colitis flares should have stool testing for amoebiasis, stool cultures for bacterial infections, and possible *Clostridioides difficile* infection based on region-specific prevalence [EL3]. Supporting text In resource-limited countries, infectious causes should be excluded from each disease relapse. The risk of enteric infections is higher in patients with IBD than in controls. *Entamoeba histolytica* [EH] is an intraluminal parasite that can cause symptomatic disease characterized by abdominal pain, cramps, and bloody diarrhoea. Amebiasis can also exacerbate IBD symptoms, as demonstrated in a report by Babic et al., who found EH or *Entamoeba dispar* in 19/119 [16.0%] cases of IBD flares.(118) A systematic review that identified 24 cases of severe and fulminant amoebic colitis treated with corticosteroids revealed that 14/24 patients were mistaken for IBD and incorrectly treated with corticosteroids, leading to rapid disease progression; nearly half of these patients underwent surgery, and a quarter of them died.(119) The authors concluded that infection with EH should be considered before administering corticosteroids to patients residing or travelling in endemic areas.(119)

Evidence: [EL3]

Statement 8c

Stool PCR is the gold standard for diagnosing EH due to its high sensitivity and specificity [EL3]. However, in many resource-limited settings where PCR may not be available, stool microscopy and serology can provide some diagnostic utility, although their sensitivity and specificity are considerably lower [EL3]. Supporting text Amoebiasis is endemic in developing parts of Central and South America, Africa, and Asia. Amoebiasis is caused by infection with the protozoan parasite EH. Millions of people are infected with EH, making amoebic colitis a leading cause of diarrhoea, and EH is estimated to kill more than 55 000 people each year.(124) The hallmark of amoebic colitis is diarrhoea, which may be watery or bloody. The disease can be limited to the ascending colon or caecum, but may also present similarly to IBD. Corticosteroid use is a risk factor for the development of fulminant forms of amoebic colitis.(124) Several diagnostic modalities, including microscopy, antigen detection, molecular tests, and serology, can assist with diagnosis. Stool microscopy allows for visualization of cysts and trophozoites by an experienced observer. While readily available, stool microscopy has poor sensitivity and specificity and cannot differentiate between *Entamoeba* spp. Microscopy should not be used when other modalities are accessible.(125) Serology is a valuable adjunct to stool studies and has high sensitivity [65–92%] and specificity [> 90%]. Stool antigen detection tests have variable sensitivity and specificity, particularly in

Evidence: [EL3], [EL3]

Statement 9a

Improving access to and expertise in ileocolonoscopy in LMIC settings will facilitate early and accurate diagnosis [EL5]. Supporting text There are issues related to endoscopy access, especially advanced endoscopy, across LMICs. The appropriate use of these limited resources is required to achieve a timely diagnosis of IBD. While sigmoidoscopy may be useful in making an initial diagnosis of UC, mapping disease extent and dysplasia surveillance require colonoscopy. Furthermore, sigmoidoscopy may be inadequate to assess disease activity in UC, although it may be sufficient in those with limited disease extent.(128-130) There is a lack of access to endoscopy in some regions of LMICs that may result in a diagnostic delay in IBD.(131, 132) Sub-Saharan Africa has low endoscopic availability, especially in outlying rural areas. There are very few trained endoscopists and gastroenterologists, and faulty equipment often cannot be repaired due to limited resources. Most endoscopy units are in urban areas with long waiting lists for access, even for patients with suspected IBD.(133) Additional resources include access to endoscopic balloon dilation for CD-related strictures, high-definition white light endoscopy for dysplasia detection, and video capsule endoscopy [VCE] and device-assisted enteroscopy [DAE] for evaluating small-bowel involvement in CD.(134-137) Access to small-bowel endoscopy is required to diagnose and manage small-bowel disease. The options for small-bowel endoscopy include VCE and DAE. Centres caring for patients with small-bowel CD should have access to one form of DAE [single- or double-balloon enteroscopy or spiral enteroscopy].(138) There are challenges in using these modalities; DAE requires access to the specific device, anaesthesia and

Evidence: [EL5]

Statement 9b

The practices and availability of sedation during endoscopy in LMICs are variable. Sedation might be a preferable option to encourage increased and better endoscopic examinations, and it should be guided by patient preferences, pre-endoscopic assessment, and local resources [EL5]. Supporting text Sedation during gastrointestinal endoscopy is a well-established practice with a focus on patient safety and comfort. The choice of sedative agents and the level of sedation (deep sedation, moderate [conscious] sedation, or procedures without sedation) should be individualized, considering patient-specific factors and procedural requirements.(144) However, even in developed countries there is no consensus on the type and depth of sedation. Arguments against sedation include cost-effectiveness, safety, patient turnaround, and patient preference.(145-147) There is a significant gap in knowledge in LMICs regarding sedation practices. Sedation availability is variable in LMICs.(148) There is also variability in access to anaesthetist- or nurse-driven sedation practices in endoscopy. Offering sedation often requires the availability of anaesthesiologist support in case of complications or delayed recovery from anaesthesia. Patient awareness and fear of the procedure in general and sedation itself are also significant issues that may impact patient willingness to undergo a procedure, which may lead to delayed or missed diagnosis. Updated surveys on patient awareness and preferences are advised.

Evidence: [EL5]

Statement 10

There are several infectious diseases that can mimic IBD, and histological features may not be pathognomonic. Granulomas, although suggestive, are not diagnostic of CD and have numerous differential diagnoses apart from CD [EL3]. A comprehensive evaluation of clinical, radiological, histological, and microbiological evaluation helps discriminate IBD mimics from IBD [EL3]. Supporting text Although IBD has no specific features, histology is an important parameter for diagnosis. The histological assessment of IBD is based predominantly on disease activity [epithelial damage, neutrophilic infiltration of epithelium, and crypts] and changes of chronicity [architectural distortion of crypts, crypt loss]. Various scores used in both UC and CD typically consider these two parameters along with some additional parameters [such as granuloma in CD, eosinophils, or lamina propria infiltrate].(149) However, none of these parameters is specific to IBD. Several conditions can mimic IBD clinically [presenting with chronic diarrhoea] and histologically. The presence of non-caseating granulomas favours the diagnosis of CD. However, it is important to recognize that there are many differential diagnoses of granulomas.(150, 151) (152-156) Granulomas can occur in CD and infectious colitis [GITB, Schistosoma, Yersinia, Paracoccidioides, histoplasmosis]. The frequency of granulomas reported in CD is variable; granulomas are more likely to be detected in resection specimens than in endoscopic biopsies.(150) Furthermore, granulomas may be less common in delayed-onset CD.(157) Fissuring ulcers and focal mucosal disease are suggestive of CD.(158) GITB, which can mimic CD, can also present with granulomas. While caseating granulomas are specific to GITB, epithelioid granulomas can be found in both conditions.(154) Large granulomas [$> 400 \mu\text{m}$], confluent or coalescing granulomas, and a higher frequency of granulomas favours GITB.(153, 154, 159-162) Ulcers lined by a band of epithelioid histiocytes also suggest GITB.(160) Other features that may suggest GITB are granulomas in the caecum and at multiple sites.(160) Small granulomas, focally enhanced colitis, pericryptal granuloma, and deep ulcers suggest CD diagnosis over GITB.(160) Schistosomal colitis can cause epithelioid granulomas, with excess mucus, nearly normal crypts [occasionally crypt distortion], eosinophilic infiltration of the mucosa and submucosa, and occasionally the presence of Paneth cell metaplasia. Sometimes, calcified or uncalcified ova may be visualized in mucosal biopsies.(163, 164) Histoplasmosis can mimic IBD as it can cause an intestinal mass or ulcer-like lesions and histology may reveal granulomas and histoplasma-invaded macrophages. Special stains, such as silver stain, best

Evidence: [EL3], [EL3]

Statement 11a

In TB-endemic areas, misdiagnosis can be the most important single factor in diagnostic delay of CD [EL3]. Diagnostic delay is more common in LMICs and is associated with low awareness and limited resources, leading to a more complicated disease course [EL3]. Supporting text The time to diagnose IBD in LMICs is greater than in HICs, which may be due to differences in healthcare provision. In addition, the difficulty in differentiating between IBD and more prevalent infectious diseases is a barrier to IBD diagnosis in LMICs.(179) A further reported barrier to accurate diagnosis of IBD in LMICs is the perceived rarity of IBD and the consequent lack of clinical awareness, leading to a lower index of suspicion.(179) Data from a multicentre study in Korea revealed that the mean interval of diagnostic delay was 16.0 ± 33.1 months; however, this gap decreased significantly over time.(180) In many resource-limited countries, several cultural barriers exist for colonoscopy. Disease is usually ignored for a long time by patients before they seek medical attention due to the social stigma of chronic disease; accordingly, diagnosis is delayed.(10, 181) Diagnostic delay in CD is associated with significantly higher stenotic complications and the need for surgery. (181) Diagnostic delay in LMICs is usually multifactorial and related to inadequate disease awareness and poor access to and low availability of diagnostic services. A study from India revealed that up to a third of patients with CD are misdiagnosed as having GITB, and empirical anti-tuberculosis treatment [ATT] is the single most significant contributor to diagnostic delay in TB-endemic areas.(181) Extreme delay in CD diagnosis can occur due to the assumption of a GITB diagnosis and improvement of stricturing disease with surgery. These patients likely present with complicated phenotypes of postoperative CD.(182) Increasing disease awareness and improving local gastroenterology care by collaborations between gastroenterologists from LMICs can accelerate development of comprehensive gastrointestinal care. The availability of and maintenance service for endoscopy equipment are major challenges. Collaboration between the local community and non-profit organizations can be a sustainable source of continual funding for gastrointestinal units.(182, 183)

Evidence: [EL3], [EL3]

Statement 11b

Differentiating between GITB and CD is a complex challenge due to overlapping clinical, radiological, and histopathological features. A multifaceted approach using clinical, immunological, molecular, endoscopic, radiological, histopathological, and microbiological assessments is essential to improve diagnostic accuracy [EL3].

Evidence: [EL3]

Statement 11c

Screening for active or latent TB before starting an anti-TNF agent, other biologic therapies, or a small molecule [immunosuppression] is essential in LMICs due to the high prevalence of TB in these regions. These therapies can significantly increase the risk of reactivating latent TB, which can lead to severe complications. Although tests such as interferon-gamma release assays and tuberculin skin test have limitations, these tests play a crucial role in detecting latent TB, with interferon-gamma release assays offering more specificity in TB-endemic areas [EL2].

Evidence: [EL2]

Statement 11d

In TB-endemic regions where diagnostic uncertainty between GITB and CD exists after colonoscopy, histology, microbiology, and a trial of ATT followed by a repeat colonoscopy after 2 months to assess response may be considered [EL3]. Supporting text Screening for active or latent TB before starting treatment is essential. There is currently no data to determine the utility of TB screening prior to initiation of corticosteroids. Distinguishing GITB from CD is a significant clinical challenge due to the similarities between the two conditions and the high prevalence of TB in LMICs. GITB and CD are both chronic granulomatous diseases with overlapping symptomatology and radiological, endoscopic, and histopathological findings. (22)A South Korean study revealed that approximately 11% of patients with GITB were misdiagnosed as having CD, while approximately 18% of patients with CD were misdiagnosed with GITB.(184) Misdiagnosis and delayed treatment can have serious consequences with a risk of disease progression, and starting ATT in patients with CD can have serious adverse effects. Furthermore, TB dissemination may occur in patients receiving immunosuppressive treatment who have GITB misdiagnosed as CD.(185) Making an accurate diagnosis as early as possible is essential to prevent complications. Differentiating between GITB and CD requires a multifaceted approach using detailed history [travel history, recent antibiotic use, exposure to TB] and a comprehensive clinical evaluation [including a chest X-ray] combined with an array of auxiliary tests

Evidence: [EL3]

Statement 12

The increased risk of TB reactivation in IBD patients treated with steroids and advanced therapies is of particular concern in LMIC settings [EL2]. Supporting text Ileocolonic resection could be considered a viable first-line treatment for CD, particularly in areas with a high prevalence of TB.(196) In clinical practice, approximately 1–3% of IBD patients treated with antiTNF agents develop active TB even after a negative screening test. These findings have been confirmed by reports from several Asian countries, including India, Japan, and South Korea. (197) The risk is notably higher with anti-TNF agents when compared with other treatments, such as 5-ASA, corticosteroids, and immunomodulators. Overall, there have been very few reported cases of TB associated with vedolizumab clinical trials have reported only a few cases and post-marketing data also indicate a low risk, with seven cases of TB in over 110 000 patient-years of exposure. There are no available data comparing the TB risk between antiTNF agents and anti-integrin agents. Data on TB risk with newer biologics remain limited.(197) Data from non-IBD indications also suggest that ustekinumab has a low risk of TB. (197) Data from non-IBD indications suggest an increased risk of TB with tofacitinib, especially in TB-endemic regions.(198) Distinguishing GITB from CD remains challenging in India, with 10–15% of cases misdiagnosed. Historically, patients were treated with ATT for years before a CD diagnosis. Although improved diagnostics now allow for more accurate differentiation, nearly a third of patients with CD still receive ATT before a correct diagnosis. TB is confirmed if endoscopic and histological improvements are observed after ATT, while poor response may indicate CD or drug-resistant TB.(199)

Evidence: [EL2]

Statement 13

Vaccinations against infectious diseases should be administered, preferably prior to starting immunosuppressive therapy in LMICs [EL3]. However, the cost of vaccines, their availability, and local infection prevalence are particular considerations in LMICs [EL5]. Supporting text In India, the vaccination rate of common vaccines [BCG, hepatitis B, diphtheria-tetanus-pertussis, oral polio vaccine, and measles] is 81% and coverage with elective vaccines is lower.(201) A study from India revealed that only 9.2% of patients with IBD had completed the hepatitis B vaccine.(202) This suggests that vaccination in adult IBD patients is low in LMICs, such as India.(203) There are only limited data in the literature to estimate the risk of infections in patients with IBD in LMICs. A study of 908 patients in north India revealed HBsAg positive status in 2.6%, 2.4% and 3% of patients with IBD, UC, and CD, respectively.(204) A region-specific meta-analysis found that the prevalence of HBsAg positive status in patients with IBD in India is 2.4% [range 1.4–3.4%].(205) Median anti-HBs titres were lower in patients with UC than controls [67 IU/L vs 105 IU/L; $p < 0.01$].(206) Adequate and effective immune responses were significantly lower in patients with UC than controls [82% vs 96%, $p = 0.001$; 41% vs 66%, $p < 0.001$]. Multivariate analysis revealed that independent predictors of adequate immune response were non-exposure to therapies like steroids, thiopurines, and biologicals, male gender, and early disease [disease duration < 5 years]. This suggests that all patients with new diagnoses of IBD in LMICs should be offered vaccination against hepatitis B. A study from India on 41 patients with CD revealed that all patients were positive for anti-hepatitis A antibodies, with similar titres as in controls.(207) A meta-analysis in patients with IBD showed that the rates of hepatitis A seropositivity were the highest in Asia.(208) Nearly a third of the patients had antibodies to SARS-CoV-2 in non-vaccinated patients with IBD from northern India.(209) This suggests that although vaccination is warranted against COVID-19, a subset of patients from India are already exposed to SARS-CoV-2. In Iran, rates of measles vaccination were 72.5% and 73.0% for patients with CD

Evidence: [EL3], [EL5]

Statement 14

The evidence for the use of empiric antibiotics during IBD flares in LMICs is currently not available [EL5]. Supporting text In a study from Iran, *C. difficile* and *Staphylococcus aureus* were found in the intestinal microbiota of 5.7% and 15.8% of patients with IBD flares, respectively.(217) Patients with UC who have intestinal colonization with extended-spectrum β -lactamase producers had more severe disease.(218)

Evidence: [EL5]

Statement 15

Access to modalities to monitor therapeutic response in IBD is suboptimal in LMIC settings [EL3]. Adoption of non-invasive modalities, such as routine tests, faecal biomarkers, and intestinal ultrasound are suggested [EL3]. Supporting Text

Evidence: [EL3], [EL3]

Statement 16

Complementary and alternative medicine [CAM] for IBD treatment is frequently used in LMIC settings [EL4]. Further evidence is required to characterize the efficacy and safety of CAM for IBD treatment [EL4]. Supporting Text Traditional Chinese medicine [TCM] is one of the most developed and diversified complementary and alternative medicine [CAM] approaches. Chinese herbal medicine [CHM] administered orally or by retention enema has shown promise in managing IBD through various compounds that exhibit antiinflammatory effects in preclinical and clinical studies. Mode of therapy, combining both systemic and

Evidence: [EL4], [EL4]

Statement 17a

Dietary therapies are a valid component of IBD management in LMIC settings [EL2] where expertise and therapies are available [EL4].

Evidence: [EL2], [EL4]

Statement 17b

Malnutrition and its impact should be addressed while instituting dietary therapies [EL3]. Supporting text Special diets, such as exclusive enteral nutrition [EEN], specific carbohydrate diet, anti-inflammatory diet for IBD, and fermentable oligosaccharides, disaccharides, monosaccharides, and polyols [FODMAP], have positive effects on IBD management.(3) Dairy restriction is a controversial approach for patients. In a study on 1888 participants [IBD: n = 872, non-IBD: n = 1016] on the effect of dairy in IBD, dairy intolerance and lactase deficiency were more common in participants with IBD [65.5% vs 46.1%; p = 0.0001]. The rates of gastrointestinal complaints among participants with IBD who had no family history of lactase deficiency, history of food sensitivity, or both were 59.91% [p = 0.68], 52.87% [p = 0.98], and 50.33% [p = 0.99] respectively, and were similar to participants without IBD.(246) EEN has been widely studied and is recommended as first-line therapy for inducing remission of luminal CD in paediatric population. EEN is a nutritionally complete liquid diet, either polymeric or oligomeric [semi-elemental or elemental], administered for 6–8 weeks through a nasogastric tube or orally.(247) The European Society for Pediatric Gastroenterology, Hepatology, and Nutrition recommends at least 6 weeks of EEN, even though the mucosa usually heals after 8 weeks. Studies from India have shown positive results for EEN for CD and acute severe UC.(248, 249) One-third of IBD patients experience irritable bowel syndrome-like symptoms. A recent meta-analysis confirmed that a low-FODMAP diet reduced functional gastrointestinal symptoms, such as bloating. In healthy volunteers, the mechanisms of action behind a low-FODMAP diet include decreased Bifidobacterium and reduced breath hydrogen.(250) Elimination diets are not recommended and may lead to several nutritional deficiencies. Considering malnutrition in IBD patients, nutrition education by

Evidence: [EL3]

Statement 18

Non-adherence to medication is an underestimated problem in LMIC settings and has important consequences [EL3]. Supporting text Poor adherence to medication is a major global issue. A World Health Organization report revealed that the average medication adherence rate among patients with chronic diseases worldwide is approximately 50%. The report emphasizes that as the number of individuals with chronic diseases increases, the consequences of poor medication adherence become more severe. Non-adherence is a primary contributor to preventable morbidity, mortality, and increased healthcare costs.(259)

Evidence: [EL3]

Statement 19

Access to advanced therapies is limited in LMIC settings and should be improved [EL3].

Evidence: [EL3]

Statement 20a

There are unique considerations, such as availability and cost, that favour wide use of thiopurines in LMIC settings compared with HICs [EL3].

Evidence: [EL3]

Statement 20b

Although frequently used in LMICs, there is no evidence to support use of 5-ASA for CD and should be avoided [EL3]. Supporting text The MIRACLE trial revealed that thiopurines along with 5-ASA are an effective maintenance therapy for clinical remission after infliximab induction in steroid-refractory UC.(270) Corticosteroid-free sustained clinical remission rates were at 68% at 1 year, 59% at 2 years, and 42% at 4 years. A study from India on 81 patients with fistulizing CD [most patients had complex fistulae] revealed a complete response rate of 25% [IQR: 24–66] at 45 months of follow up after thiopurine monotherapy in a setting with limited biological availability.(271) A study on 115 patients with stricturing CD revealed that thiopurine monotherapy has a good shortterm response with a modest long-term response.(272) The cumulative probability of maintaining response was 73% at 1 year, 40.7% at 2 years, and 18.5% at 5 years. A quarter of patients with stricturing CD developed leukopenia. Overall, thiopurines had a good sustained long-term response in a cohort of 988 patients with IBD.(273) Overall effectiveness of thiopurine at 5 and 10 years was 79% and 72% in UC, and 69% and 63% in CD, respectively. A study of 1093 patients with IBD with 7 [range 4–12]

Evidence: [EL3]

Statement 21

IBD patients from LMICs are underrepresented in clinical trials of advanced therapies [EL1].

Evidence: [EL1]

Statement 22

In LMICs, a significant proportion of abdominal operations for IBD are performed in emergency settings, which may indicate delays in referral or economic constraints and may lead to poor outcomes [EL4]. Supporting text A systematic review on the clinical characteristics of patients with UC in Asia revealed cumulative colectomy rates at 1 and 10 years of 2.4% and 7.6% in Hong Kong and 3.4% and 15.6% in Malaysia, respectively.(20) However, the study did not distinguish between colectomies performed as emergency or elective procedures. A retrospective study from a Brazilian centre reported that 68.9% of surgical procedures on patients with CD were performed in an emergency setting [both abdominal and perianal procedures], with the most common indications being ileal strictures and anorectal fistulae. Although this study did not present differences in short-term results based on setting, it did show that emergency surgery was a predictor for surgical recurrence [Odds Ratio: 2.1, 95% CI: 1.6–6.6].(279) A retrospective study from India assessed outcomes of 72 patients with UC who had emergency surgery and revealed an in-hospital postoperative mortality of 16.7%. The authors reported wound sepsis in 37.5% of patients, fever in 27.8%, adhesive intestinal obstruction in 8.3%, pelvic sepsis in 6.6%, ileostomy-related complications in 6.6%, blow out of Hartmann's pouch in 6.6%, and continued rectal bleeding in 3.3% of patients. The median duration of postoperative hospital stay was 10 days.(32) Another retrospective study from India presented data on 223 patients with CD of which 32.7% had at least one surgical procedure. Although the study did not distinguish between elective and emergency procedures, the analysis revealed male sex, small-bowel disease, perianal disease, and disease behaviour as independent predictors for surgery.(280)

Evidence: [EL4]

Statement 23

The length of resected and remaining bowel segments in CD surgery is rarely reported in LMICs [EL5]. Structured and standardized reporting of the length of resected and remaining bowel segments during CD surgery in LMICs is a simple yet very useful improvement [EL4]. Supporting text In LMICs, where healthcare resources are often limited, the precision of surgical interventions for IBD is paramount. Accurate measurement of resected and remaining bowel segments plays a crucial role in reducing the risk of disease recurrence, particularly in patients with CD,(283, 284) where inflammation can return at the site of anastomosis or other parts of the gastrointestinal tract.(285) Although the role of surgeon experience and training is critically important in LMICs, many surgeons may not have access to the same level of training and continuous professional development as their counterparts in HICs.(286) Resource limitations [e.g. lack of electronic documentation] may further negatively affect the surgeon's performance [e.g. the meticulous measurements required during IBD surgery], leading to suboptimal outcomes. A systematic review that included studies from HICs highlighted the critical nature of segmental resections in managing IBD-related colorectal cancer and dysplasia. The review found that inadequate resection margins, often due to the absence of advanced imaging and measurement tools, were associated with higher rates of metachronous cancer recurrence.(287) This underscores the importance of precise measurements during surgery to reduce risk of recurrence. A systematic review on surgery in CD concluded that an adequate report of resected small-bowel segments and length of remaining small bowel is of critical importance [e.g. for any planning of repetitive surgical interventions and subsequent management].(288) As there is no direct evidence from LMICs, recommendations must be carefully developed based on existing guidelines.

Evidence: [EL5], [EL4]

Statement 24

Surgical-site infections are the most frequent postoperative complications in LMICs [EL3]. The aetiology of these complications is multifactorial [e.g. pre-existing malnutrition, ignorance of disease behaviour, or delay in surgery] [EL3]. Supporting text Postoperative complications following IBD surgery remain a significant concern in LMICs.(8, 22, 289) A retrospective study from Iran al. used records from six tertiary care hospitals to explore in the outcomes of adult patients with UC undergoing colectomy; of the 2933 admitted patients with UC, 4% needed colectomy and as many as 40% experienced postoperative complications, with bowel obstruction and pouchitis being the most common late complications.(290) Another retrospective study from a tertiary centre in China analysed the outcomes of 35 adult patients with IBD [CD: n = 21, UC: n = 14] undergoing laparoscopic surgery. The study reported a 26.3% morbidity rate for CD and 28.6% for UC, with complications such as wound infections [CD: 15.8%, UC: 16.7%] and intestinal obstructions [CD: 5.2%, UC: 8.3%].(291) Another study from China examined the prevalence and risk factors [e.g. male gender and disease behaviour] in the population studied. The authors recommended key strategies in managing complications [e.g. improving preoperative nutritional status and conservative surgical techniques].(292) A study from Brazil on 37 patients with CD (44 surgical procedures) revealed that 40.9% [18/44] of surgeries resulted in complications, including fistulae in 34.1% [15/44], abscesses in 22.7% [10/44], and anastomotic dehiscence in 11.3% [5/44]. Significant risk factors for these complications included longer disease duration [p = 0.04], penetrating disease behaviour [p = 0.013], extended surgical duration, [mean 181.1 minutes, p = 0.016], and larger resected specimens [median 25 cm, p = 0.014].(293) A paediatric study on early and long-term surgical outcomes and complications in a referral children's hospital in Iran revealed that 21 paediatric IBD patients underwent a variety of surgical procedures, primarily due to anaemia, intestinal bleeding, or failure to thrive. A high rate of early postoperative complications was observed, including wound infections [23.8%] and pelvic abscesses [14.3%], with some cases requiring reoperation. The long-term complication rate was 28.6% during follow up. Predictors for long-term complications included lack of prior drug treatment and bleeding as surgical indications.(294)

Evidence: [EL3], [EL3]

Statement 25

Adoption of Enhanced Recovery After Surgery [ERAS] programs remain a challenge in LMICs [EL4]. Wider adoption of ERAS programs could significantly improve postoperative outcomes, especially in settings with limited resources and high postoperative morbidity rates [EL5]. Supporting text Enhanced Recovery After Surgery [ERAS] programs are multimodal perioperative care pathways designed to minimize physiological disturbance, thus facilitating early recovery following major surgery. A systematic review of ERAS programs for IBD surgery revealed a benefit from consistent adoption of this pathway.(300) While direct evidence of implementing such programs in LMICs is still lacking, insights can be drawn from specific studies. For example, a study in India on surgical management of CD in a tertiary care centre over an approximate 8-year period analysed the outcomes of 28 patients [male = 18, mean age 31.2 years] who underwent surgical resection for small-bowel involvement. Patients presented with complications, such as subacute intestinal obstruction, fistulae, and perforations, which required either elective or emergency surgeries. Notably, preoperative conditions such as anaemia, malabsorption, and growth retardation, along with use of steroids and immunosuppressants, were identified as significant risk factors that adversely affected postoperative outcomes, leading to complications such as anastomotic leak.(301) While this study did not specifically evaluate ERAS protocols, it underscores the challenges faced in LMIC settings, particularly the high rates of postoperative morbidity and the lack of structured recovery protocols. The findings highlight the critical need for improved postoperative care and patient education in such settings. Investment in education and training is needed in LMICs to implement ERAS programs. This will be economically beneficial in LMICs, as it will improve postoperative outcomes and reduce costs.(302)

Evidence: [EL4], [EL5]

Statement 26

In refractory moderate-to-severe UC in LMICs, transitioning from medical to surgical management is indicated by non-response to intensive medical therapies. Early surgical intervention can improve outcomes and reduce mortality, especially in the presence of life-threatening complications such as colonic perforation, toxic megacolon, or severe bleeding [EL3]. Close collaboration between gastroenterologists and surgeons is essential to improve outcomes.

Evidence: [EL3]

Statement 27

In LMICs, physical examination as a standalone diagnostic tool is commonly practiced to evaluate perianal CD. Endoanal ultrasound is suggested as a less expensive alternative to pelvic magnetic resonance imaging, provided there is sufficient availability and specific expertise [EL5]. Supporting text Of the few studies on available imaging modalities for assessing perianal CD in LMICs, a multicentre audit from Colombia on 65 patients with perianal CD revealed physical examination as the most common method for the initial diagnosis [61.5%]. Pelvic magnetic resonance imaging [MRI] was performed in 27.7%, endorectal ultrasound [US] in 6.2%, and clinical examination under anaesthesia in only 1 patient.(305) Two additional studies, one describing clinical and demographic characteristics of IBD patients in an underdeveloped region in Brazil (306), and one reporting methods to differentiate intestinal TB from CD in India,(199) did not discuss the available imaging methods to assess perianal disease. A study from north India reported MRI features of 27 patients with perianal CD, focusing on clinical and radiological characteristics and patient management; MRI was suggested as a mandatory imaging modality for all patients with perianal CD.(307) Although a study on perianal fistulizing CD in India notes that endoanal US is less expensive, this study did not provide any data on actual access to MRI or endoanal US in this setting.(308) Pelvic MRI was the imaging modality used in a recent cross-sectional study from Brazil that evaluated the effect of infliximab serum concentrations on disease activity in perianal fistulizing CD,(309) which was consistent with the Brazilian Consensus Statement as of 2022. Based on a Delphi consensus considering expert and stakeholder opinions, pelvic MRI was recommended to evaluate fistulizing perianal CD,

Evidence: [EL5]

Statement 28

In LMICs, sphincter-cutting techniques, such as fistulectomy, fistulotomy, and cutting seton are still widely used for surgical management of complex perianal CD. However, recent studies have shown good results from the adoption of sphincter-preserving techniques [EL4]. Supporting text Several studies from LMICs examined the use of cutting seton and fistula plugs in anal fistula surgery. One study from a tertiary referral centre in Egypt described 7 patients undergoing anal surgery for CD without providing further detailed information.(311) More detail was provided in a relatively recent study from India (271) on 65 patients with CD, of whom 58% received at least one surgical procedure, namely fistulectomy or fistulotomy, followed by seton placement. An alternative approach, videoassisted anal fistula treatment, was used for only 18% of these patients. A multicentre study from Colombia on 65 patients with perianal fistulizing CD revealed that 46% of these patients had a surgical procedure, namely mucosal flap surgery [18.5%], ileostomy [13.8%], ligation of the internal tract [6.2%], closure of the internal orifice [4.6%], temporary colostomy [4.6%], and definitive colostomy with proctectomy [4.6%].(305) A retrospective assessment from a tertiary referral centre in Brazil,(312) including 625 patients with IBD [416 with CD], also provided some information on patients with perianal CD who received relatively conventional procedures [e.g. fistulotomies and setons]. Recently, there have been a few reports on newer methods. An expert group from a tertiary referral centre in Malaysia used laser ablation of fistula tracts in 63 patients, of which 3 [4.3%] had IBD.(313) In a clinical interventional study at an IBD referral centre in Iran, 5 adult patients with refractory disease and draining perianal fistulae due to CD received mesenchymal stromal cells, injected locally and combined with fibrin glue, which was a potentially safe and effective therapeutic approach. (271) Lastly, a narrative overview on current IBD therapies made a specific point in identifying challenges for CD surgery in limited-resource settings in South Asia but did not address surgical management of perianal CD in detail.(308)

Evidence: [EL4]

Optimizing 5-Aminosalicylate for Moderate Ulcerative Colitis: Expert Recommendations from the Asia-Pacific, Middle East, and Africa IBD Coalition

File: 12. APMA_Moderate UC_Int research 2024.pdf

Akyuz F et al. *Intest Res.* 2024. doi:10.5217/ir.2024.00089

14 rules extracted

APMA Rec 1 (Moderate UC definition)

Moderate UC is defined as: 4–6 bowel movements/day, temperature $\leq 37.8^{\circ}\text{C}$, pulse ≤ 90 bpm, haemoglobin ≥ 10.5 g/dL, and CRP ≤ 30 mg/L (modified Truelove & Witts criteria). Clinical presentation, patient-reported symptoms, and objective markers (FC, endoscopy) should also be considered in the full assessment.

APMA Rec 2 (Optimal 5-ASA dose for moderate UC: 4 g/day)

Optimal 5-ASA dose for induction of remission in moderate UC: 4 g/day oral 5-ASA [81% APMA expert consensus]. Doses less than 4 g/day are unlikely to achieve adequate response in moderate disease.

APMA Rec 3 (Topical 5-ASA for distal moderate UC)

Topical 5-ASA at ≥ 1 g/day is recommended for inducing remission in patients with DISTAL moderate UC (proctosigmoiditis or proctitis). Combination of oral + topical 5-ASA is superior to either alone for distal disease.

APMA Rec 4 (5-ASA response: assess at 2–4 weeks)

Anticipated timeframe for initial clinical response to 5-ASA in moderate UC: 2–4 weeks. Formally assess response at this point — do not wait beyond 4 weeks to classify as non-responder before considering therapy adjustment.

APMA Rec 5 (Corticosteroids: not added based on risk factors alone)

Oral corticosteroids should NOT be added to 5-ASA therapy based solely on one or more accepted risk factors for aggressive disease in moderate UC. A full clinical risk assessment including disease burden, endoscopic severity, and biomarkers is required before adding corticosteroids.

APMA Rec 6 (5-ASA optimisation steps before escalation)

Before escalating beyond 5-ASA in moderate UC, optimise therapy in a stepwise manner: (1) Maximise oral 5-ASA dose to ≥ 4 g/day; (2) Add topical 5-ASA ≥ 1 g/day if distal disease; (3) Optimise patient adherence and extend treatment duration to minimum 6 weeks; (4) Switch 5-ASA formulation (e.g., pH-release vs. time-release) if tolerability issue.

APMA Rec 7 (Bridging if 5-ASA fails: budesonide MMX or corticosteroid)

If no response after optimised 5-ASA therapy in moderate UC: add-on oral budesonide MMX (9 mg/day) OR systemic oral corticosteroid (≥ 40 mg/day prednisolone) as bridging therapy while planning further management. [APMA Rec 13]

APMA Rec 8 (Predictors of aggressive disease: early immunomodulator or biologic)

Patients with predictors of aggressive disease in moderate UC should be considered for early immunomodulator or biologic/advanced therapy, rather than extended corticosteroid use. Predictors of aggressive disease: early age at diagnosis, extensive colitis, frequent flares requiring corticosteroids or hospitalisation, male gender, smoking, concurrent PSC or other EIM.

APMA Rec 9 (Moderate UC relapse definition)

Relapse definition for moderate UC: recurrence of disease symptoms after achieving remission, including failure to maintain rectal bleeding score = 0, stool frequency score = 0, or MES < 3 (Mayo Endoscopic Score).

Statement 1

Assessing Moderate UC The APMA IBD Coalition members reached a unanimous consensus (100%) that moderate UC is a unique phenotype of UC and warrants clinical attention. A proportion of the coalition further indicated that the lack of timely and accurate risk stratification for patients may lead to consequences such as (1) suboptimal treatment (95%) or a delay in treatment initiation or treatment escalation (89%) if the disease activity is un-

Statement 2

■Treatment of Moderate UC: Challenges and Unmet Needs The APMA IBD Coalition outlined key challenges to the appropriate treatment of moderate UC, including a lack of clear treatment recommendations leading to delayed intervention, particularly among gastroenterologists with limited exposure to UC in practice. In the APMA region, variations in treatment accessibility, affordability, reimbursement policies, and eligibility criteria hinder a universal approach to moderate UC.^{21,28} The panel indicated that ambiguous treatment pathways for moderate UC may lead to (1) early and long-term exposure to adverse effects of advanced therapies (83%)^{20,21,26,47}; (2) early and long-term costs of biologics (72%)^{21,28}; (3) treatment fatigue for patients (44%)⁴⁸; and (4) early exhaustion of biologics options due to loss of response (33%), as described earlier.^{32,33} The experts emphasized the need to establish a treatment pathway for moderate UC that provides clear guidance on (1) the selection of induction treatment, including dosage and duration, and administration route based on the location and extent of disease; (2) risk stratifying strategies to identify the need to escalate therapy; and (3) de-escalation strategies upon achievement of remission. However, such treatment guidelines must also account for patient heterogeneity in terms of treatment history, clinical presentation, disease progression and treatment response. 1) Treatment Goals in Moderate UC

Statement 3

■Treatment of Moderate UC: Optimization of 5-ASA for Patients with Moderate UC Current guidelines recommend conventional 5-ASAs as induction therapy for remission in patients with mild-to-moderate UC, followed by advanced therapies if conventional therapies are no longer effective.^{1,2,59,60} Before a treatment is considered “ineffective,” the physician should first ensure that the treatment has been correctly and fully utilized at the maximum dose and optimum duration.¹⁷ Ultimately, these measures to improve the utilization of conventional medicines for moderate UC may effectively conserve the use of advanced therapies for patients experiencing severe UC.^{17,45} The optimization of 5-ASA therapy can be achieved by implementing measures suggested by the APMA IBD Coalition, which have been modified based on strategies proposed in previous literature^{17,45}: maximizing 5-ASA dosage, incorporating topical therapy, enhancing patient adherence through education and by streamlining the dosing strategy and delivery, and prolonging treatment duration. 1) Maximizing the 5-ASA Dose

Statement 4

■Treatment Escalation Beyond 5-ASA for Patients with Moderate UC When all available options for optimizing 5-ASA treatment have been exhausted, escalating to immunomodulatory agents and/or advanced therapies may be considered,^{1,47} in alignment with guidelines that recommend escalation for patients with moderate-to-severe UC with inadequate response or intolerance to conventional therapy.² As per the STRIDE-II consensus recommendations, 94% of the APMA IBD Coalition would define “inadequate response” to 5-ASAs for moderate UC as a failure to attain a reduction $\geq 50\%$ in rectal bleeding and stool frequency.⁴⁹ Risk factors that were identified to contribute to “inadequate response” to 5-ASA include (1) frequent symptom flares requiring corticosteroids or hospitalization (89%); (2) extensive colitis (67%); (3) moderate endoscopic activity (39%); (4) young age at diagnosis (33%); (5) concurrent PSC or other EIM (17%); (6) smoking (11%); and (7) male gender (6%).

Statement 5

A proposed treatment algorithm for moderate UC is presented in Fig. 1, which takes into account current practices and recommendations of the APMA IBD Coalition.

Colitis and Crohn's Foundation (India) Consensus Statements on Use of 5-Aminosalicylic Acid in Inflammatory Bowel Disease

File: 13. Indian Consensus statement on 5ASA.pdf

Sood A et al. *Intest Res.* 2020;18(4):355-378. doi:10.5217/ir.2019.09176

58 rules extracted

Statement 5 (Assess response at Day 14)

Assess response to 5-ASA therapy at DAY 14. If no improvement in rectal bleeding and stool frequency by Day 14, the patient should be classified as a non-responder and therapy should be escalated. Do not wait beyond 2 weeks to classify as non-responder. [Grade B | EL I]

Statement 6 (Dose escalation for non-response)

Dose escalation to maximum 5-ASA dose of 4–4.8 g/day may be performed for patients not responding to lower initiating doses before classifying as 5-ASA non-responders. [Grade B | EL I]

Statement 7 (Sulfasalazine preferred with peripheral arthritis)

Sulfasalazine (2–3 g/day) may be PREFERRED in IBD patients with concomitant PERIPHERAL ARTHRITIS — sulfasalazine has DMARD (disease-modifying antirheumatic drug) activity. Avoid sulfasalazine for axial arthritis (ankylosing spondylitis) — it is ineffective for this indication. [Grade C | EL III]

Statement 8 (5-ASA withheld in acute severe colitis)

In ACUTE SEVERE COLITIS: 5-ASA has NO proven role as adjunct therapy. 5-ASA should be WITHHELD during IV corticosteroid treatment — approximately 3% of patients develop paradoxical worsening of diarrhoea on 5-ASA. Reintroduce after clinical improvement. [Grade C | EL III]

Statement 10 (Induction dose for initial maintenance)

In mild-to-moderate UC, continue the induction dose of 5-ASA for INITIAL MAINTENANCE of remission. Do not reduce dose prematurely before confirmed mucosal healing. [Grade B | EL I]

Statement 11 (Maintenance dose reduction in complete mucosal healing)

5-ASA maintenance dose REDUCTION can be considered in patients with a mild clinical course AND complete mucosal healing confirmed by endoscopy (MES 0). Do not reduce dose in patients with ongoing endoscopic activity. [Grade B | EL II-1]

Statement 14 (Proctitis: suppositories preferred; 1 g once daily)

For ulcerative proctitis: suppositories are preferred over enemas or foams for targeted drug delivery. Recommended dose: 5-ASA 1 g once daily as a suppository.

Statement 15 (Proctitis maintenance: 3 g/week topical or low-dose oral)

Maintenance therapy for proctitis: 3 g weekly divided dose of topical 5-ASA OR low-dose oral 5-ASA (2–2.4 g/day). Weekly topical dosing is as effective as daily dosing for maintenance.

Statement 16 (Rectal 5-ASA: first-line maintenance in proctitis)

Rectal 5-ASA is FIRST-LINE maintenance therapy in ulcerative proctitis. Oral 5-ASA alone is less effective than rectal 5-ASA for proctitis.

Statement 17 (Proctitis not responding: add topical corticosteroids)

Proctitis not responding to combined oral + rectal 5-ASA: add topical corticosteroids. Combination of beclomethasone dipropionate (BDP) + 5-ASA enema is superior to either agent alone.

Statement 18 (5-ASA withdrawal: significant relapse risk)

Withdrawal of 5-ASA is associated with significant relapse risk. Complete withdrawal of 5-ASA is NOT recommended unless the patient is intolerant. Some evidence suggests 5-ASA reduces colorectal cancer (CRC) risk in extensive colitis. [Grade B | EL I]

Statement 19 (Proctitis with mucosal healing: oral can be stopped)

In ulcerative proctitis patients who have achieved complete mucosal healing: oral 5-ASA may be discontinued with continuation of rectal suppositories alone. [Grade C | EL III]

Statement 20 (Sulfasalazine in mild-moderate COLONIC CD)

Sulfasalazine may be considered for induction of remission in mild-to-moderate COLONIC Crohn's disease [Grade B | EL I]. Sulfasalazine is NOT effective for ileal or ileocolonic CD and should not be used for these disease locations.

Statement 21 (5-ASA NOT indicated for ileal/proximal CD)

5-ASA formulations are NOT indicated for induction of remission in ileal, ileocolonic, or proximal small intestinal Crohn's disease. [Grade A | EL I]

Statement 22 (No role for 5-ASA maintenance in most CD)

There is NO role for 5-ASA or sulfasalazine in MAINTENANCE of remission in the MAJORITY of Crohn's disease patients. [Grade A | EL I]

Statement 23 (5-ASA postoperative prophylaxis in CD after ileal resection)

5-ASA has a role in POSTOPERATIVE PROPHYLAXIS after ileal resection in Crohn's disease — it reduces clinical and endoscopic recurrence rates compared with placebo. [Grade A | EL I]

Statement 24 (Paediatric: 5-ASA first-line in UC; indefinite use)

5-ASA is FIRST-LINE therapy for induction and maintenance of remission in children with UC [Grade A | EL I]. Continue indefinitely unless intolerant. Avoid withdrawing 5-ASA in paediatric UC without clear indication.

Statement 25 (Paediatric dosing)

Paediatric 5-ASA dosing: oral 5-ASA 60–80 mg/kg/day (maximum 4.8 g/day); rectal 5-ASA 25 mg/kg/day (maximum 1 g/day); sulfasalazine 40–70 mg/kg/day (maximum 4 g/day). [Grade B | EL I]

Statement 29 (Paradoxical worsening: 5-ASA hypersensitivity)

PARADOXICAL WORSENING (~3% of patients): some patients develop paradoxical worsening of colitis on 5-ASA (increased abdominal pain, diarrhoea, rectal bleeding) — this is a drug hypersensitivity reaction, NOT disease progression. Requires IMMEDIATE discontinuation of 5-ASA. Patient is allergic to 5-ASA — do NOT rechallenge. [Grade A | EL I]

Statement 30 (5-ASA renal toxicity monitoring)

5-ASA may cause RENAL TOXICITY (interstitial nephritis). Monitor renal function (serum creatinine, eGFR) AT LEAST ONCE A YEAR in all patients on long-term 5-ASA. Incidence: approximately 1/4,000 patients per year. Can occur as early as 29 days or as late as 5 years after initiation. [Grade A | EL I]

Statement 31 (5-ASA in chronic renal failure)

5-ASA dose modification is NOT routinely required in chronic renal failure, but renal function must be monitored MORE CLOSELY (every 3–6 months rather than annually). Weigh the benefit of 5-ASA against the risk of further renal deterioration. [Grade C | EL III]

Statement 33 (Folate supplementation required with sulfasalazine)

Folate supplementation is REQUIRED with SULFASALAZINE — sulfasalazine inhibits dihydrofolate reductase, causing folate deficiency. Standard: folic acid 5 mg/day. Pregnant women on sulfasalazine: folic acid 2 mg/day minimum. Other 5-ASA formulations do NOT require routine folate supplementation. [Grade B | EL II-3]

Statement 34 (Long-term 5-ASA may reduce CRC risk in UC)

Long-term 5-ASA therapy in extensive/distal colitis is POSSIBLY associated with a reduced risk of colorectal cancer (CRC) in UC — approximately 37–54% relative risk reduction [Grade B | EL II-1]. This is a further reason to maintain long-term 5-ASA and avoid complete withdrawal.

Statement 35 (Adherence critical: non-adherence 3x relapse risk)

Patient ADHERENCE to 5-ASA therapy is critically important. Non-adherent patients are approximately 3 times MORE LIKELY to relapse than adherent patients [Grade B | EL II-2]. Patient education about the importance of continuous therapy is MANDATORY at every clinical encounter.

Statement 1

5-ASA containing formulations are effective for induction of remission in patients with mild to moderate UC of any extent and are the first line of treatment for this indication. • ■Grade of recommendation: A, Level of evidence: I, Voting: 87.5% strongly agree, 12.5% agree Two RCTs conducted in 1960s have shown that sulfasalazine (2–4 g/day) was more effective than placebo in induction of remission. However, it was not well tolerated due to adverse events and high rates of discontinuation were observed.^{22,23}

Statement 2

In mild to moderate UC, 5-ASA (2–4.8 g/day) or sulfasalazine (3 g/day) should be given for induction of remission. • ■Grade of recommendation: A, Level of evidence: I, Voting: 78.3% strongly agree, 21.7% agree The Assessing the Safety and Clinical Efficacy of a New Dose of 5-ASA (ASCEND) trials were a series of dose-finding trials for 5-ASA in UC. A pooled analysis of 3 ASCEND trials showed no significant difference between 4.8 g/day and 2.4 g/day in

Statement 3

In distal or extensive mild to moderate UC, combination of topical and oral 5-ASA formulations is preferred over oral 5-ASA alone. • ■Grade of recommendation: A, Level of evidence: I, Voting: 91.7% strongly agree, 4.2% agree, 4.2% disagree A systematic review and meta-analysis of 4 RCTs found combined topical and oral 5-ASAs to be superior to oral 5-ASA alone in inducing remission in mild to moderate active leftsided UC, proctosigmoiditis, and proctitis (RR of no remission, 0.65; 95% CI, 0.47–0.91).³¹ Another systematic review and network meta-analysis (n = 8,020 patients with extensive or leftsided mild to moderate UC) reported that combined oral and rectal 5-ASA agents were superior to standard dose oral 5-ASA (2–3 g/day) for induction of remission (failure to induce remission with combined oral and rectal 5-ASAs; odds ratio [OR], 0.41; 95% CI, 0.22–0.77).²⁴ Active comparisons of efficacy and tolerability showed that combined oral and rectal 5-ASAs were significantly superior to all other interventions for inducing remission in mild to moderate UC, except budesonide multi-matrix.²⁴ In the surface under the cumulative ranking curve analysis (SUCRA), combined oral and rectal 5-ASAs ranked highest in inducing remission (SUCRA, 0.99) followed by high-dose 5-ASA (SUCRA, 0.82).²⁴ Considering 13% median remission rate for placebo across trials, approximately 63%, 33%, and 26% of patients treated with combined oral and rectal 5-ASA, high-dose and standard-dose 5-ASA, respectively, are expected to achieve clinical remission.²⁴

Statement 4

Once daily dose of oral sustained release preparation should be considered as it is equally effective as divided doses. • ■Grade of recommendation: A, Level of evidence: I, Voting: 72.0% strongly agree, 28.0% agree New, once daily (OD) formulations of 5-ASA use multi-matrix technology which involves incorporating 5-ASA into a lipophilic matrix, which itself is dispersed within a hydrophilic matrix, to delay and prolong the dissolution.³² Two Cochrane systematic reviews have demonstrated no significant difference between OD versus conventional dosing for induction of

Statement 5

A period of 2 weeks of 5-ASA therapy should be given before considering a patient as a nonresponder. • ■Grade of recommendation: B, Level of evidence: I, Voting: 80.0% strongly agree, 20.0% agree While selecting 5-ASA for induction of remission in patients with UC, the timing at which clinical response is evident also needs to be considered. Nonresponders to the recommended 5-ASA induction doses should be identified early, and therapy escalation should be considered in such cases. Based on ASCEND (4.8 g/day, 800 mg tablet) I and II trials, symptoms of rectal bleeding and stool frequency either improved or resolved by day 14 in the majority of patients on both 4.8 g/day and 2.4 g/day of 5-ASA (73% vs. 61%, respectively). By day 14, symptom resolution was seen in 43% versus 30% of patients on 4.8 g/day versus 2.4 g/day dose ($P = 0.035$). Moreover, relief in symptoms after 2 weeks correlated with high rate of relief in symptoms after 6 weeks.³⁶ Hence, patients should be assessed at day 14 for therapeutic decision making.

Statement 6

Dose escalation to a maximum dose of 4–4.8 g may be done for patients who do not respond to lower initiating doses of 5-ASA compounds. • ■Grade of recommendation: B, Level of evidence: I, Voting: 76.0% strongly agree, 24.0% agree In the ASCEND I trial, subgroup analysis of patients with moderately active disease revealed that higher dose of 5-ASA was better by an absolute difference of 15% in achieving therapeutic success (72% vs. 57%, $P = 0.038$).³⁷ ASCEND II confirmed a statistically superior effect of the 4.8 g/day over the 2.4 g/day dose in moderately active disease (72% overall response vs. 59%, respectively)³⁸ and ASCEND III showed a significant benefit of 4.8 g/day versus 2.4 g/day in patients with difficult to treat UC, specifically those who required cortico-

Statement 7

Sulfasalazine may be preferred in patients with concomitant arthralgia/arthritis. • ■Grade of recommendation: C, Level of evidence: III, Voting: 100.0% strongly agree Articular involvement is the most common extraintestinal manifestation in IBD, with a prevalence ranging between 17% and 39%.⁴⁰ It frequently involves the axial joints but can also be associated with peripheral arthritis. Peripheral arthritis has 3 subtypes: type 1, pauciarticular, joint activity parallels intestinal disease activity, type 2, polyarticular form, affecting small joints with symptoms running independent from IBD course, and type 3, which includes both axial and peripheral forms.^{41–43} The approach to management of arthritis in patients with IBD is similar to that in treatment of spondyloarthritis. While the treatment of axial arthritis focuses on combination of exercise and nonsteroidal anti-inflammatory drugs (NSAIDs), selective conventional non-biologic disease-modifying antirheumatic drugs (DMARDs) are used for peripheral arthritis resistant to initial NSAID therapy, if biologics are not already required for axial or gastrointestinal disease manifestations.⁴⁴ Sulfasalazine is one such DMARD, which is an azo-bonded combination of 5-ASA and sulfapyridine. It is often poorly tolerated due to adverse effects (due to the sulfapyridine moiety) such as headache, nausea, vomiting, drug rash, hepatitis, and hematologic toxicity. This limits its use in IBD, for which safer 5-ASA compounds have been developed. However, an Italian Expert panel has recommended 2–3 g/day of sulfasalazine in mild to moderate IBD with peripheral arthritis and early disease that failed on intra-articular corticosteroids.⁴⁵ On the other hand, the Assessment in SpondyloArthritis international Society and European League Against Rheumatism do not

Statement 8

In acute severe colitis, 5-ASA has no proven role as an adjunct therapy. 5-ASA should be introduced after clinical improvement. • ■Grade of recommendation: C, Level of evidence: III, Voting: 59.3% strongly agree, 33.3% agree, 7.4% disagree Acute severe UC is a medical emergency with high mortality. Intravenous corticosteroids remain the cornerstone of treatment, as they act rapidly and are effective in up to 70% of the patients.^{47,48} 5-ASAs can be temporarily withdrawn while the patients are being treated with intravenous corticosteroids, as there is no evidence supporting benefit of these drugs in acute severe UC. Conversely, a paradoxical increase in diarrhea has been reported in 3% patients on 5-ASAs.^{48,49} However, topical 5-ASA or corticosteroids may be added in patients who have high stool frequency.²⁰ After an initial response to the intravenous corticosteroids, patients can be switched to oral corticosteroids, and 5-ASA, azathioprine, or 6-mercaptopurine can be added.⁵⁰

2. Role of 5-ASA in Maintenance of Remission in UC

Statement 9

Long-term 5-ASA is indicated for maintenance of remission in patients with UC. • ■Grade of recommendation: A, Level of evidence: I, Voting: 100.0% strongly agree

Statement 10

In patients with mild to moderate UC, the dose that was used for induction of remission should be continued for initial maintenance of remission. • ■Grade of recommendation: B, Level of evidence: I, Voting: 53.8% strongly agree, 23.1% agree, 23.1% disagree The safety and efficacy of long-term use of 5-ASA as a maintenance therapy has been studied in several studies and its continued use is known to prevent disease relapse in patients with UC.^{9,10} A Cochrane review of 38 studies (n = 8,127) assessed efficacy, dose-responsiveness, and safety of oral 5-ASA for maintenance of clinical or endoscopic remission in UC. Forty-one

Statement 11

5-ASA maintenance dose reduction can be considered in patients with UC having mild clinical course with complete mucosal healing. • ■Grade of recommendation: B, Level of evidence: II-1, Voting: 73.1% strongly agree, 19.2% agree, 7.7% disagree Mucosal healing in UC is associated with improved clinical outcomes in terms of lower relapse rates,⁵³⁻⁵⁵ lesser requirement for immunosuppressive agents,⁵⁶ fewer hospitalizations and colectomies,^{56,57} lower risk of colorectal cancer⁵⁸⁻⁶⁰ and improved quality of life.^{61,62} Achieving complete mucosal healing should, therefore, be the primary endpoint of treatment. Besides mucosal healing, various factors including adherence to treatment, duration of disease in remission, degree of mucosal healing (complete/partial), and clinical course should be considered before reducing the maintenance dose of 5-ASA. No difference in the long-term risk of flare was noted between low (2.4–2.8 g/day) versus high (4.4–4.8 g/day) dose 5-ASA in patients who adhere to treatment.⁶³ Data from an IBD registry showed that a longer duration (> 2 years) of disease remission correlated with a lower risk of relapse (P < 0.001).⁶⁴

Statement 12

For patients with moderate to severe UC who required corticosteroids for induction of remission, 5-ASAs can be used for maintenance. • ■Grade of recommendation: B, Level of evidence: II-2, Voting: 96.4% strongly agree, 3.6% agree Use of corticosteroids for inducing remission is associated with an increased risk of subsequent clinical relapses.⁶⁹ Studies evaluating outcomes after systemic corticosteroid use have shown that up to 30%–40% patients subsequently require medical rescue therapies or colectomy.⁷⁰⁻⁷² In a cohort of UC patients treated with 5-ASA after a course of oral systemic corticosteroids, 75% patients relapsed (median, 29 months; range, 1–156 months), and 14% required colectomy (median, 11 months; range, 1–24 months). Kaplan Meier curve showed relapse and colectomy rates over 1 year

Statement 15

Patients with proctitis can be maintained with 3 g weekly divided dose of topical 5-ASA or low dose of oral 5-ASA (2–2.4 g daily). • ■Grade of recommendation: A, Level of evidence: I, Voting: 55.6% strongly agree, 40.7% agree, 3.0% strongly disagree

Statement 14

In patients with proctitis, suppositories should be preferred over enemas or foams. • ■Grade of recommendation: A, Level of evidence: I, Voting: 88.9% strongly agree, 11.1% agree Recommended dosage of topical therapy with 5-ASA suppository for induction of remission is 1 g OD. • ■Grade of recommendation: A, Level of evidence: I, Voting: 92.3% strongly agree, 7.7% agree

Statement 16

Rectal 5-ASA is first-line maintenance therapy in ulcerative proctitis. An RCT with 95 ulcerative proctitis patients found 5-ASA suppositories at a dose of 1 g thrice a week to be more effective than placebo in maintaining remission.⁷⁸ Another RCT reported a significantly higher relapse rate at 12 and 24 months with placebo as compared to OD 500-mg 5-ASA suppository ($P < 0.001$).⁷⁹ A systematic review and meta-analysis evaluating topical 5-ASA therapies for maintenance of remission in ulcerative proctitis also reported the benefit of rectal 5-ASA (pooled RR, 2.80; 95% CI, 1.21–6.45; degree of heterogeneity [I²], 66%; $P = 0.02$).⁸⁰ Another recent meta-analysis including 7 RCTs ($n = 555$), showed that topical 5-ASA was effective in preventing relapse of distal UC.⁸¹ The superiority of rectal 5-ASA over placebo for maintenance of remission at 1 year was also demonstrated in a recent Cochrane review (RR, 2.22; 95% CI, 1.26–3.90).⁸² Two RCTs have compared combination treatment with oral 5-ASA plus intermittent 5-ASA enema and oral 5-ASA alone, for maintenance of remission. One compared patients who received 1 g 5-ASA enemas twice a week plus oral 5-ASA 3 g/ day for 7 days with daily oral 5-ASA only and the other compared combined therapy with 5-ASA tablets 1.6 g/day and

Statement 17

In patients with proctitis not responding to oral and rectal 5-ASA, topical corticosteroids can be added for inducing remission. • ■ Grade of recommendation: A, Level of evidence: I, Voting: 76.0% strongly agree, 24.0% agree Two meta-analyses have shown that topical 5-ASA is more effective than topical corticosteroids, in inducing symptomatic, endoscopic, or histological remission.^{97,102} However, in patients not responding/intolerant to 5-ASA (topical, oral or a combination of both) within 2–3 weeks of initiating therapy, combination therapy of topical 5-ASA and topical corticosteroids can be considered.⁹⁴ A randomized, double-blind trial demonstrated higher efficacy of beclomethasone dipropionate (BDP) and 5-ASA enema combination therapy (BDP/5-ASA) versus BDP or 5-ASA alone. The combination was significantly superior to single-agent therapy in terms of improved endoscopic and histological scores.¹⁰³

Statement 18

Withdrawal of 5-ASAs is associated with a significant risk of relapse. Decision to withdraw should only be made after several factors are taken into consideration such as duration of remission and mucosal healing. • ■ Grade of recommendation: B, Level of evidence: I, Voting: 42.9% strongly agree, 46.4% agree, 10.7% disagree Multiple factors determine the duration of treatment with 5-ASA. A recent Cochrane review suggested that 5-ASA was superior to placebo as a maintenance therapy in UC.⁸² These findings endorse a lifelong approach to therapy with 5-ASA in UC. However, dose reduction after induction of remission has been evaluated in various studies. In a real-world OPTIMUM study, no difference was found between reduced dose and maintained dose for maintenance of remission after induction. Also, maintenance of remission was noted to be significantly better than placebo in patients in whom duration of remission was > 12 months (hazard ratio [HR] for relapse, 0.600; 95% CI, 0.486–0.740; $P < 0.0001$ for > 12 months and HR, 0.352; 95% CI, 0.289–0.431; $P < 0.0001$ for > 24 months). 5-ASA therapy was not withdrawn in any of the patient.¹⁰⁴ In a retrospective Korean study, left-sided or extensive colitis at time of diagnosis (HR, 1.46; 95% CI, 1.01–2.10; $P = 0.04$) was observed to be independent predictor for clinical relapse on 5-ASA therapy.¹⁰⁵ Although 5-ASA preparations are generally well tolerated, adverse reactions including worsening colitis; nephrotoxicity, interstitial lung disease, bronchiolitis obliterans, pancreatitis, hair loss, and Stevens-Johnson syndrome have been described.¹⁰⁶ These can also be major determinants in deciding continuation or withdrawal of the therapy. Mucosal healing (endoscopic/histological) also predicts the subsequent clinical course. Narang et al.¹⁰⁷ showed that histological remission predicts a sustained clinical remission (87.1% patients in histological remission remained asymptomatic at 1 year). Another RCT showed that 5-ASA dose reduction is more successful in sustaining remission if a Mayo endoscopic sub-score of 0 is achieved.⁶⁶ Attainment of mucosal healing might help decide upon the withdrawal or discontinuation of the drug. Complete withdrawal of 5-ASA agents is currently not recommended unless the patient is intolerant as there is some evidence that regular 5-ASA reduces the risk of colorectal cancer.⁶⁸

Statement 19

In patients with proctitis with mucosal healing, oral 5-ASA may be discontinued with continuation of rectal suppositories. • ■Grade of recommendation: C, Level of evidence: III, Voting: 92.6% strongly agree, 7.4% agree A meta-analysis involving 3,977 patients on oral 5-ASA (granules and tablets) and 2,513 patients on rectal 5-ASA (suppositories, enema, and foam) demonstrated that mucosal healing rates are superior with rectal formulations (36.9% and 50.3% for patients treated with oral and topical mesalamine, respectively).⁹¹ Another systematic review by Ford et al.⁸¹ that studied the relative efficacies of different routes of administration of 5-ASA agents also concluded that intermittent topical 5-ASAs are superior to oral 5-ASA in maintaining remission in quiescent UC. Though there is no direct evidence showing the benefit of discontinuing oral 5-ASA and continuing suppositories in patients with proctitis who have achieved mucosal healing, the expert panel recommends that oral 5-ASA may be discontinued with continuation of rectal suppositories in such patients. 5. Role of 5-ASA in CD

Statement 20

Sulfasalazine may be considered for induction of remission in mild to moderate colonic CD. • ■Grade of recommendation: B, Level of evidence: I, Voting: 37.5% strongly agree, 54.2% agree, 8.3% disagree A Cochrane meta-analysis has evaluated the role of aminosalicylates in induction of remission or response in CD. A pooled analysis of 3 studies¹⁰⁸⁻¹¹⁰ showed that sulfasalazine was not superior to placebo for induction of remission or response at 17 to 26 weeks of follow-up (RR, 1.52; 95% CI, 0.95–2.43; P = 0.08); however, because of sparse data and heterogeneity the results had low quality of GRADE evidence.¹⁰⁸ A subsequent comparison of 2 RCTs.^{109,110} after removing the source of heterogeneity revealed a favorable response with sulfasalazine over placebo after 17–18 weeks of follow-up (RR, 1.38; 95% CI, 1.00–1.89; P = 0.05; zero heterogeneity; and moderate quality of evidence). There was no difference in adverse events between sulfasalazine and placebo arms.¹⁰⁸ The benefit however was limited to patients with Crohn's colitis. Those patients with small bowel disease or those who continued to have active disease even after previous corticosteroid and sulfasala-

Statement 21

5-ASA formulations are not indicated for induction of remission in ileal/ileocolonic/proximal small intestinal CD. • ■Grade of recommendation: A, Level of evidence: I, Voting: 63.0% strongly agree, 22.2% agree, 11.1% disagree, 3.7% strongly disagree Comparison between sulfasalazine and corticosteroids showed that sulfasalazine monotherapy was inferior to corticosteroids at 17–18 weeks of follow-up (RR, 0.68; 95% CI, 0.51–0.91; P = 0.009; moderate quality of evidence due to sparse data).¹⁰⁸ Another pooled analysis showed that controlled release 5-ASA (1–2 g/day) was not superior to placebo at 16 weeks of followup (RR, 1.46; 95% CI, 0.89–2.40; P = 0.14). There was no difference in proportion of patients having adverse effects.¹⁰⁸ A pooled analysis of 3 studies comparing 5-ASA (4 g/day) with placebo showed a nonsignificant mean difference in Crohn's Disease Activity Index (CDAI) reduction (–19.8 points; 95% CI, –46.2 to 6.7; P = 0.14); another comparison of delayed release 5-ASA (3–4.5 g/day) with tapering dose conventional corticosteroids showed no significant difference in efficacy after 8–12 weeks of follow-up (RR, 1.04; 95% CI, 0.79–1.36).¹⁰⁸ Comparison of 5-ASA 4 g/day with budesonide 9 mg/day showed that 5-ASA was significantly less effective than budesonide at 16 weeks (RR, 0.56; 95% CI, 0.40–0.78; P < 0.001).¹⁰⁸ There was significant difference in proportion of patients experiencing adverse effects, however, more patients withdrew from 5-ASA arm due to adverse effects. Another trial corroborated these findings when olsalazine 2 g/day led to a higher withdrawal rate, mainly due to diarrhea, as compared with placebo (P = 0.015).¹⁰⁸ Although the Cochrane analysis has limitations of sparse data, trial heterogeneity and high risk of bias, yet, it has been sufficiently demonstrated that sulfasalazine was modestly effective, with benefit confined to Crohn's colitis; and there is no proven role of 5-ASA in induction of remission in ileal/ileocolonic/proximal small intestinal CD.

Statement 22

There is no role of 5-ASA or sulfasalazine in maintenance of remission in a majority of patients with CD. • ■Grade of recommendation: A, Level of evidence: I, Voting: 50.0% strongly agree, 37.5% agree, 12.5% disagree The efficacy of sulfasalazine or 5-ASA in maintaining remis-

Statement 23

5-ASA has a role in postoperative prophylaxis following ileal resection in CD. • ■ Grade of recommendation: A, Level of evidence: I, Voting: 56.5% strongly agree, 39.1% agree, 4.3% disagree A Cochrane systematic review of 23 RCTs that investigated medical therapies for postoperative CD recurrence reported that clinical recurrence and severe endoscopic recurrence reduced significantly with 5-ASA as compared with placebo (RR for clinical recurrence, 0.76; 95% CI, 0.62–0.94; number needed to treat [NNT] = 12; RR for severe endoscopic recurrence, 0.50; 95% CI, 0.29–0.84; NNT = 8). Risk of serious adverse events was lower with 5-ASA versus azathioprine/6-mercaptopurine (RR, 0.51; 95% CI, 0.30–0.89) but risk of any endoscopic recurrence was higher (RR, 1.45; 95% CI, 1.03–2.06). 5-ASA therapy did not differ significantly from azathioprine/6-mercaptopurine for any other outcome.⁴⁹ Another Cochrane systematic review of 9 RCTs found that 5-ASA was significantly more effective than placebo in postoperative prophylaxis of CD (OR, 0.68; 95% CI, 0.52–0.90), but there was no difference in efficacy between 5-ASA and thiopurines (OR, 1.08; 95% CI, 0.63–1.85).¹¹⁴

Statement 24

5-ASA agents are the first line therapy for induction and maintenance of remission in children with UC. • ■ Grade of recommendation: A, Level of evidence: I, Voting: 70.8% strongly agree, 29.2% agree, 4.2% disagree 5-ASAs are recommended as first-line therapy for induction and maintenance of remission of mild to moderate UC in children.^{115,116} 5-ASA therapy should be used for maintenance indefinitely unless intolerant, as it is highly effective and has a good safety profile. A prospective cohort study successfully reported clinical remission at week 3 in 42% pediatric patients treated with the combination of 5-ASA oral and enema (1 g/day).¹¹⁷ In a randomized, double-blind, active control study, pediatric UC patients treated with high and low-dose 5-ASA for a period of 6 weeks demonstrated similar efficacy in improving Pediatric UC Activity Index (55% vs. 56% of the patients in high- and low-dose of 5-ASA, respectively) ($P = 0.924$) as well as Truncated Mayo score (70% vs. 73%).¹¹⁸ An RCT comparing efficacy of once and twice daily dosing of 5-ASA in inducing remission in children with UC did not find any significant difference in the rates of induction of remission (60% vs. 63%, $P = 0.78$).¹¹⁹ A meta-analysis including 37 RCTs revealed that 5-ASA is highly effective for induction of remission (RR of no remission 0.79 with 5-ASAs; 95% CI, 0.73–0.85; NNT = 6). Higher effect was observed at ≥ 2.0 g/day dose (RR, 0.91; 95% CI, 0.85–0.98).¹²⁰ However, there is no increase in remission rates at doses of > 2.5 g/day. 5-ASA is also effective in preventing relapse of quiescent UC as compared to placebo (RR of relapse, 0.65; 95% CI, 0.55–0.76; NNT = 4). Doses ≥ 2.0 g/day were more effective in preventing relapse as compared to < 2.0 g/day (RR, 0.79; 95% CI, 0.64–0.97).¹²⁰ A multicenter, double-blind study on 68 patients with mild-to-moderate active UC in the age range 5 to 17 years, reported an achievement of clinical improvement in 45% and 37% of patients who received 5-ASA agent (oral balsalazide) 6.75 and 2.25 g/day, respectively while clinical remission was seen in 12% and 9% of patients respectively. The study supported the tolerability and safety profile of 5-ASA and its role in clinical remission in pediatric UC patients.¹²¹ In another multicenter, randomized, double-blind study, 39% pediatric mild to moderate UC patients clinically improved with olsalazine and were asymptomatic after 3 months, com-

Statement 25

Suggested dosing is oral 5-ASA 60 to 80 mg/kg/day to 4.8 g daily; rectal 5-ASA 25 mg/kg up to 1 g daily; sulfasalazine 40–70 mg/kg/day up to 4 g daily. • ■ Grade of recommendation: B, Level of evidence: I, Voting: 72.7% strongly agree, 27.3% agree An RCT enrolled 83 pediatric patients (age group: 5–17 years) who were treated with low dose delayed release 5-ASA viz. 27–71 mg/g/day and high dose delayed release 5-ASA viz. 53–118 mg/g/day. Both low- and high-dose arms demonstrated similar efficacy in 81 of 83 modified intent to treat population in short term treatment of mild to moderate active UC.¹¹⁸ In another Phase I RCT, 52 children with UC, aged 5 to 17 years, and stratified by weight (18–82 kg), received multi-matrix 5-ASA at doses of 30, 60, or 100 mg/kg/day OD (maximum 4,800 mg/day) for 7 days. Children and adolescents with UC had pharmacokinetic profiles of 5-ASA and metabolite acetyl-5-ASA similar to historical adult cohort.¹²⁵ Children can therefore receive a dose that is similar to the adult dose. In an open-label trial 5-ASA 500 mg suppository daily at bedtime was found to be efficacious and safe in children with ulcerative proctitis.¹²³ 5-ASA enemas in the dose of 25 mg/kg (up to 1 g) daily for 3 weeks along with oral dose, can induce clinical remission and response in 42% and 71% pediatric UC patients, respectively, after 3 weeks.¹¹⁷

Statement 26

Treatment with 5-ASA or sulfasalazine, if effective, should be continued during pregnancy as available evidence does not suggest adverse fetal outcomes. • ■Grade of recommendation: A, Level of evidence: I, Voting: 77.3% strongly agree, 22.7% agree

Statement 27

5-ASA is safe in lactating mothers. • ■Grade of recommendation: C, Level of evidence: II-3, Voting: 68.0% strongly agree, 32.0% agree Breast feeding might be protective against early onset IBD.¹³¹ Sulfasalazine is of low risk during breastfeeding as confirmed by multiple prospective clinical trials.¹³² Though sulfapyridine moiety is absorbed in minimal amounts and is excreted in breast milk, the milk/serum ratio is acceptable.^{131,133} 7. Side Effects and Monitoring of 5-ASAs

Statement 28

5-ASA is well tolerated by most patients. • ■Grade of recommendation: A, Level of evidence: I, Voting: 100% strongly agree A systematic review and meta-analysis showed that the likelihood of experiencing any adverse event with 5-ASAs was not statistically different when compared with placebo both for induction of remission (10 RCTs: RR, 1.02; 95% CI, 0.81–1.29) and in prevention of relapse (5 RCTs: RR, 0.98; 95% CI, 0.84–1.15).¹²⁰ A Cochrane systematic review with 53 RCTs of parallel design, with minimum treatment duration of 4 weeks (n = 8,548) concluded that the difference in incidence of adverse events with 5-ASA agents and placebo did not reach statistical significance. However, sulfasalazine was not as well tolerated by the patients as 5-ASA (29% vs. 15% experiencing adverse events:

Statement 29

Paradoxical worsening of colitis may occur on initiation of 5-ASA due to drug hypersensitivity and requires discontinuation of the drug. • ■Grade of recommendation: A, Level of evidence: I, Voting: 66.7% strongly agree, 33.3% agree A systematic review of 46 randomized trials on short-term adverse effects of 5-ASA agents in UC reported that approximately 3% of patients on oral 5-ASA agents develop paradoxical worsening of their symptoms such as abdominal pain, diarrhea and blood in stools, and cramping.¹³⁴ Symptoms usually disappear when 5-ASA agents are discontinued. These patients should be considered allergic to 5-ASAs, drug should be withdrawn, and no 5-ASA preparation should be used in these patients.¹³⁴ The precise mechanism of exacerbation of colitis induced by 5-ASA is not clear. Scheurlen et al.¹³⁵ mentioned that diarrhea caused by 5-ASA was attributed to a secretory mechanism secondary to inhibition of ileal and colonic Na + K + ATPase. Another proposed mechanism is an alteration of arachidonic acid metabolism manifesting as secretory diarrhea and malabsorption.

Statement 30

5-ASA may cause renal toxicity and therefore monitoring of renal function should be carried out at least once in a year. • ■Grade of recommendation: A, Level of evidence: I, Voting: 63.6% strongly agree, 31.8% agree, 4.5% disagree Nephrotoxicity is a rare adverse effect of 5-ASA therapy. Muller et al.¹³⁶ reported the incidence of nephrotoxicity in patients with IBD taking 5-ASA to be about 1/4,000 patients/year. Another systematic review reported less than 0.5% incidence of nephrotoxicity in patients with IBD consuming 5-ASA.¹³⁷ 5-ASArelevant nephrotoxicity includes interstitial nephritis, glomerulonephritis, and minimal-change nephropathy with nephrotic syndrome.¹³⁸ Renal impairment is reported to occur as early as within 29 days and as late as 5 years;^{136,139} 50% of cases present within 1 year of treatment initiation.¹³⁷ No relationship between type/dose of 5-ASA therapy and development of renal disease has been reported. The time of diagnosis of 5-ASA nephrotoxicity and subsequent discontinuation of drug determines the course of renal impairment.¹⁴⁰ A review showed that if 5-ASA-associated

Statement 31

Dose modification is not required in chronic renal failure except that renal function should be monitored more closely. • ■Grade of recommendation: C, Level of evidence: III, Voting: 52.0% strongly agree, 44.0% agree, 4.0% disagree Patients with preexisting renal dysfunction may be more likely to suffer 5-ASA nephrotoxicity than those with normal renal function.¹⁴³ There is lack of RCT on the safety of 5-ASA agents in IBD patients suffering from chronic kidney disease.¹³⁷ IBD patients are more susceptible to renal damage during acute exacerbations, such as during an acute episode of diarrhea.¹³⁷ Therefore, IBD patients with chronic kidney disease need close monitoring,¹⁴⁴ especially in acute exacerbations of disease, in case of 5-ASA induced nephrotoxicity, and in patients with comorbid conditions known to affect renal functions, such as diabetes and hypertension.^{137,145-147}

Statement 32

Although sulfasalazine has the potential to cause decreased fertility in men, 5-ASA formulations do not affect fertility in men or women. • ■Grade of recommendation: B, Level of evidence: II-1, Voting: 47.6% strongly agree, 42.9% agree, 9.5% disagree Various studies have shown the reversible adverse effect of sulfasalazine therapy on fertility in males with IBD. However, no such effects were observed on fertility of female patients. Infertility due to sulfasalazine in males is attributed to its sulfapyridine component rather than 5-ASA, the therapeutically active component. Sulfasalazine leads to changes in sperm morphology, decreased sperm motility and sperm count.^{127-129,131,133,148} Birnie et al.¹⁴⁹ reported abnormal semen production and oligospermia in 86% and 72% patients respectively, treated with sulfasalazine. However, infertility in males is reversible and preventable by switching sulfasalazine to other 5-ASA formulations.^{6,150} Absence of sulfapyridine moiety in the newer 5-ASAs

Statement 33

Folate supplementation is required with use of sulfasalazine, other 5-ASA formulations do not require it. • ■Grade of recommendation: B, Level of evidence: II-3, Voting: 62.5% strongly agree, 29.2% agree, 4.2% disagree, 8.3% strongly disagree Sulfasalazine therapy is known to decrease folate synthesis by inhibiting dihydrofolate reductase and reducing folate absorption. These effects are not observed with other 5-ASA formulations.^{129,152,153} Folic acid supplementation in sulfasalazine treated pregnant women, decreases the incidence of oral clefts, fetal neural tube defects and cardiovascular anomalies in newborns.^{129,152} Pregnant women with IBD being treated with sulfasalazine should receive higher doses of folic acid (2 mg/day) than pregnant women without IBD.^{128,131,133,150} 8. Chemopreventive Role of 5-ASA

Statement 34

Long-term 5-ASA therapy in extensive/distal colitis is possibly associated with reduced risk of colon cancer in UC. • ■Grade of recommendation: B, Level of evidence: II-1, Voting: 26.9% strongly agree, 57.7% agree, 15.4% disagree A systematic review (6 case-control studies) showed that 5-ASA use in IBD patients may have chemopreventive effect. Four of the 6 studies exhibited significant risk reduction with 5-ASA ($P < 0.05$), while risk reduction in the other 2 studies did not reach significance.¹⁵⁴ Qiu et al.¹⁵⁵ in their systematic review and meta analysis (26 observational studies; 15,460 patients) also reported significant decrease in colorectal cancer (CRC) in UC patients (OR, 0.46; 95% CI, 0.34–0.61) but not in CD patients (OR, 0.66; 95% CI, 0.42–1.03). 5-ASA use correlated significantly with CRC (OR, 0.54; 95% CI, 0.39–0.74) but not with dysplasia (OR, 0.47; 95% CI, 0.20–1.10). The meta-analysis further showed significant protective effect of 5-ASA only in clinical studies (OR, 0.51; 95% CI, 0.39–0.65) and not in population-based studies (OR, 0.71; 95% CI, 0.46–1.09). Another systematic review and meta-analysis of 17 observational case-control, retrospective cohort, population- and hospital-based studies showed that the use of 5-ASA correlated with reduced risk of CRC in UC (OR, 0.63; 95% CI, 0.48–0.84).¹⁵⁶ The chemopreventive efficacy of 5-ASA increased with higher average daily dose (sulfasalazine ≥ 2 g/day and

Statement 35

Adherence to 5-ASA therapy improves outcome in patients with IBD. Hence, patient education is important. •
■Grade of recommendation: B, Level of evidence: II-2, Voting: 56.5% strongly agree, 39.1% agree, 4.3% disagree

Adherence to medication plays a significant role in the management of IBD as patients need long-term therapy. Non-adherence lowers the effectiveness of treatment and also increases cost of therapy. Non-adherence to therapy is a major concern. A self-reported survey from India revealed that 81% of patients were non-adherent to treatment, defined as taking 80% or less of the dose advised. The reasons for non-adherence (not mutually exclusive) were forgetfulness (77%), felt better (14.2%), high frequency of doses (10.1%), no effect of medications (7.9%), and non-availability of medications (2.3%). Non-adherent patients were three times more likely to develop a relapse as compared to those with adherence (OR, 3.38; 95% CI, 1.29–8.88; $P = 0.012$).¹⁵⁹ In contrast, another study from northern India (266 IBD patients) showed that more than 80% of patients were adherent to their medications, adherence being the least for topical therapy. Higher education, professional occupation, and upper socio-economic status were associated with lower adherence to medications.¹⁶⁰ Two cohort studies found that non-adherent patients are likely to experience greater risk of disease relapse compared to those who are adherent (HR, 5.5; 95% CI, 2.3–13; $P < 0.001$ and RR, 1.44; 95% CI, 1.08–1.94; $P = 0.014$, respectively).^{161,162} Although there is paucity of data on the impact of health ed-

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Singh A et al. *Intest Res.* 2021;19(3):269-285. doi:10.5217/ir.2020.00155

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Statement 8a (NHL: commonest malignancy; risk in elderly)

Non-Hodgkin's lymphoma (NHL) is the commonest malignancy associated with thiopurine therapy — mainly in elderly patients. Risk decreases gradually after thiopurine withdrawal. Risk-benefit discussion is mandatory in patients over 65 years.

Statement 8b (HSTCL: combination therapy + young males)

HEPATOSPLENIC T-CELL LYMPHOMA (HSTCL): a peripheral T-cell lymphoma occurring in YOUNG MALES receiving COMBINATION therapy with thiopurines + anti-TNF agents for ≥ 2 years. This is a rare but typically FATAL malignancy. Counsel patients on this risk before initiating or continuing combination therapy.

Statement 8c (Thiopurines in EBV-seronegative patients: avoid)

Thiopurines increase the risk of primary EBV infection in seronegative patients, which can progress to aggressive and fatal lymphoma. THIOPURINES SHOULD BE AVOIDED IN EBV-SERONEGATIVE PATIENTS — alternative therapies should be used. Check EBV serostatus before starting thiopurines.

Statement 8d (Other malignancy risks: NMSC, cervical cancer)

Other malignancy risks associated with thiopurines include: increased risk of non-melanoma skin cancer (NMSC) — particularly squamous cell carcinoma — and uterine cervical cancer. Annual skin surveillance and regular cervical screening are required.

Statement 3 (Thiopurines in elderly: effective but higher risk)

Thiopurines in the ELDERLY (≥ 60 years): well-tolerated and as efficacious as younger patients for maintenance of IBD remission. However, comorbidities limit use (drug-drug interactions) and elderly patients are at INCREASED risk of infections, lymphoma, and hepatotoxicity with thiopurines. Benefit-risk discussion is essential. Monitor closely.

Statement 4 (Thiopurines in children: second-line; adverse effects common)

Thiopurines in PAEDIATRIC IBD are RESERVED AS SECOND-LINE therapy due to an unfavourable safety profile. Adverse effects occur in 15–28% of paediatric IBD patients; up to 18% require discontinuation. Paediatric patients are at increased risk of HSTCL, particularly males on combination therapy.

Statement 5a (Combination IFX+AZA superior to monotherapy)

Combination therapy of thiopurines + infliximab (IFX) is SUPERIOR to either therapy alone in inducing remission in both UC and CD. Thiopurines reduce anti-drug antibody (ADA) formation against IFX and improve IFX trough drug levels.

Statement 5b (AZA with adalimumab: evidence not convincing)

Evidence for combining azathioprine with ADALIMUMAB is NOT convincing. Concomitant use of thiopurines with vedolizumab and ustekinumab is NOT currently recommended outside of exceptional circumstances.

Statement 5c (Stopping IFX while continuing thiopurines: high relapse)

In patients on combination IFX + thiopurine who achieve corticosteroid-free remission for 6–12 months: withdrawal of one agent may be considered. If IFX is STOPPED while continuing thiopurines: high cumulative relapse rates are expected. If IFX is stopped, complete mucosal healing at time of IFX withdrawal is associated with lower relapse risk.

Statement 1 Pregnancy (Continuation safe and recommended)

Continuation of pre-conception thiopurine therapy for maintaining remission during PREGNANCY is SAFE and RECOMMENDED. Meta-analyses show no increase in low birth weight or congenital anomalies attributable to thiopurines. Risk of preterm birth: slightly increased, likely related to underlying disease activity.

Pregnancy: thiopurine + biologic combination

Infants born to mothers on combination thiopurines + biologics are at increased risk of infections in the first year of life. Consider cessation of one agent (usually thiopurines) in the combination during pregnancy to reduce infant infection risk, balanced against maternal disease control.

Statement 2 Lactation (Breastfeeding allowed on AZA)

Breastfeeding is ALLOWED on azathioprine maintenance therapy. Meta-analyses show no increased risk of infections in infants breastfed by mothers on thiopurines. Thiopurines concentrate in breast milk during the first 4 hours after drug intake — advise patients to BREASTFEED at least 4 hours after drug ingestion to reduce infant exposure.

NUDT15 hypermethylator: low-dose thiopurine + allopurinol

A 6-MMPR:6-TGN ratio >11 indicates a thiopurine hypermethylator phenotype — these patients preferentially produce hepatotoxic 6-MMP metabolites rather than therapeutic 6-TGN. Management: switch to low-dose thiopurine + allopurinol combination to shift metabolite ratio towards 6-TGN.

Demographics and risk: gender and age considerations

Males are at increased risk of lymphoma with thiopurines. HSTCL and EBV-related T-cell lymphoma are more common in young adults. Thiopurine-induced pancreatitis is more common in FEMALES. Factor patient demographics (age, sex, EBV status) into the benefit-risk assessment when prescribing thiopurines.

Social history before thiopurine: alcohol and smoking

Obtain a detailed social history before initiating thiopurine therapy. Concomitant ALCOHOL ABUSE increases hepatotoxicity risk with thiopurines. SMOKING is associated with thiopurine-induced pancreatitis. Patients with these risk factors require closer monitoring and counselling.

COVID-19 and thiopurines (continued use appropriate)

Continuation of thiopurines does not appear to increase COVID-19 severity or risk of poor outcomes. Thiopurine-mediated cytokine suppression may in fact be beneficial for pulmonary complications of COVID-19. Thiopurines should not be routinely discontinued during COVID-19 infection unless there is severe systemic illness.

Statement 1

The efficacy and safety of thiopurines depend on several factors. Recognition of patient demographics is important. Increasing age is associated with increased risk of adverse effects including infections and malignancies. Certain malignancies such as hepatosplenic T cell lymphoma (HSTCL) and systemic Epstein-Barr virus (EBV; infectious mononucleosis) related T cell lymphoma are more common in young adults. While males are at an increased risk of developing lymphoma; other adverse effects, pancreatitis in particular, are more common in females. Concomitant alcohol abuse increases the probability of hepatotoxicity whereas smoking is associated with pancreatitis. Elaborate clinical history and examination and laboratory evaluation (complete blood count [CBC], liver function tests, renal function tests, and blood glucose) for as-

Statement 2

The efficacy and safety of all thiopurine agents (AZA, 6-MP, and 6-TG) is similar and the choice of agent depends on local prescribing preferences, physician and patient preference, availability and financial implications. The majority of physicians in India use AZA.^{23,28} A designer thiopurine analogue; B-ON, has also been developed using rational drug design.⁴⁹ It mediates earlier and stronger induction of T cell apoptosis compared to 6-MP and is associated with fewer adverse effects in vitro, however further studies consolidating these

Statement 4

■Combination Therapy of Thiopurines and Anti-TNF Agents Combination therapy of thiopurines has been extensively

Statement 1

Leucopenia Myelosuppression is the commonest (2%–12%) dose-dependent adverse effect and the most common cause of drug discontinuation. The presentation is usually with leucopenia, although other cell lines may also be suppressed in conjunction (anemia, thrombocytopenia). A decline in leukocyte count within 1–2 weeks of therapy may be a marker of impending leucopenia. Although myelosuppression can occur at any time, severe myelosuppression is less frequent after the first 8 weeks of therapy. The TPMT and NUDT15 variants are associated with increased risk of leucopenia. As discussed earlier, the frequency

Statement 2

Hepatotoxicity The spectrum of thiopurine-induced liver injury includes hypersensitivity reaction (acute hepatitis), cholestasis (mixed or bland) and endothelial injury (veno-occlusive disease, peliosis hepatis and nodular regenerative hyperplasia). While hypersensitivity and cholestasis are idiosyncratic reactions, occurring early; injury mediated by endothelial cell damage is considered to be dose-dependent and a late event.⁷² The most common form of thiopurine-induced liver injury is transient asymptomatic elevation of liver enzymes, which requires continuation of therapy with close monitoring. At times persistently rising liver enzymes warrant dose reduction and/or discontinuation of therapy.⁷³ Split-dosing and LDTA are other possible strategies to overcome asymptomatic TPMT-related elevation of liver enzymes. The acute cholestatic injury due to AZA usually presents with fatigue and jaundice, pruritis is uncommon. Cholestatic hepatic injury usually resolves on stopping the drug, but rarely it can be prolonged or culminate into vanishing bile duct syndrome.⁷⁴ Endothelial injury to liver manifests as fatigue with signs and symptoms of hepatic parenchymal injury and portal hypertension. In cases of veno-occlusive disease an acute presentation with abdominal pain and ascites can be seen. Bio-

Statement 3

Acute Pancreatitis Acute pancreatitis is an idiosyncratic complication reported in about 3% of patients. The episodes are usually mild; female sex and smoking being the probable risk factors.^{76,77} 6-MP should not be used in patients with AZA-induced pancreatitis. However, 6-TG (at doses of 20–40 mg) maybe a safer alternative to AZA/6-MP for patients who are at high risk of AZA/6MP induced acute pancreatitis.⁷⁸

Statement 4

Gastrointestinal Intolerance Nausea, vomiting and, rarely, diarrhea and pain abdomen are side effects of thiopurines encountered during initial few weeks. These are usually self-limiting. Successful shifting from AZA to 6-MP or 6-TG in such patients has been reported.⁷⁹

Statement 5

Infections Immunosuppressive actions of thiopurines affect both innate and acquired immune systems, thus increasing the risk of opportunistic as well as non-opportunistic infections. Concomitant use of other immunosuppressive drugs further accentuates the risk.⁸⁰ The risk of opportunistic infections was markedly increased if thiopurines were used in combination with systemic steroids as compared to monotherapy with thiopurines (odds ratio [OR], 17.5; 95% CI, 4.5–68 and OR, 3.8; 95% CI, 2.0–7.0, respectively).⁸¹ A meta-analysis showed that the combination of thiopurine and anti-TNF therapy increased the risk of serious infections compared to monotherapy with thiopurines (hazard ratio, 2.11; 95% CI, 1.80–2.48).⁸⁰ Serious infections mainly involved the respiratory tract, gastrointestinal tract, and skin. Opportunistic infections with viruses (varicella, cytomegalovirus, EBV, etc.), bacteria and mycobacteria have been encountered with increased frequency with use of thiopurines.⁸⁰ Pre-therapy screening for HBV, HCV, HIV, varicella and tuberculosis is therefore recommended.⁸² Patients should be sensitized to seek help promptly in case of fever or suspected infection to ensure early detection, diagnosis and treatment.

Statement 1

Pregnancy Thiopurines, being teratogenic in animals (previously labelled as Food and Drug Administration [FDA] class D), are of concern for their use in pregnancy and lactation. AZA crosses placenta and its metabolites are present in fetal RBCs. Meta-analyses have found no increase in the risk of low birth weight or congenital anomalies with use of thiopurines in pregnancy; however the risk of preterm birth was higher.^{90,91} The risk of preterm birth was proposed to be associated with IBD activity rather than thiopurines in a large Swedish registry.⁹² Therefore, continuation of pre-conception therapy with thiopurines for maintaining remission during pregnancy is safe and recommended. Infants born to mothers receiving thiopurines in combination with biologics have an increased risk of infections and cessation of one of the 2 drugs, usually thiopurines, should be considered.⁹² Also LDTPA should not be used because of teratogenic risks of allopurinol. Initiating thiopurines for the first time in pregnancy is not recommended due to uncertainty of complications and delayed therapeutic response.⁷²

Statement 2

Lactation Assessment of long-term outcomes in babies born to mothers taking AZA during pregnancy or lactation showed no increased risk of infections. Therefore, breast feeding can be allowed during maintenance therapy with AZA. The major part of thiopurines in breast milk is excreted within the first 4 hours after drug intake and the maximum exposure of the drug to the infant is less than 1% of the maternal dose. Therefore the lactating mother should breastfeed at least 4 hours after the drug ingestion to reduce the infant's drug exposure. The practice of "pumping and dumping" 4 hours breast milk is neither necessary nor likely to be effective and should be discouraged.⁹³

Statement 3

The number of elderly with IBD is increasing worldwide due to longevity of survival of patients with adult-onset IBD or due to development of elderly-onset IBD. Among the elderly IBD patients, UC (E2 > E3) is more common than CD (colonic > ileal). Thiopurines, when used for maintenance of remission in elderly, are well tolerated and have been shown to be as efficacious as in younger IBD patients. However, associated comorbidities limit their use (drug-drug interactions). Elderly patients are also at an increased risk of adverse effects, including infections, and malignancies. Therefore, the long-term use of thiopurines in elderly should be considered carefully, weighing risks versus benefits.⁹⁴

Statement 4

Children The Paediatric IBD Porto Group of European Society for Paediatric Gastroenterology Hepatology and Nutrition has differentiated pediatric IBD into 5 categories: typical UC, atypical UC, IBD-unclassified (IBDU), Crohn's colitis, and CD.⁹⁵ While pancolitis is the predominant phenotype for UC, inflammatory ileo-colonic disease is the commonest presentation in CD. Pediatric-onset IBDU is twice as common as adult-onset disease. Prolonged maintenance of corticosteroid-free remission and mucosal healing aids in improved growth in children with IBD. Clinical remission in pediatric IBD can be maintained with 5-ASA, thiopurines or biologics. Use of thiopurines is associated with decreased rates of hospitalization, corticosteroid use, and surgery in both UC and CD; but they are reserved as a second line therapy in children due to unfavorable safety profile. Adverse effects may occur in 15%–28% of pediatric IBD patients, and up to 18% may require discontinuation of therapy due to the same. In addition, there is an increased risk of infections, HSTCL, EBV positive mononucleosis associated T cell lymphoma and NMSC in young patients. Therefore, thiopurines should be used with caution with regular moni-

Statement 5

COVID-19 During the current pandemic of coronavirus disease 2019 (COVID-19), continuation of thiopurines and other immunosuppressive agents which can theoretically increase the risk of infections is a major concern. However, the currently available evidence does not suggest increased risk of COVID-19 in patients with IBD as compared to non-IBD patients.⁹⁹ It is noteworthy that COVID-19 disease severity is characterized by increased levels of various cytokines (interleukins [IL-2, IL-6, IL7], granulocyte-colony stimulating factor, TNF- α , etc.).¹⁰⁰ This cytokine storm in COVID-19 resembles the cytokine profile of inflamed intestine in IBD. Hence, suppression of cytokines by thiopurines may in fact be beneficial for pulmonary complications. Also, as thiopurines act by incorporating into DNA/RNA

Statement 6

■Vaccination for Patients with IBD Receiving Thiopurines As there is an increased risk of infections in patients on thiopurines, vaccination is recommended for all patients with IBD prior to initiation of immunosuppression (Tables 1, 2). Live

Statement 1

Thiopurines are effective for maintenance of remission for CD and UC.

Statement 2

An elaborate clinical workup is required for initiating and monitoring therapy with thiopurines.

Statement 3

Screening for infections like HBV, HCV, VZV, and EBV is recommended.

Statement 4

■Daily doses of AZA, 6-MP and 6-TG are 2–3 mg/kg, 1–1.5 mg/kg, and 0.2–0.3 mg/kg respectively. Lower doses of AZA (< 2 mg/kg/day) are effective in Asians. Combination of low dose thiopurines and allopurinol can be used in patients who are hypermethylators with 6-MMPR: 6-TGN ratio > 11. Genetic testing for polymorphisms in TPMT and NUDT15 can guide about the starting dose and predict dose-dependent adverse effects.

Statement 5

■Combination of thiopurines with IFX is superior to either therapy alone in inducing remission in both UC and CD. Thiopurines decrease the formation of anti-drug antibodies and thereby improve the clinical efficacy of IFX. Evidence for combining AZA with adalimumab is not convincing. Concomitant use of thiopurines with vedolizumab and ustekinumab is not recommended at the moment.

Statement 6

■ Thiopurines have a narrow therapeutic window, therefore monitoring of treatment is required for optimization of therapy and prevent adverse effects. 6-TGN and 6-MMPR are the 2 metabolites measured in clinical practice to guide therapy. Mean corpuscular volume may be considered a surrogate marker for thiopurine metabolites where drug metabolite levels are not available/affordable.

Statement 7

■ Thiopurine use is associated with development of both idiosyncratic and dose-dependent adverse effects. Idiosyncratic adverse effects include fatigue, flu like illness, myalgias, gastrointestinal intolerance, headache, and pancreatitis whereas leucopenia, myelosuppression and hepatotoxicity are dose-dependent adverse effects.

Statement 8

■ Thiopurines carry a risk of development of malignancy. Lymphoma (non-Hodgkin lymphoma, NHL) is the commonest malignancy, usually seen in elderly patients receiving thiopurines. The risk decreases gradually after withdrawal of therapy. Hepatosplenic T cell lymphoma is a peripheral T cell lymphoma that occurs in young males receiving combination therapy with thiopurines and anti-TNF agents for ≥ 2 years. Thiopurines increase the risk of development of primary EBV infection (especially in seronegative patients) which can progress to an aggressive and fatal lymphoma. Thiopurines should therefore be avoided in EBV seronegative patients. There is also an increased risk of nonmelanoma skin cancer and uterine cervical cancer.

Statement 9

■ Pregnancy in patients with IBD should be planned when the disease is in clinical remission. Thiopurines can be continued during pregnancy and lactation.

Statement 10

■ Thiopurine use in elderly (≥ 60 years) is associated with increased risk of adverse effects including infections and malignancies and therefore should be used with caution.

Statement 11

Thiopurines can be continued in patients with IBD at risk for COVID-19.

Statement 12

Vaccination with live (except MMR) and inactivated vaccines is recommended. CD, Crohn's disease; UC, ulcerative colitis; HBV, hepatitis B virus; HCV, hepatitis C virus; VZV, varicella zoster virus; EBV, Epstein-Barr virus; AZA, azathioprine; 6-MP, 6-mercaptopurine; 6-TG, 6-thioguanine; 6-MMPR, 6-methyl mercaptopurine ribonucleotides; 6-TGN, 6-thioguanine nucleotides; TPMT, thiopurine-S-methyltransferase; NUDT15, nucleoside diphosphate-linked moiety X-type motif 15; IFX, infliximab; TNF, tumor necrosis factor; IBD, inflammatory bowel disease; COVID-19, coronavirus disease 2019; MMR, measles mumps rubella.

Statement 1

GBD 2017 Inflammatory Bowel Disease Collaborators. The global, regional, and national burden of inflammatory bowel disease in 195 countries and territories, 1990-2017: a systematic analysis for the Global Burden of Disease Study